

**VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATION SYSTEMS**



DATA, PROCESS AND OBJECT MODULING

TOPIC:

WEB APPLICATION FOR PRACTICING ENGLISH

Instructor:

Dr. Cao Thi Nhan
Msc.Duong Phi Long

Class: IEM5723.Q11.CTTT

Student:

Nguyen Chu Nguyen Chuong	22520165
Pham Thi Minh Hien	22520422
Pham Quang Dai Phuc	22521137
Le Minh Sang	22521240

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SELF ASSESSMENT

No	Name	Role	Contribution	Contribution
1	Nguyen Chu Nguyen Chuong	Leader	- Designed the Entity Relationship Diagram (ERD). - Mapped the Entity Class Diagram to the Relational Database Model. - Analyzed and designed the Create Test functionality (Use-case, Activity, Sequence, and Class diagrams). - Overall report compilation and final review.	100/100
2	Pham Thi Minh Hien	Member	- Analyzed and designed administrative functions: Update Test and Delete Test. - Analyzed and designed retrieval functions: Search Test, Sort Test, and View Test Details. - Created comprehensive UML diagrams for the above modules.	100/100
3	Pham Quang Dai Phuc	Member	- Analyzed security flows and user identification. - Specific functions designed: Login, Register, and Google Login integration. - Created comprehensive UML diagrams (Use-case, Activity, Sequence, Class) for the authentication system.	100/100
4	Le Minh Sang	Member	- Analyzed and designed the core student functionality: Take the Test. - Created UML diagrams regarding the test-taking process. - Conducted System Testing, defined Test Cases, and wrote the Implementation & Testing documentation.	100/100

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PROJECT SUMMARY

This report presents the analysis and design of Enly – an English learning and test preparation website. The project aims to build a structured platform that allows students to practice English tests online, while providing teachers and administrators with tools to manage content and user activities. The system is designed following the principles of object-oriented analysis and design (OOAD), and all major UML models required by the course are developed.

The requirements analysis identifies three main actors: Student, Teacher, and Admin—and defines their interactions with the system through use-case model. Key functionalities include user management, viewing and managing courses, creating and performing practice tests, automatic scoring, and communication through a contact page. Detailed use-case specifications describe the main flows, alternative flows, and pre/postconditions of critical functions such as “Take Practice Test.”

The behavioral modeling section presents activity diagrams, which illustrate the flow of actions within major use cases, highlighting decision points and parallel activities. Sequence diagrams demonstrate how system components interact over time to complete tasks such as test creation, test execution, and scoring. A state diagram is also developed to show changes in the lifecycle of a practice test.

The structural modeling includes a detailed class diagram, describing the system’s logical architecture, class relationships, attributes, and methods. An entity class diagram is also constructed to represent data entities used for database design. Based on this model, a complete relational database schema is derived with clearly defined primary and foreign keys.

Overall, the report outlines a fully analyzed and well-structured system design for the Enly platform. The models and diagrams serve as a solid foundation for future implementation, ensuring that the system is scalable, maintainable, and aligned with user needs.

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LIST OF ABBREVIATIONS

No.	Abbreviation	Content
1	IELTS	International English Language Testing System
2	TOEIC	Test of English for International Communication
3	OOAD	Object-Oriented Analysis and Design
4	UML	Unified Modeling Language
5	API	Application Programming Interface
6	UI/UX	User Interface / User Experience
7	AI	Artificial Intelligence
8	ACID	Atomicity, Consistency, Isolation, Durability
9	ORM	Object-Relational Mapping
10	FK	Foreign Key

CHAPTER 1. GENERAL INTRODUCTION

1. Introduction

In recent years, the demand for online English learning and standardized test preparation has increased significantly, especially for international examinations such as IELTS and TOEIC. Learners require an accessible, reliable, and user-friendly digital platform where they can practice multiple test types, track their progress, and receive timely feedback. At the same time, organizations providing test-preparation services require modern information systems to manage exam content, monitor learner activities, and support instructors in evaluating Writing and Speaking submissions.

To address these needs, the project introduces Enly, an online web application designed to assist students in practicing English through structured test modules. The system integrates essential functionalities for learners, teachers, and administrators. By applying principles of Object-Oriented Analysis and Design (OOAD), the project aims to model the system using UML diagrams - including use-case, activity, sequence, state, and class diagrams - to ensure a well-organized and scalable software structure. This chapter provides an overview of the project context, the challenges motivating system development, the objectives of the new platform, and the scope limitations of the study.

2. Problems of project

Through interviews with students, teachers, and administrators, as well as an examination of the organization's current system, several problems were identified:

2.1. Limited Functionality for Learners

The current platform still exhibits several limitations. It does not fully support structured online test-taking for complete IELTS or TOEIC modules, causing gaps in the learning experience. Test searching and filtering functions are inconsistent and often inefficient, making it difficult for users to locate suitable practice materials. Additionally, learners lack a centralized interface to review their test history, performance data, and detailed answer analysis. The authentication system is also limited, as it does not offer convenient sign-in options such as Google OAuth, resulting in reduced accessibility and user convenience.

2.2. Inefficiencies in Exam Content Management

Exam content management within the existing system is highly inefficient. The process of creating tests is largely manual, especially for complex sections such as Reading and Listening that require multiple question types and structured content. The absence of necessary validation rules frequently leads to incomplete, inconsistent, or incorrect test configurations, which can negatively affect the learner experience. Moreover, administrators do not have adequate tools to review, update, or verify test content efficiently. This lack of support increases the risk of errors and requires significant manual effort to maintain test quality.

2.3. Fragmented Workflow for Writing and Speaking Evaluation

The workflow for handling Writing and Speaking submissions is fragmented and lacks centralization. Learners often submit their responses through external platforms or manual channels, resulting in scattered and difficult-to-track data. Teachers face challenges in managing and monitoring these submissions, and inconsistencies in the evaluation process may arise due to the absence of a unified system. Additionally, without standardized data structures for collecting and organizing responses, delays in grading and feedback delivery frequently occur, affecting the timeliness and overall effectiveness of the evaluation process.

2.4. System Scalability and Maintenance Limitations

The existing platform also faces significant limitations in scalability and long-term maintainability. It was not designed to support an expanding range of test types or a growing number of users, making it unsuitable for future organizational needs. The data models currently in use do not accommodate flexible question formats, multimedia resources, or large-scale data storage, which restricts the system's ability to evolve. Furthermore, technical inconsistencies within the system hinder potential upgrades or the integration of advanced features such as automated scoring, thereby limiting the platform's developmental potential.

3. Objectives of project

3.1. Provide a Complete Digital Testing Experience

The first objective of the project is to deliver a complete and structured digital testing experience for learners. The system aims to support practice tests across IELTS Academic, IELTS General, TOEIC, and TOEIC Speaking - Writing, ensuring users can simulate real exam conditions. It will also provide intuitive navigation for selecting test categories, viewing detailed descriptions, and choosing between full-test or part-based practice modes. Additionally, the platform will incorporate automatic scoring for objective questions and offer a well-organized interface for learners to review their answers and understand their performance more effectively.

3.2. Improve Exam Content Management for Administrators

Another key objective is to enhance the efficiency and accuracy of exam content management. The system will provide administrators with modern tools to create, edit, view, sort, and delete tests with ease. Strong data validation mechanisms will be implemented to prevent incomplete or inconsistent test configurations, reducing errors caused by manual work. Furthermore, the platform will allow the use of flexible multimedia resources - including audio files, passages, and prompts - making it possible to construct diverse and high-quality test materials that align with standardized exam formats.

3.3. Support Teacher Workflows for Writing and Speaking Evaluation

The project also seeks to improve the workflows for Writing and Speaking evaluation by centralizing the submission and grading process. Learners will be able to upload their responses directly through the system, and submissions will be forwarded to teachers using a standardized and consistent format. This approach ensures that educators can manage and track submissions more efficiently. After evaluation, graded results and feedback will be presented to learners in a clear and structured manner, enabling them to better understand their strengths and weaknesses and monitor their learning progress over time.

3.4. Establish a Scalable, Maintainable System Architecture

A further objective is to develop a scalable and maintainable architecture that supports long-term growth. By applying Object-Oriented Analysis and Design (OOAD) principles, the system will be modeled using UML diagrams to describe its behavior, interactions, and structural components. The database will be designed to accommodate future expansion, including the introduction of new exam types or modules. This ensures that the platform remains adaptable, stable, and efficient as user demands evolve and organizational needs expand.

4. Limitations of the study

4.1. No Implementation of Advanced AI or Automatic Scoring

A major limitation of the study is that it does not include the implementation of advanced AI-powered features. Automated scoring for Writing and Speaking tasks, personalized learning recommendations, or performance analytics dashboards fall outside the scope of this project. The focus remains primarily on core system functionalities and conceptual modeling rather than intelligent automation.

4.2. Reduced Focus on Performance Testing

Performance testing, such as evaluating load capacity, stress limits, or response times under heavy traffic, is not conducted within this study due to academic constraints. Therefore, the system's behavior in real-world high-demand environments is not assessed, and any required optimization strategies are not explored.

4.3. UML Diagrams May Not Capture All Real-World Scenarios

The UML diagrams created in this project represent idealized system workflows based on available information. In practical development, additional complexities or edge cases may emerge that require refinement or adjustment to these models. Consequently, the diagrams should be viewed as foundational rather than fully exhaustive representations of the final system.

CHAPTER 2. REQUIREMENT SPECIFICATION

This section identifies the software requirements needed to support the organization's online test-preparation activities. Based on interviews and analysis of current operations, the system requirements are divided into functional requirements, which specify the core behaviors the system must perform, and non-functional requirements, which define the quality attributes necessary for reliable and efficient operation. These requirements provide the foundation for designing a modern, integrated system that addresses existing limitations and supports future growth.

1. Situation survey/ Survey of current situation

1.1. Interview

To gain a comprehensive understanding of the users' needs and the limitations of the existing system, a series of structured interviews were conducted with key stakeholder groups. These included students who frequently take online tests, administrators responsible for managing exam content, teachers involved in evaluating Writing and Speaking submissions, and operational staff who support day-to-day system activities. Each group provided unique insights into how the system is currently used and what improvements they expect to see in the new platform.

During the interviews, students emphasized the need for a more intuitive interface, faster access to test materials, and reliable tracking of test history and performance. They also expressed a strong preference for flexible login options, particularly Google authentication, as well as clearer navigation when selecting test categories or specific test components. Many students reported difficulties when searching for tests or filtering by category in the existing system, noting that the process was not always responsive or accurate.

Administrators highlighted several pain points related to test management. They noted that creating new exam content is time-consuming, especially for Reading and Listening tests with multiple question types. Missing validation checks often lead to incomplete or inconsistent test entries, which can disrupt the user experience.

Administrators also expressed the need for better support tools to edit, update, and review test structures before publishing them to users.

Teachers, particularly those responsible for evaluating Writing and Speaking tasks, shared concerns about the inefficiency of receiving and managing submissions. They emphasized the importance of having a structured workflow that sends students' responses directly to the teaching center for grading, along with complete and organized test details. Their feedback highlighted the need for better backend consistency to ensure that grading tasks are not delayed due to missing information or poorly structured test data.

Overall, the interviews provided essential insights into user expectations and system shortcomings. These findings form the foundation for defining the software requirements in the subsequent sections, ensuring that the new system aligns with real usage needs and supports both learning and administrative workflows more effectively.

1.2. Survey of the organization/ Current organizational situation

The organization currently provides an online examination and test-preparation platform supporting IELTS Academic, IELTS General, TOEIC, and TOEIC Speaking–Writing. Its primary user groups include students, administrative staff, and teachers involved in evaluating written and spoken responses. The organizational objective is to deliver a consistent, accurate, and user-friendly online testing experience.

At this stage, the core business workflow focuses on enabling students to search for tests, complete practice or full-exam sessions, and review their performance history. While functional, the existing system reveals several limitations, such as inconsistent data structures, insufficient validation during test creation, and the absence of advanced features like Google OAuth authentication. These shortcomings occasionally result in incomplete test configurations and reduced user satisfaction.

From an administrative perspective, exam content management remains largely manual. Administrators must input passages, audio files, question sets, and answers without automated validation support. This manual process is particularly challenging for Reading and Listening tests, which require multiple question types and structured data.

Consequently, frequent re-checking is needed to ensure the accuracy and completeness of test materials.

Teachers responsible for grading Writing and Speaking also encounter inefficiencies due to unstandardized submission formats and inconsistent data flow. These issues delay the evaluation process and highlight the need for a more integrated and coherent backend system.

Overall, the current organizational context demonstrates the necessity for system modernization. A more scalable, automated, and standardized platform is required to improve operational efficiency and meet the growing demand for high-quality online exam preparation.

1.3. Survey of business processes, major activities

The organization operates in the education and test-preparation sector, providing training services and standardized exam materials for learners preparing for international tests such as IELTS and TOEIC. Its core business processes focus on the development, delivery, and quality assurance of learning content, alongside the management of learner progress and instructional resources. The major activities include designing exam-preparation curricula, curating test materials, coordinating teaching staff, and supporting learners throughout their study journey.

A significant part of the organization's operations involves the creation and maintenance of exam content. Academic staff are responsible for developing passages, questions, and listening materials that reflect authentic exam conditions. These materials undergo internal review to ensure accuracy, alignment with exam standards, and appropriate difficulty levels. Additionally, teachers play a central role in evaluating learners' Writing and Speaking tasks, providing feedback, and tracking individual progress across training modules.

From an operational perspective, the organization manages daily workflows such as scheduling classes, distributing learning resources, responding to student inquiries, and ensuring timely communication between instructors and learners. Administrative personnel also oversee record-keeping, learner registration, and the monitoring of student outcomes

to assess training effectiveness. These activities collectively support the organization's educational objectives and uphold the quality of its training programs.

Overall, the business processes emphasize content quality, instructional coordination, and learner support—forming the foundation upon which digital systems and tools must be developed to enhance operational efficiency and improve service delivery.

1.4. Survey of existing system

The organization currently relies on a partially digitalized system to support exam preparation and administrative activities. The existing tools primarily consist of basic online forms, static learning materials, and manually curated test documents. While these resources enable learners to access practice content, the system lacks an integrated platform for managing test workflows, tracking learner progress, or automating administrative tasks.

Most processes include the creation of test materials, distribution of exam content, and collection of student responses which are handled manually by staff. This manual approach often leads to inconsistencies in test formatting, delays in updating content, and difficulties in maintaining data accuracy. Furthermore, the system does not provide centralized storage for test items or submission records, resulting in fragmented information and increased administrative workload.

From a user perspective, the current system offers limited interactivity. Learners can access practice materials but cannot perform structured online tests, receive automated scoring, or review performance data over time. For subjective components such as Writing and Speaking, submissions are typically collected through external channels (e.g., email or messaging platforms), making it difficult to track progress or ensure standardized evaluation.

Overall, the existing system provides only minimal digital support and lacks the functionality required to handle growing demand, diverse test formats, and large-scale learner management. These limitations highlight the necessity of developing a modern, centralized, and automated platform to improve operational efficiency and enhance the learning experience.

2. Software requirements

This section defines the software requirements for the proposed system, including both functional and non-functional specifications. These requirements are derived from stakeholder interviews, organizational workflow analysis, and an assessment of the limitations of the existing system.

2.1. Functional requirements

The functional requirements describe the specific behaviors and operations the system must support to meet organizational needs and user expectations. The major functional requirements include:

- **User Authentication and Account Management:** The system shall support user registration and authentication through both traditional email–password credentials and third-party login via Google OAuth. During registration and login, all input fields must be validated to ensure correct syntax and prevent invalid submissions. Once authenticated, the system will verify user credentials against stored records and assign appropriate access rights based on predefined roles, such as Student or Administrator.
- **Test Search and Retrieval:** The system shall enable users to search for available tests by entering keywords or filtering according to specific categories such as IELTS Academic, IELTS General, TOEIC, and TOEIC SW. When a user selects a particular test, the system must retrieve and display its full details to support informed decision-making. In cases where no matching results are found, the system shall provide a clear notification to ensure users understand that the search returned no available tests.
- **Test Execution and Submission:** The system shall support both part-based practice sessions and full-test execution across all supported exam formats. When a test is initiated, the system must retrieve all relevant content—including passages, audio files, questions, and answer keys—from the database. Users shall be able to input responses, submit their tests, and confirm submission actions. All test attempts, scores, and timestamps must be recorded automatically in the database, and upon completion, users should be able to review their results as well as the correct answers for further learning.
- **Test Management (Employee):** The system shall allow administrators to manage exam content by creating new tests, including the addition of passages, audio files, questions, and answer keys. It must support various question types such as multiple choice, fill-in-the-blank, and table-matching formats. Before saving, the system

shall validate the completeness of the test to ensure that all components—including questions and associated materials—are fully provided. Administrators must also be able to edit existing tests, delete outdated ones with confirmation prompts, and access detailed views of test structures for quality assurance.

- Writing and Speaking Workflow: The system shall support the submission of Writing and Speaking tasks by allowing learners to upload their responses for review. These submissions shall be forwarded to the teaching center for manual evaluation by academic staff. After grading, the system must store teacher feedback and scoring results, enabling learners to review their performance and progress over time.

2.2. Non-functional requirements

The system must meet several non-functional requirements to ensure stable, secure, and scalable operation. From a performance perspective, the system is expected to maintain efficient response times, with core API operations completing within two seconds and all web pages or test content loading within three seconds under normal conditions. To support increasing user demand, the platform should accommodate at least 500 concurrent users without degradation. In terms of usability, the interface must be intuitive, consistent, and accessible to users with different levels of technical experience. Clear navigation flows, meaningful validation messages, and user-friendly error notifications are essential to maintaining a smooth and effective user experience.

Security is also a critical requirement. All user passwords must be securely hashed, and authentication should rely on robust token-based mechanisms such as JWT, with third-party logins implemented in compliance with OAuth 2.0 standards. The system must also enforce strict role-based access control to prevent unauthorized access to administrative functionalities. Reliability and data integrity are equally important; the system should ensure that no information is lost during critical operations such as test creation, updates, deletions, or submissions. Comprehensive validation and confirmation steps must be applied to all high-risk actions to maintain consistent and accurate database records.

Finally, scalability is a key consideration for long-term sustainability. The system architecture should allow for future expansion, including the addition of new test formats,

increased data storage capacity, and potential integration of advanced features such as automated scoring or analytic dashboards. This ensures that the platform can evolve alongside organizational growth and changes in user demand.

CHAPTER 3. ANALYSIS AND DESIGN

1. Use-case Diagram

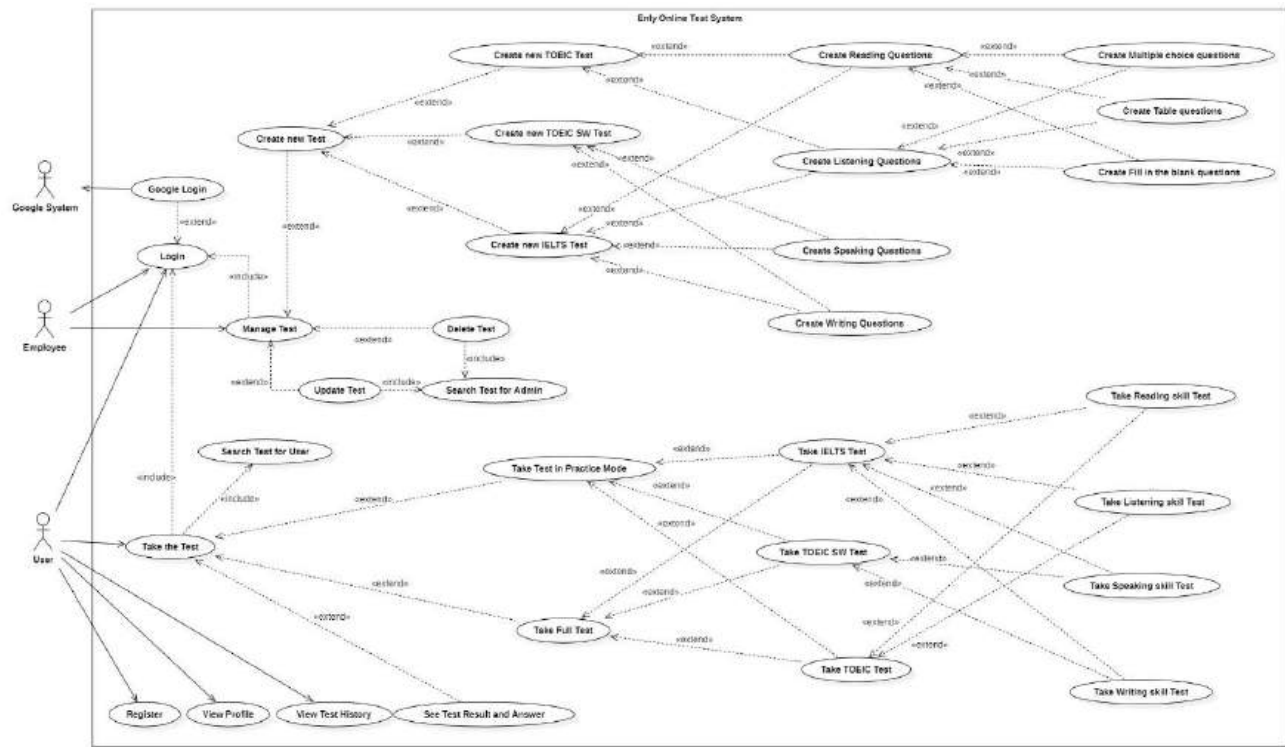


Figure 3.1. Use-case Diagram

1.1. List of actors

- User: Person who access Only Online Test System to practice English
- Employee: Person who have the authority to manage Tests
- Google System: Handles Login by Google for User

1.2. List of Use-case

- Register
- Login
- Google Login
- Manage Test
- Take the Test
- View Profile
- View Test History

2. Use-case specification and activity diagram (Activity Diagram)

2.1. Register:

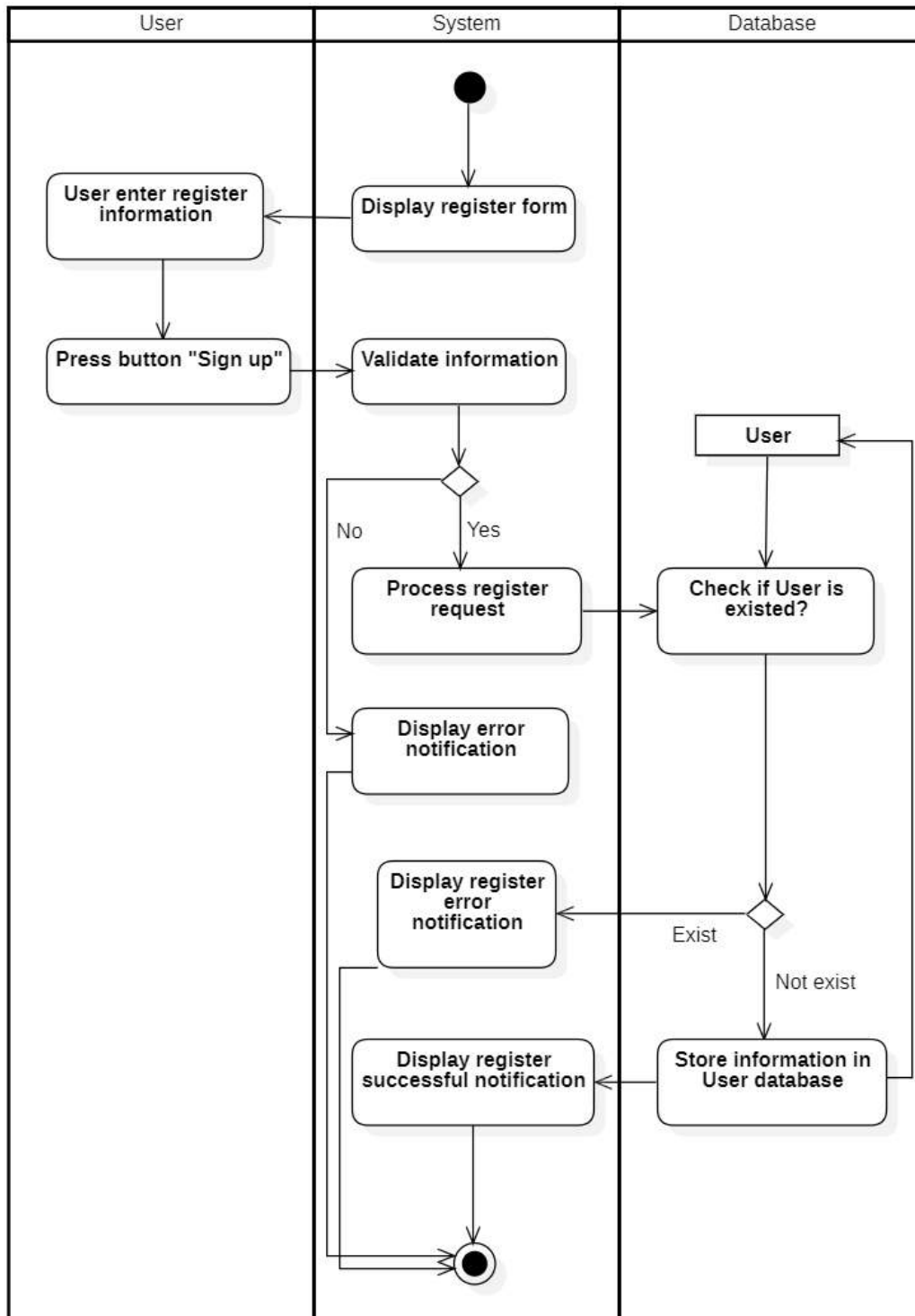


Figure 3.2. Register - Activity diagram

Table 3.1. Register - Activity description table

Use Case Name	Register
Primary Actor	User
Description	This use case describes the process in which a user enters registration information, submits the registration form, and the system validates and stores the account information in the database if the user does not already exist.
Trigger	The user clicks the “Sign up” button after entering registration information.
Preconditions	1. The user is on the registration page. 2. The system is connected to the database.
Postconditions	Success: User account is created and stored in the database; a success notification is displayed. Failure: Registration fails and an error notification is displayed.
Main flow	1. The system displays the registration form. 2. The user enters their registration information. 3. The user presses the “Sign up” button. 4. The system validates the submitted information. 5. The system sends a request to check whether the user already exists in the database. 6. The database checks for existing user records. 7. If no existing user is found, the system stores the new user information in the database. 8. The system displays a successful registration notification.
Alternative Flows	4.a. The system displays an error notification due to invalid information. 5.a. The system displays a “Registration error” notification because the user already exists.

2.2. Login:

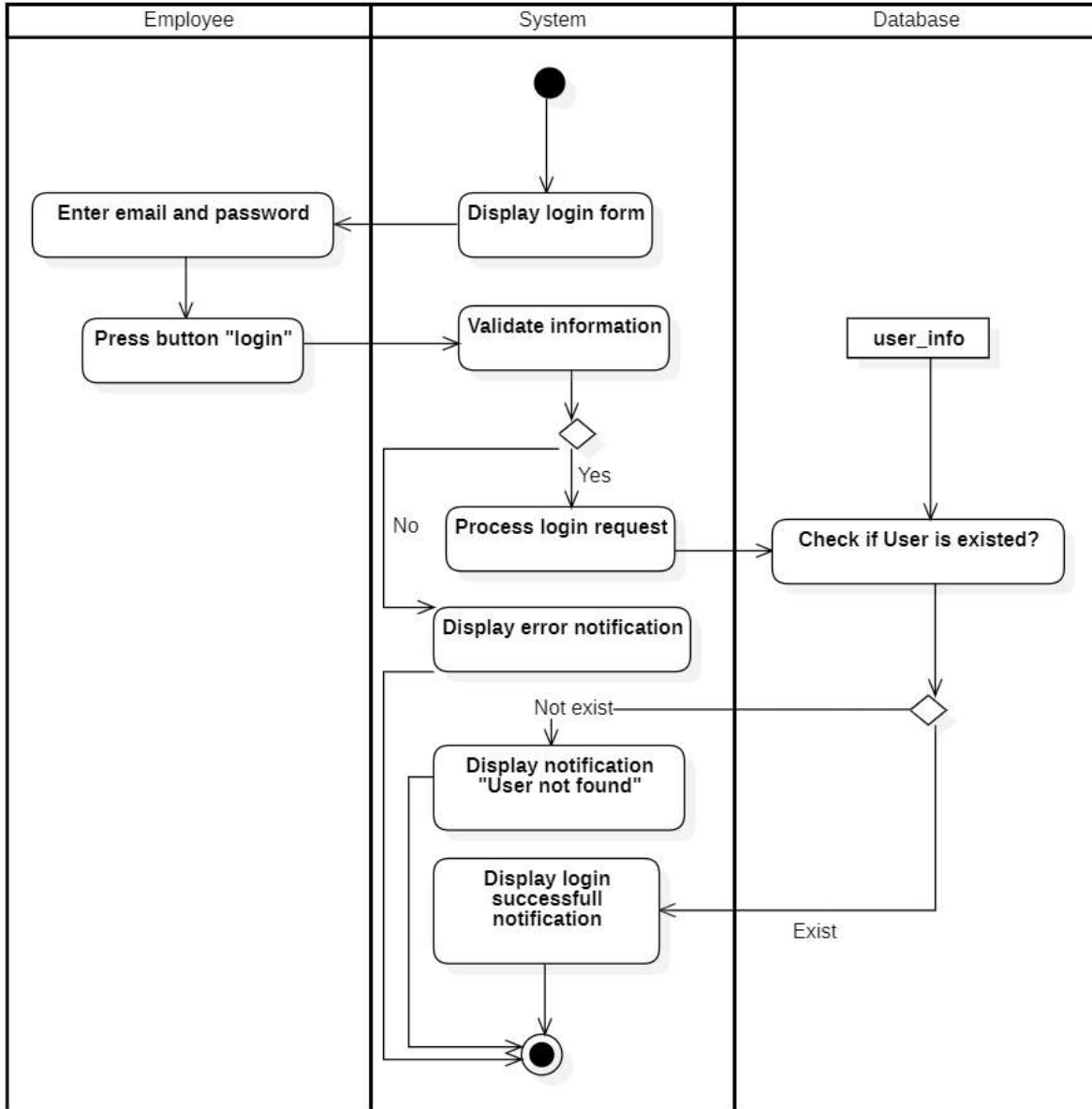


Figure 3.3. Login - Activity diagram

Table 3.2. Login - Activity description table

Use Case Name	Login
Primary Actor	Employee, User
Description	This use case allows a user to log into the system by entering their email and password. The system validates the provided

	credentials and checks whether the user exists in the database. If the user is found and the information is correct, the system logs the user in and displays a successful login notification. If the credentials are invalid or the user does not exist, the system displays an appropriate error message.
Trigger	User clicks the “Login” button after entering email and password.
Preconditions	1. The user is on the Login page. 2. The system is connected to the database.
Postconditions	Success: The user is logged into the system. Failure: The system displays a login error notification.
Main flow	1. The system displays the login form. 2. The user enters their email and password. 3. The user presses the “Login” button. 4. The system validates the entered information. 5. The system checks the database to determine if the user exists. 6. The system logs the user in and displays a “Login successful” notification.
Alternative Flows	4.a. The system displays an error notification due to invalid input information. 5.a. The system displays a “User not found” notification because the account does not exist.

2.3. Google Login:

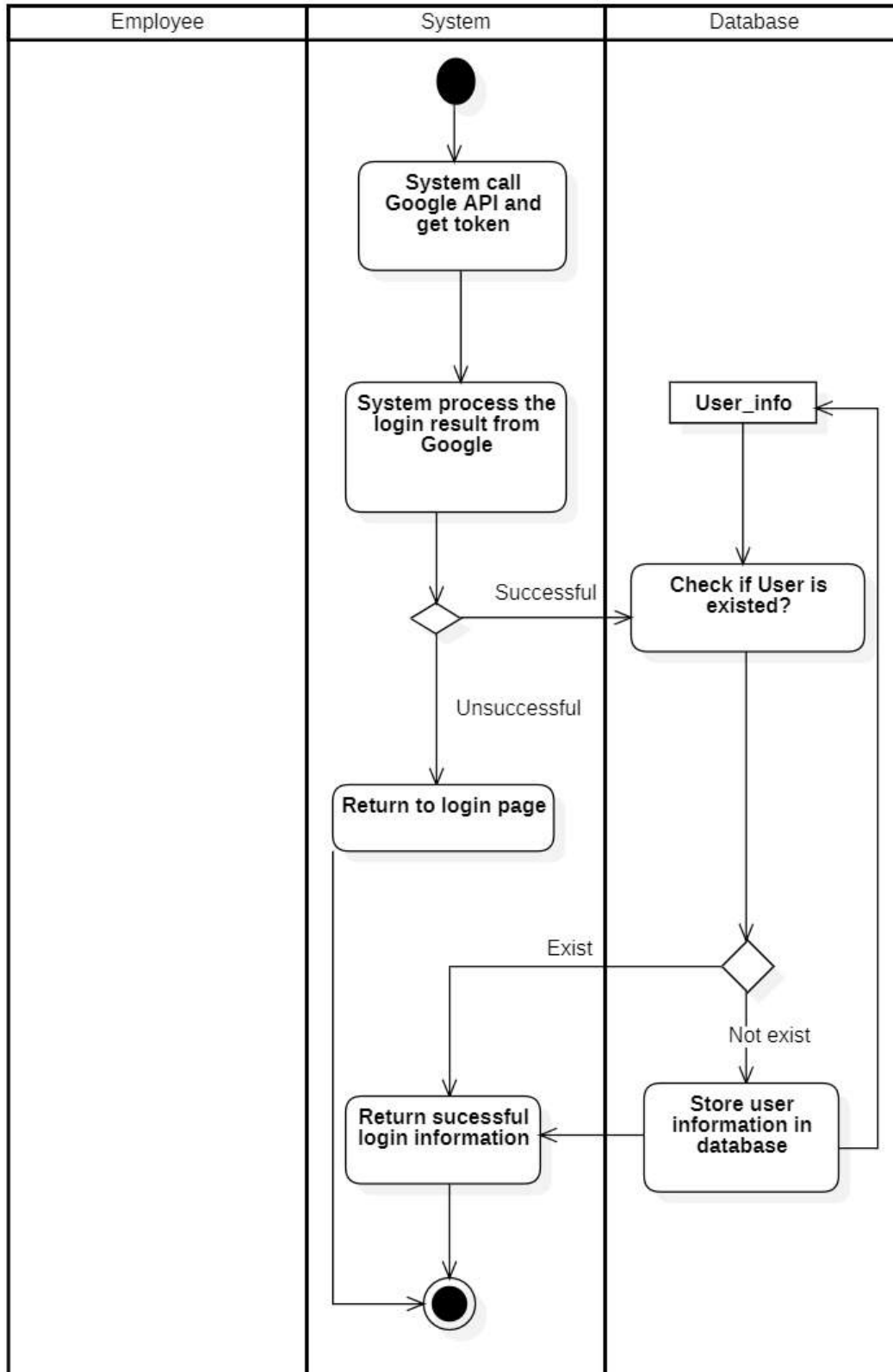


Figure 3.4. Google login - Activity diagram

Table 3.3. Google login - Activity description table

Use Case Name	Google Login
Primary Actor	Employee, User
Description	This use case allows a user to log into the system using Google OAuth. The system calls the Google API to obtain a token, processes the login result, and checks whether the user already exists in the database. If the login is successful and the user exists or is successfully created, the system returns successful login information. If the Google login fails or the user information cannot be processed, the system redirects the user back to the login page.
Trigger	The user selects the “Login with Google” option.
Preconditions	The user is on the Login page. The system is connected to Google OAuth services and the internal database.
Postconditions	Success: The user is logged into the system via Google. Failure: The system returns to the login page and displays an appropriate error.
Main flow	1. The system calls the Google API and retrieves the authentication token. 2. The system processes the login result returned from Google. 3. The system validates the result to determine if the Google login attempt is successful. 4. The system checks the database to determine if the user already exists. 5. If the user does not exist, the system stores the new user information in the database. 6. The system returns successful login information to the user.

Alternative Flows	3.a. The system returns to the login page because the Google login attempt is unsuccessful. 4.a. The system returns to the login page because user information cannot be processed.
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2.4. Search Test For Employee:

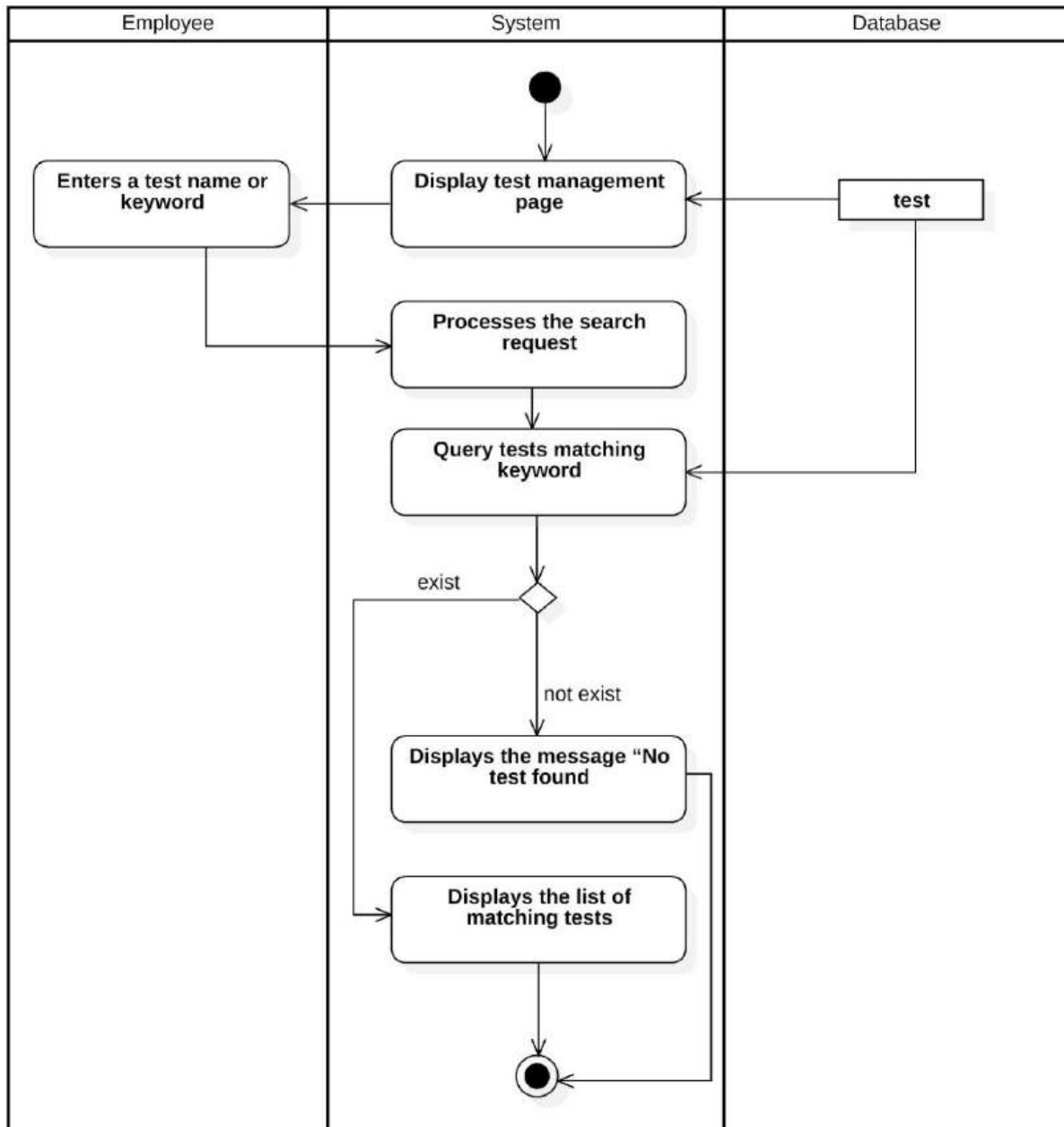


Figure 3.5. Search Test For Employee - Activity diagram

Table 3.4. Search Test For Employee - Activity description table

Use Case Name	Search Test For Employee
Primary Actor	Employee
Description	This use case allows an employee to search for tests by entering a test name or keyword. The system retrieves matching test records from the database. If results exist, they are displayed on the test management page. If no matches are found, the system shows a “Test not found” notification.
Trigger	Employee clicks into search bar
Preconditions	1. The employee is logged into the system. 2. The employee is currently on the Test Management Page.
Postconditions	Success: Matching tests are displayed. Failure: The system displays a “Test not found” message.
Main flow	1. The system displays a test management page with a list of tests. 2. The employee enters a test name or keyword into the search field. 3. The system processes the search request. 4. The system queries the database for tests matching the keyword. 5. The database returns a list of tests. 6. The system displays the list of matching tests.
Alternative Flows	3.a. The system displays the message “No test found.”

2.5. Search Test For User:

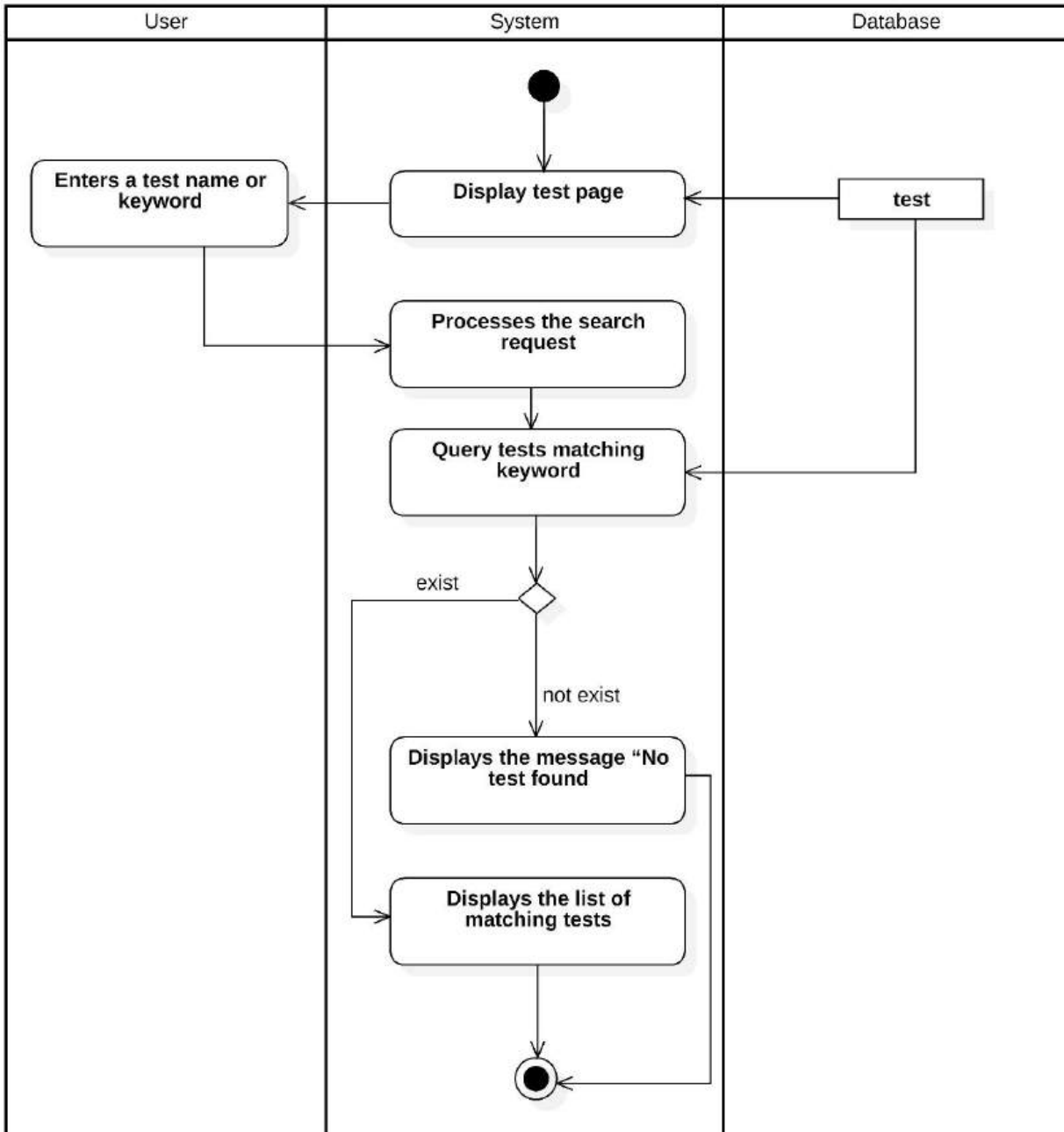


Figure 3.6. Search Test For User - Activity diagram

Table 3.5. Search test for user - Activity description table

Use Case Name	Search Test For User
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Primary Actor	User
Description	This use case allows an user to search for tests by entering a test name or keyword. The system retrieves matching test records from the database. If results exist, they are displayed on the test page. If no matches are found, the system shows a “Test not found” notification.
Trigger	User clicks into search bar
Preconditions	1. The User is logged into the system. 2. The User is currently on the Test Page.
Postconditions	Success: Matching tests are displayed. Failure: The system displays a “Test not found” message.
Main flow	1. The system displays a test page with a list of tests. 2. The user enters a test name or keyword into the search field. 3. The system processes the search request. 4. The system queries the database for tests matching the keyword. 5. The database returns a list of tests. 6. The system displays the list of matching tests.
Alternative Flows	3.a. The system displays the message “No test found.”

2.6. Sort Test For Employee:

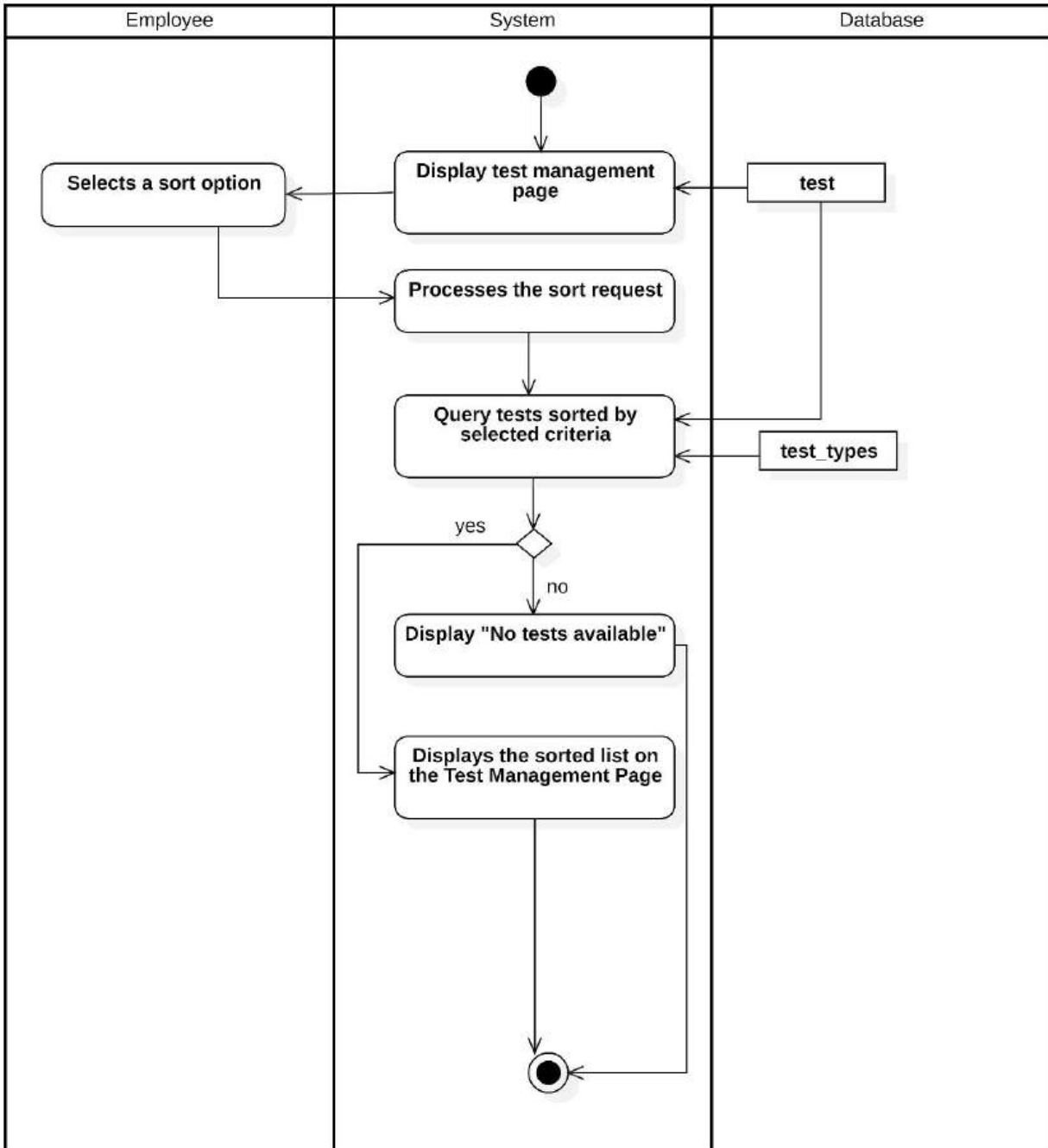


Figure 3.7. Sort Test For Employee - Activity diagram

Table 3.6. Sort test for employee - Activity description table

Use Case Name	Sort Test For Employee
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Primary Actor	Employee
Description	This use case allows the employee to sort the list of tests based on a test type. The system retrieves and displays the sorted list according to the employee's chosen sort option.
Preconditions	1. The employee is logged into the system. 2. The employee is currently on the Test Management Page with the list of tests displayed.
Postconditions	Success: The system displays the list of tests sorted according to the selected criterion.
Main Flow	1. The system displays a test management page with a list of tests. 2. The user selects a sort option from the dropdown. 3. The system receives the sort request with the selected sorting parameter. 4. The system queries the database for tests sorted by the given parameter. 5. The database returns the sorted list of tests. 6. The system displays the sorted list on the Test Management Page.

2.7. Sort Test For User:

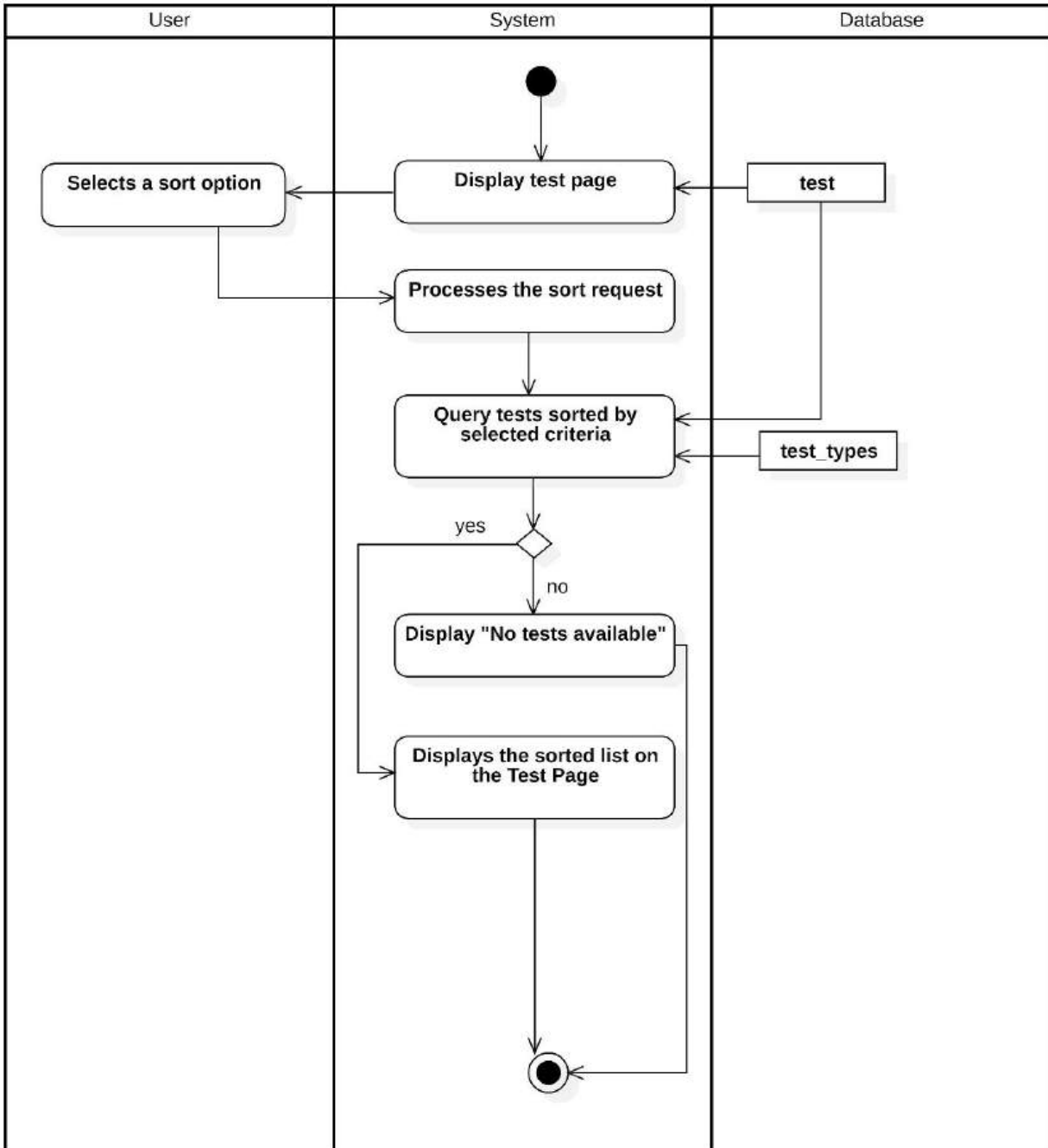


Figure 3.8. Sort Test For User - Activity diagram

Table 3.7 Sort test for user - Activity description table

Use Case Name	Sort Test For User
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Primary Actor	User
Description	This use case allows the user to sort the list of tests based on a test type. The system retrieves and displays the sorted list according to the user's chosen sort option.
Preconditions	1. The user is logged into the system. 2. The user is currently on the Test Page with the list of tests displayed.
Postconditions	Success: The system displays the list of tests sorted according to the selected criterion.
Main Flow	1. The system displays a test page with a list of tests. 2. The user selects a sort option from the dropdown. 3. The system receives the sort request with the selected sorting parameter. 4. The system queries the database for tests sorted by the given parameter. 5. The database returns the sorted list of tests. 6. The system displays the sorted list on the Test Page.

2.8. View Test Detail:

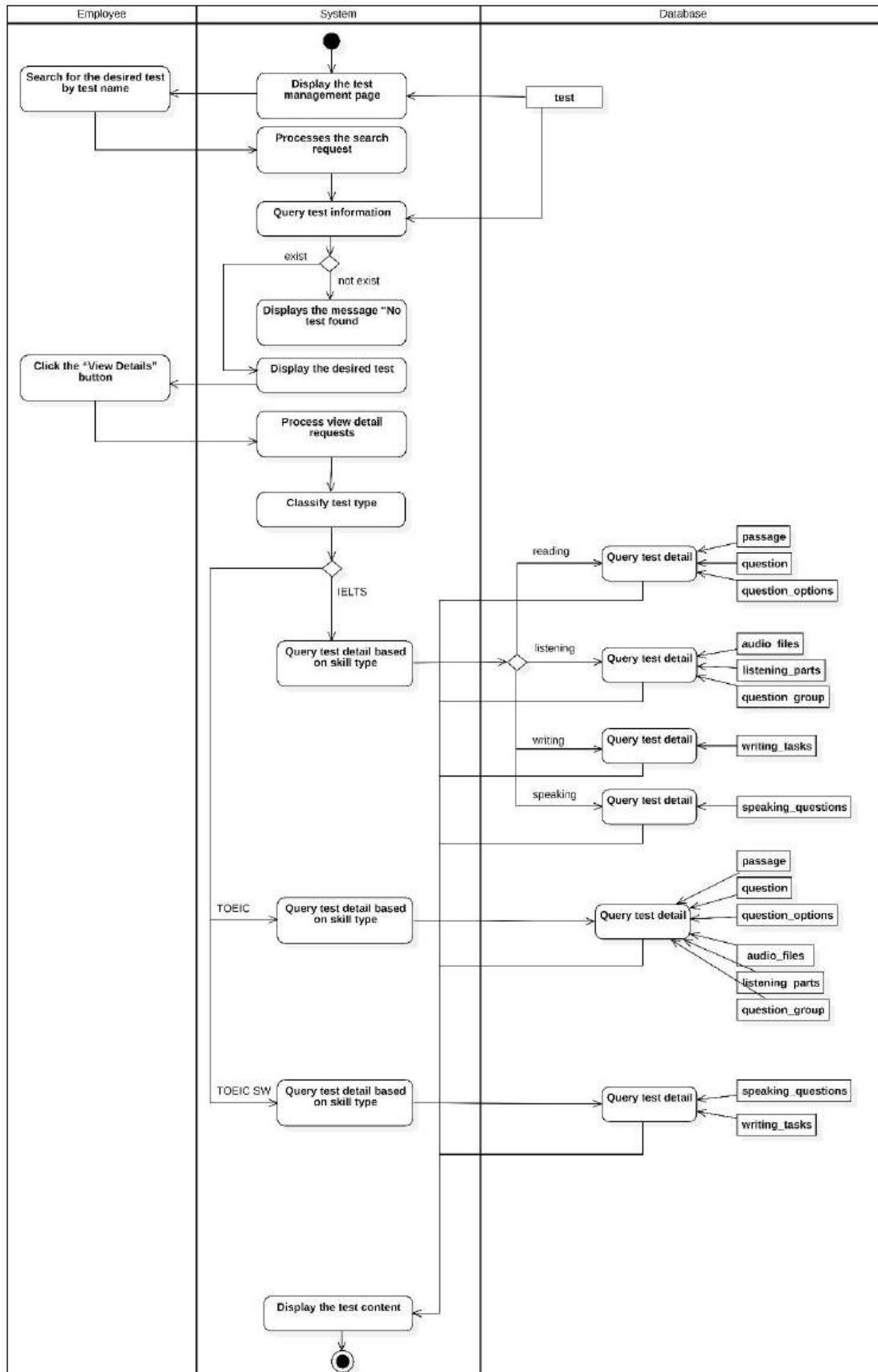


Figure 3.9. View test detail - Activity diagram

Table 3.8. View test detail - Activity description table

Use Case Name	View Test Detail
Primary Actor	Employee
Description	Displays full details of the selected test, including its parts, questions, answers, and related resources
Trigger	Employee clicks into Test Management Page
Preconditions	1. The user is logged in. 2. The test exists in the database.
Postconditions	Test information is displayed; no data is modified.
Main Flow	1. The system displays a test management page with a list of tests. 2. The employee searches for the desired test by test name. 3. The system processes the search request. 4. The system queries the test information from the database. 5. The system displays the desired test. 6. The employee clicks the “View Detail” button next to the selected test. 7. The system processes requests. 8. The system retrieves all test details from the database. 9. The system displays the test content based on the test type
Alternative Flows	4a. The system displays a message that the test is not found. 4b. The employee modifies the test name being searched.

2.9. View All Test History:

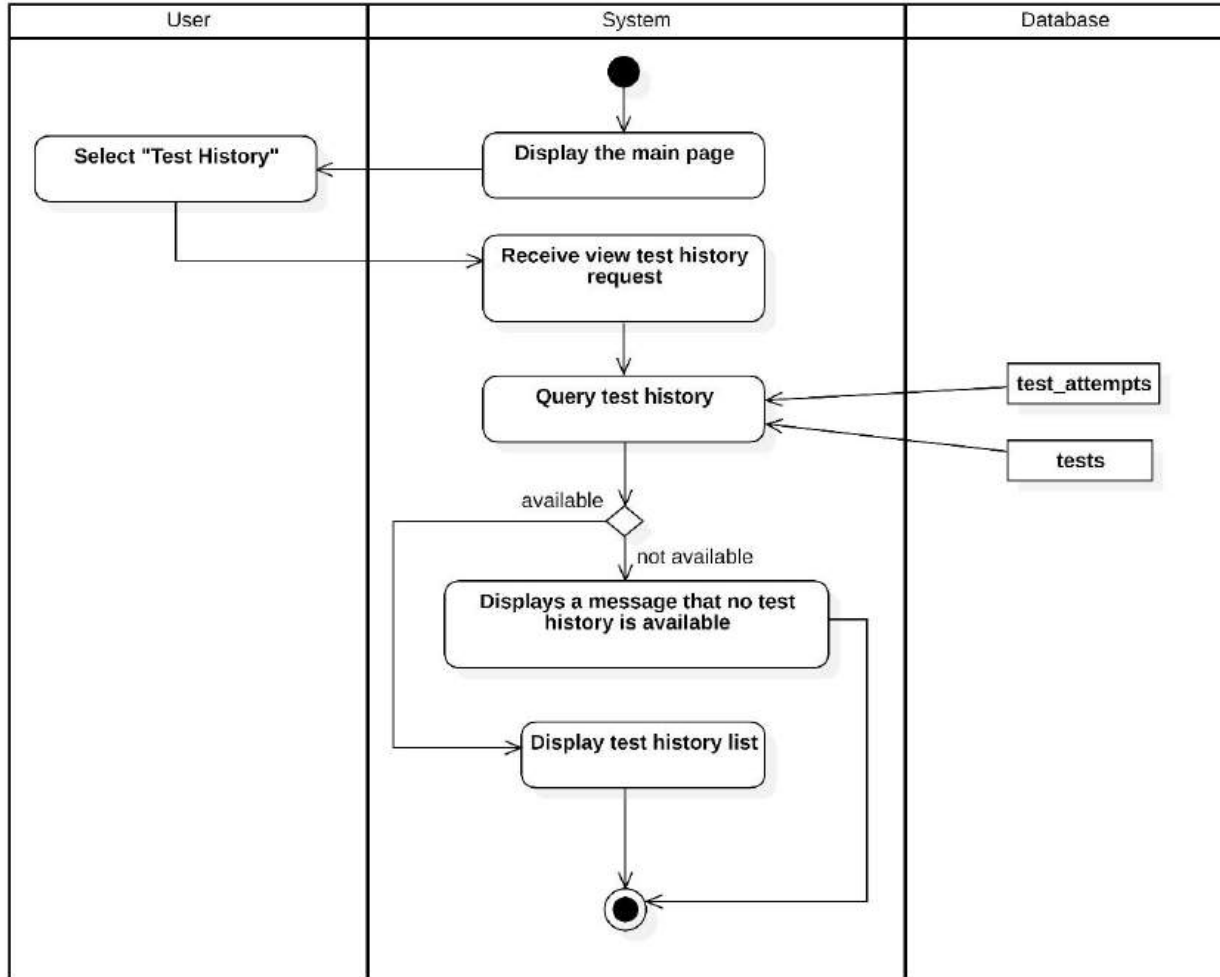


Figure 3.10. View all test history - Activity diagram

Table 3.9. View All Test History - Activity description table

Use Case Name	View All Test History
Primary Actor	User
Description	Allows the user to view a list of their past test attempts and related history.
Trigger	User clicks into View Test History Button.

Preconditions	1. The user is logged in. 2. The user clicks the user settings dropdown button.
Postconditions	User Settings Dropdown is displayed to the user
Main Flow	1. The system displays the main page. 2. The user selects the "Test History" option. 3. The system receives the view test history request. 4. The system queries the test history from the database. 5. The system verifies that test history data is available. 6. The system displays the test history list.
Alternative Flows	4a. The system displays a message stating that no test history is available.

2.10. View User Profile:

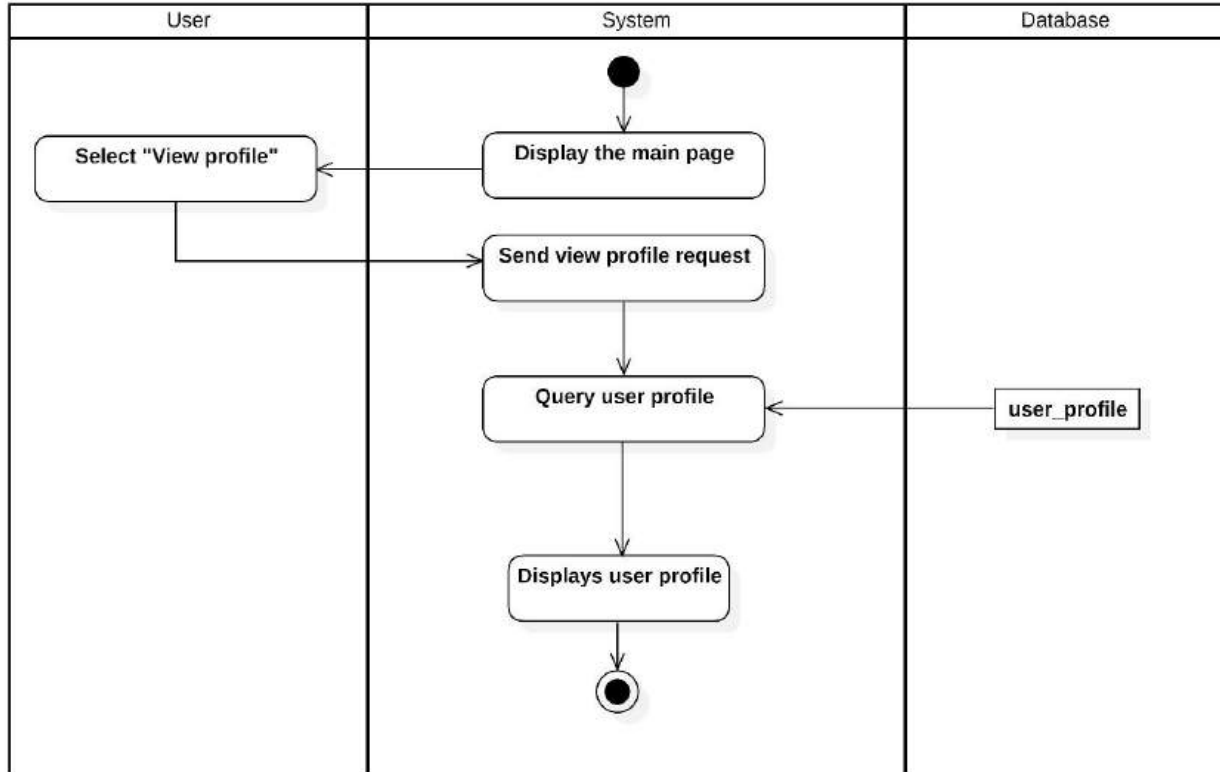


Figure 3.11. View user profile - Activity diagram

Table 3.10. View User Profile - Activity description table

Use Case Name	View User Profile
Primary Actor	User
Description	Allows the user to view their personal account information and profile details.
Trigger	User clicks into Profile Page
Preconditions	1. The user is logged in. 2. The system is displaying the main page.
Postconditions	The user's profile is displayed.

Main Flow	1. The system displays the main page. 2. The user selects the "View profile" option. 3. The system receives the view profile request. 4. The system queries the user profile information from the database. 5. The system displays the user profile details.
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2.11. Create Test:

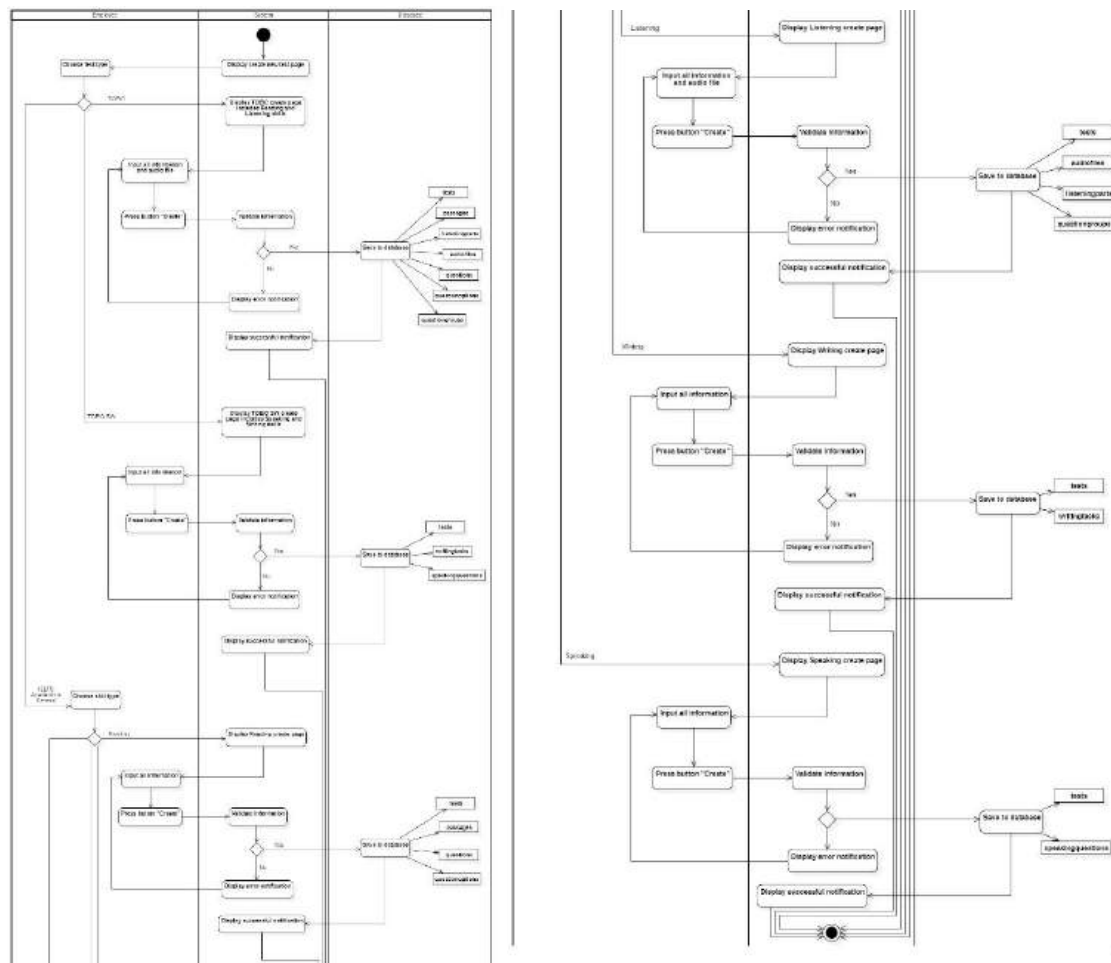


Table 3.11. Create New Test - Activity description table

Use Case name	Create New Test
Primary Actor	Employee

Description	This use case allows an employee to create a new, complete test for the online exam platform. The process includes selecting a test type (IELTS, TOEIC), and selecting the specific skill to build (e.g., Reading, Listening). The employee enters general test information, adds content (such as passages or audio files), and creates associated questions.
Trigger	The employee selects button “Create new test” from the Test Management Page
Pre-condition	- The employee must be authenticated and authorized as an Admin. - The employee is on the Test Management Page.
Post-condition	- A new test record is created and saved to the database, along with all its associated components (passages, audio files, questions, answers). - The employee is redirected to the Test Management Page, where the new test is now visible.
Basic flow	1. Select Test Type: The System displays the test creation interface. The Employee chooses the Test Type (TOEIC or IELTS). 2. Branch - TOEIC Selected: 2.a. The System displays the TOEIC module input form (Listening & Reading). 2.b. The Employee inputs test information and uploads the audio file. 3. Branch - IELTS Selected: 3.a. The System displays the skill selection options. 3.b. The Employee selects a specific skill (Reading, Listening, Writing, or Speaking). 3.c. The System displays the specific input page for the selected skill.

	3.d. The Employee inputs test information (Title, Duration, etc.) and uploads specific assets (Passage for Reading; Audio for Listening; Prompts for Writing/Speaking). 4. Submission: The Employee presses the "Create" button. 5. Validation: The System validates the entered information. 6. Storage: The System saves the new test data to the Database. 7. Notification: The System displays a success notification to the Employee.
Alternative flows	A1. Validation Failed (Invalid Input): At step 5 of the Basic Flow, if the information provided is incomplete or invalid (e.g., missing title, unsupported file format): 1. The System displays an error notification/indicator. 2. The flow returns to the data input step (Step 2b or 3d depending on the test type). 3. The Employee corrects the information and presses "Create" again.

2.12. Update Test:

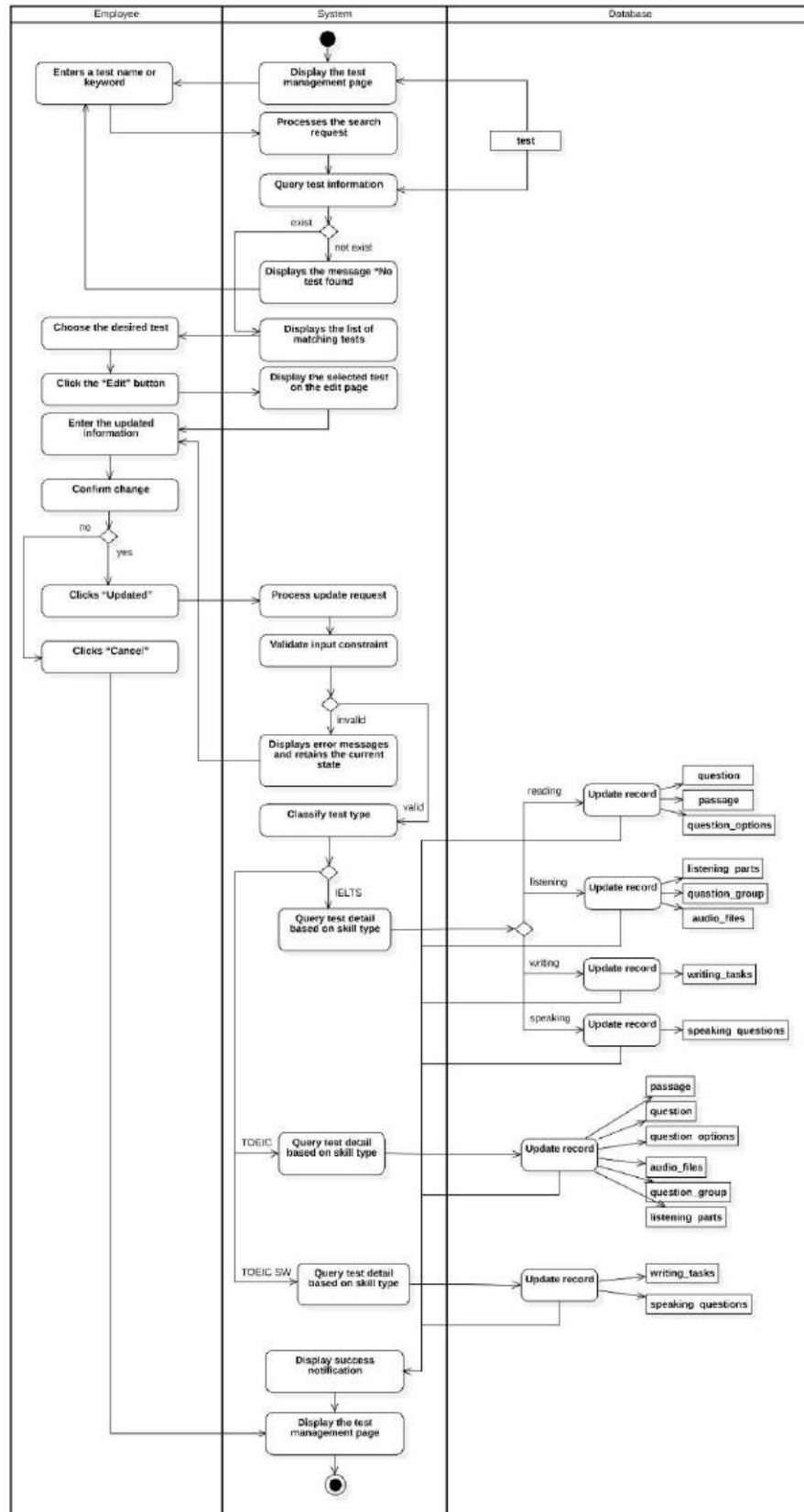


Figure 3.12. Update test - Activity diagram

Table 3.12. Update Test - Activity description table

Use Case Name	Update Test
Actor	Employee
Description	Allows employees to edit existing test information and content in the system.
Trigger	The employee clicks the “Edit” button next to the selected test.
Preconditions	1. The user is logged in. 2. The user has permission to edit tests. 3. The test exists in the database.
Postconditions	The test data is successfully updated in the system.
Main Flow	1. The system displays a test management page with a list of tests. 2. The employee searches for the desired test by test name. 3. The system processes the search query for the test. 4. The system queries the test information from the database. 5. The system displays the desired test. 6. The employee clicks the “Edit” button next to the selected test. 7. The system loads the test data from the database and displays it on the edit page. 8. The employee updates the necessary information of the test. 9. The employee clicks “Save Changes.” 10. The system validates the input data for correctness and completeness.

	11. The system updates the record in the database successfully. 12. The system displays a success notification and returns to the test management page.
Alternative Flows	4a. The system displays a message that the test is not found. 4b. The employee modifies the test name being searched. 9a. The employee clicks “Cancel” to abort the editing process. 10a. If there are any errors, the system displays appropriate error messages and retains the current state, allowing the employee to continue editing until all errors are corrected.

2.13. Delete Test:

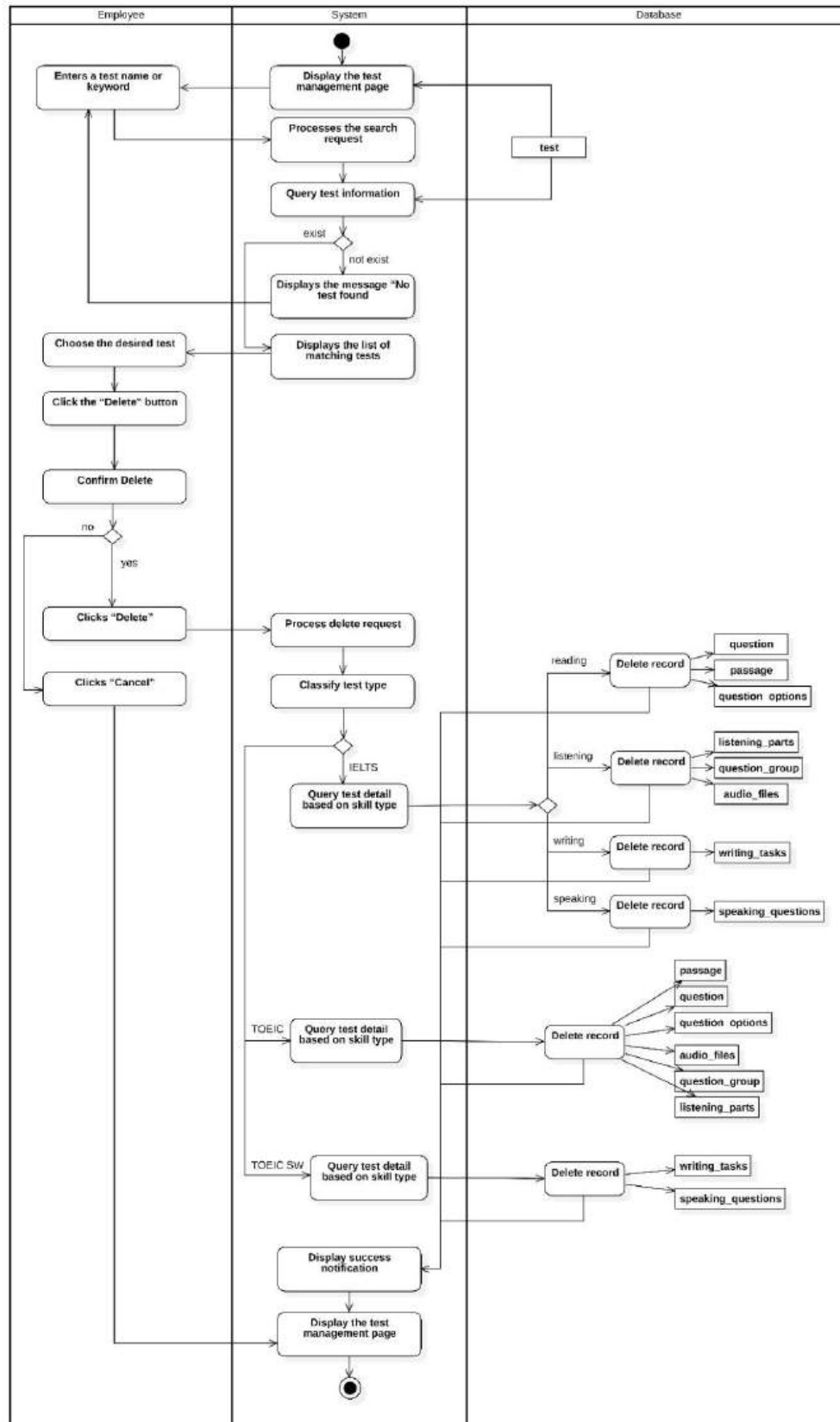


Figure 3.13. Delete Test - Activity diagram

Table 3.13. Delete Test - Activity description table

Use Case Name	Delete Test
Primary Actor	Employee
Description	Allows the employee to delete tests from the system.
Trigger	The employee clicks into the Test Management Page.
Preconditions	1. The user is logged in. 2. The user has permission to delete tests. 3. The test exists in the database.
Postconditions	The test is deleted from the system (if confirmed by the user).
Main Flow	1. The system displays a test management page with a list of tests. 2. The employee searches for the desired test by test name. 3. The system processes the search request. 4. The system queries the test information from the database. 5. The system displays the desired test. 6. The employee clicks the “Delete” button next to the selected test. 7. The system displays a confirmation dialog. 8. The employee confirms the deletion. 9. The system processes deleting and deletes the test in the database. 10. The system displays a success notification and returns to the test management page.

Alternative Flows	4a. The system displays a message that the test is not found. 4b. The employee modifies the test name being searched. 8a. The employee clicks “Cancel” to abort the delete process and return to the test management page.
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2.14. Take the Test:

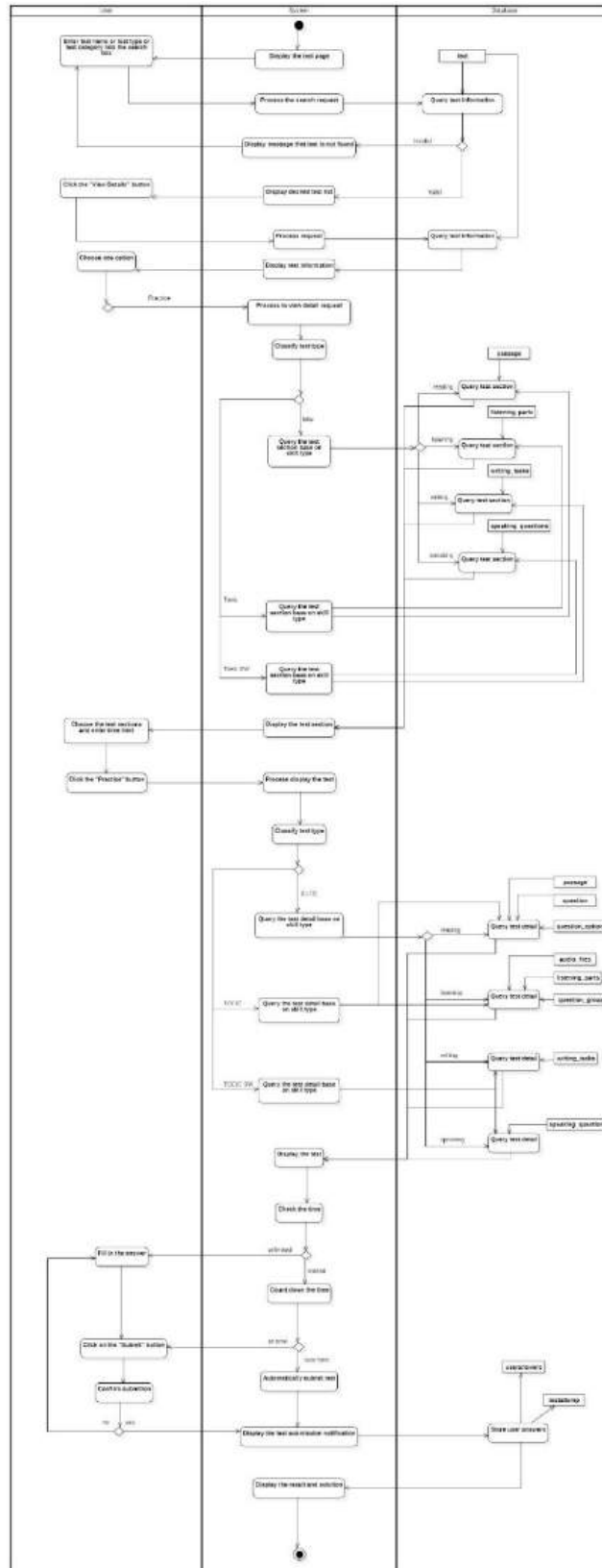


Figure 3.14. Take the test - Activity diagram

Table 3.14. Take the test - Activity description table

Use Case Name	Take the test
Primary Actor(s)	Student
Description	This use case allows the user to take the test. The system displays the test and stores the user's answers.
Preconditions	1. The user is logged into the system. 2. The user is currently on the Test Page with the list of tests displayed.
Postconditions	Success: The system displays the test and stores the user's answers.
Main Flow	1. The system displays a test page with a list of tests. 2. The user selects a test and clicks on the "View detail" button. 3. The system displays the test details. 4. The system queries the database for tests sorted by the given parameter. 5. User chooses one of following options: 6. If user chooses Practice, then S1- Practice is performed 7. If the user chooses "Do full test", then S2- Do full test is performed. 8. The system displays the test. 9. The system check the time 10. The system counts down the time.

	11. The user fills in the answer. 12. The user clicks the “Submit” button. 12. The user confirms submission. 13. The system displays the test submission notification. 14. The system stores the user’s answer. 15. The system displays the result and solutions.
Sub Flow	S1- Practice S1.1. The system displays the test section list and time limitations. S1.2. The user chooses the test sections and time limitations option. S1.3. The user clicks on the “Practice” button. S2- Do full test S2.1. The user clicks on the “Do full test” button.
Alternative Flows	7.a. The use-case continues with step 9. 8.a. The system automatically submits the test due to out of time. The use-case continues with step 12.

3. Sequence Diagram and Class Diagram

3.1. Sequence Diagram

3.1.1. Register:

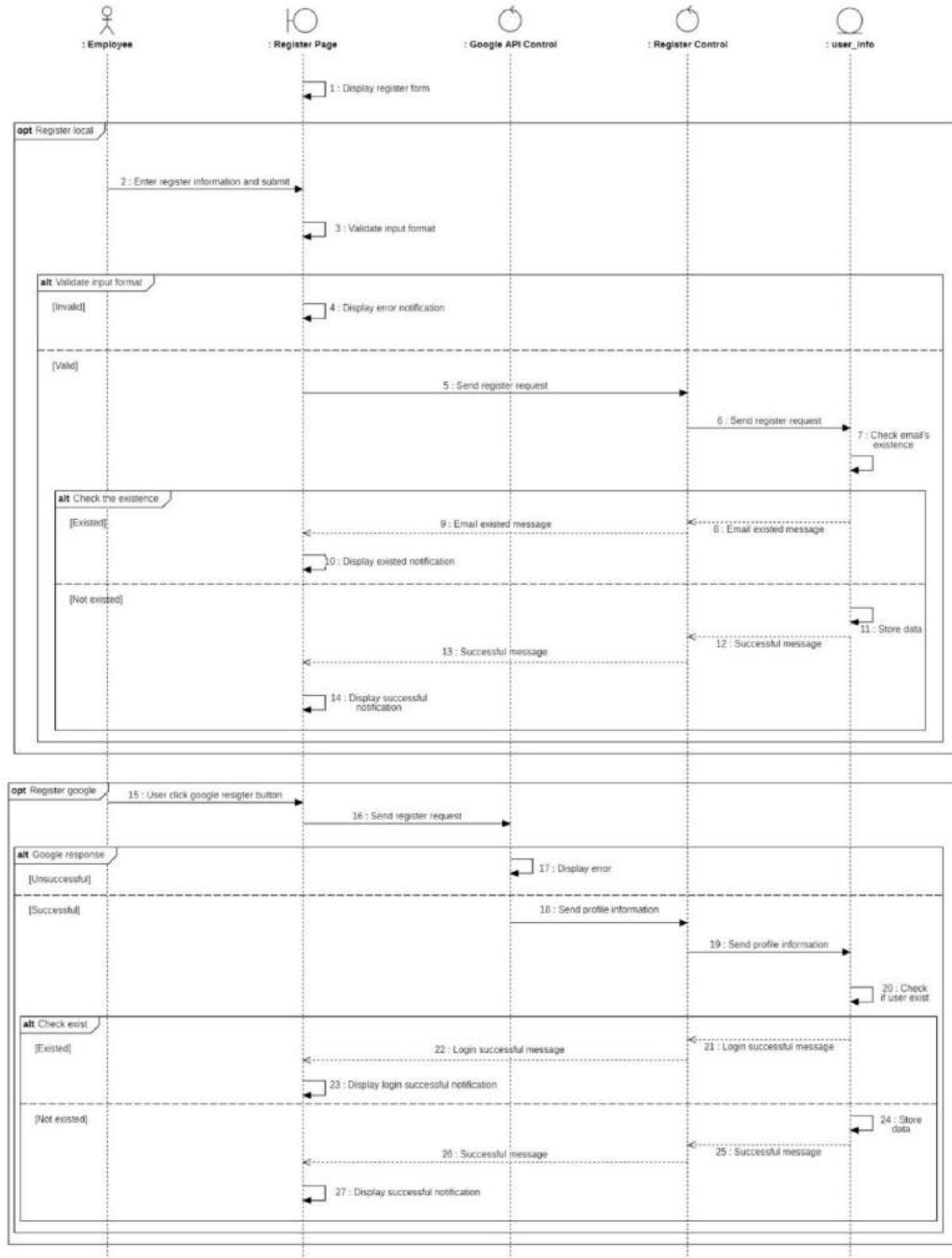


Figure 3.15. Register - Sequence Diagram

3.1.2. Login:

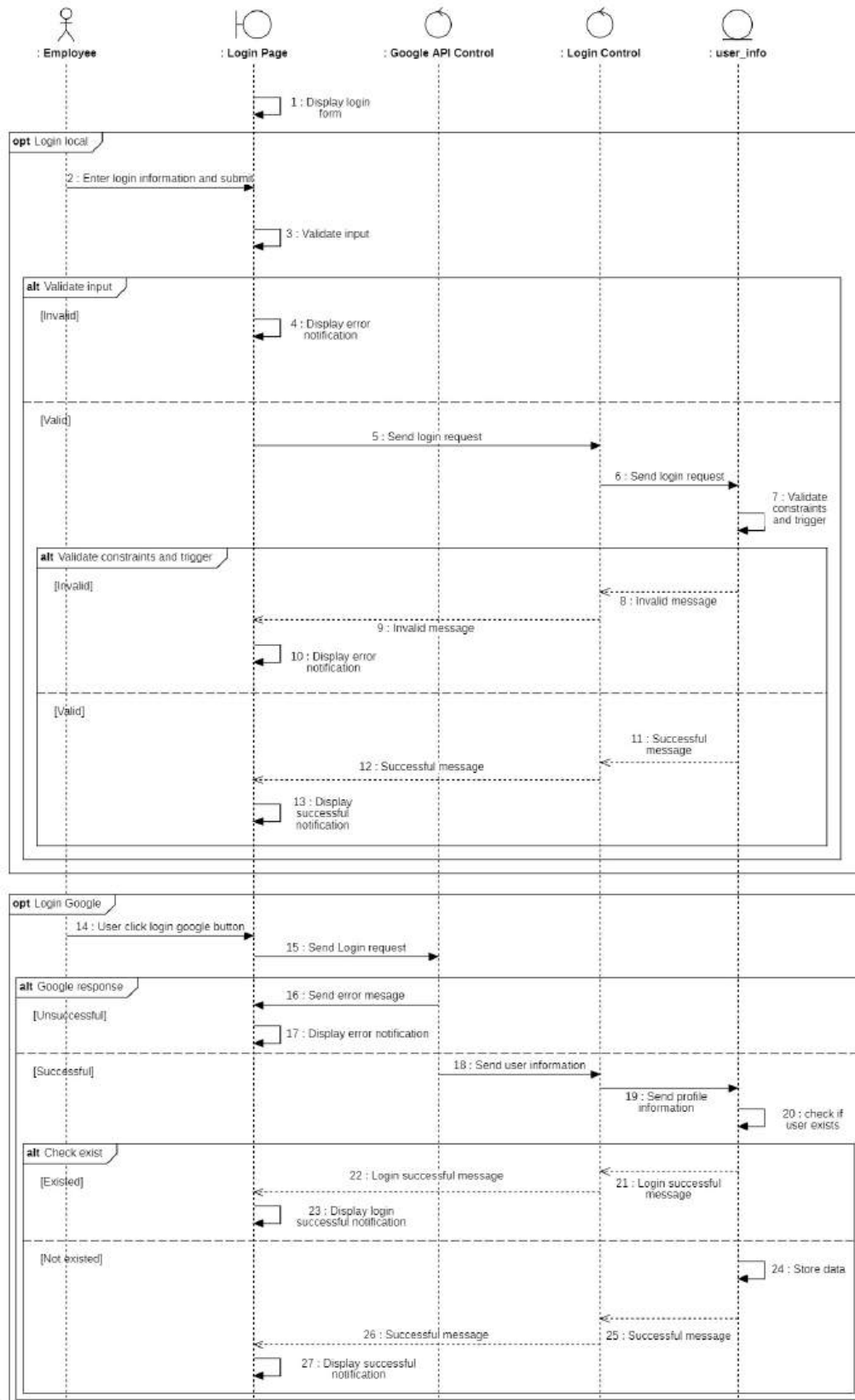


Figure 3.16. Login - Sequence Diagram

3.1.3. Manage Test:

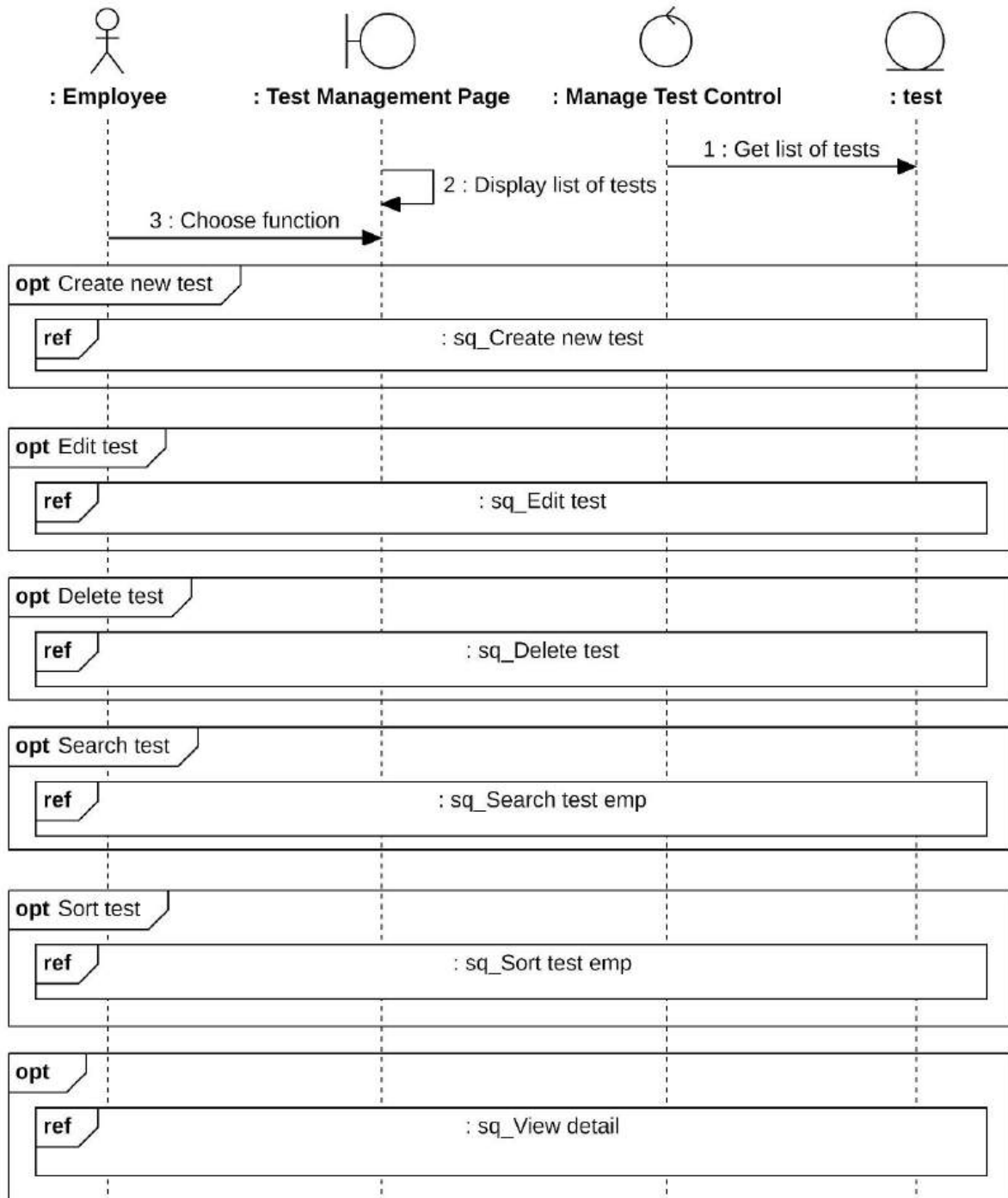


Figure 3.17. Manage test - Sequence Diagram

3.1.4. Search Test For Employee:

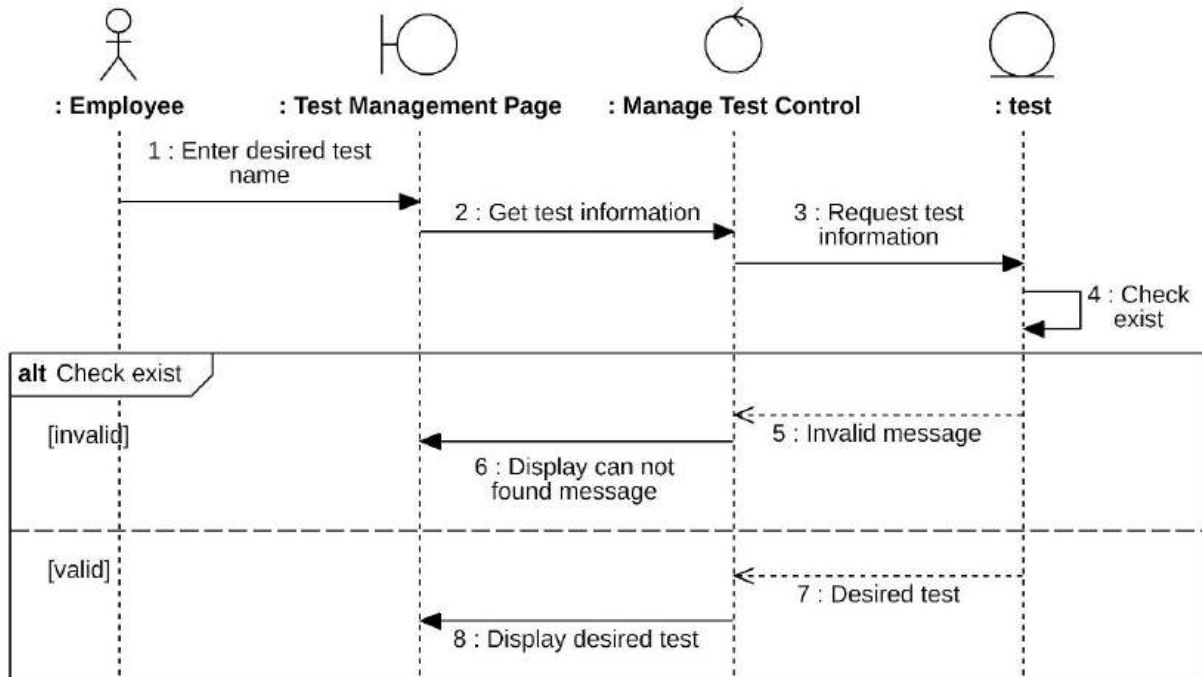


Figure 3.18. Search test for employee - Sequence Diagram

3.1.5. Search Test For User:

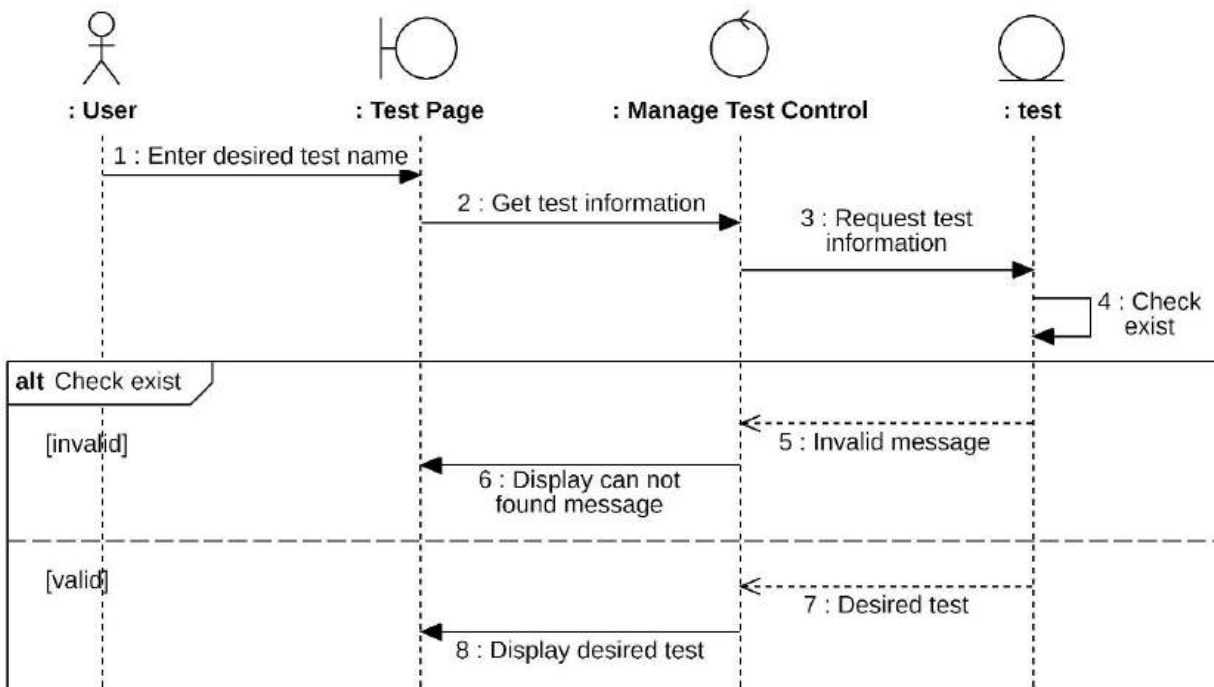


Figure 3.19. Search test for user - Sequence Diagram

3.1.6. Sort Test For Employee:

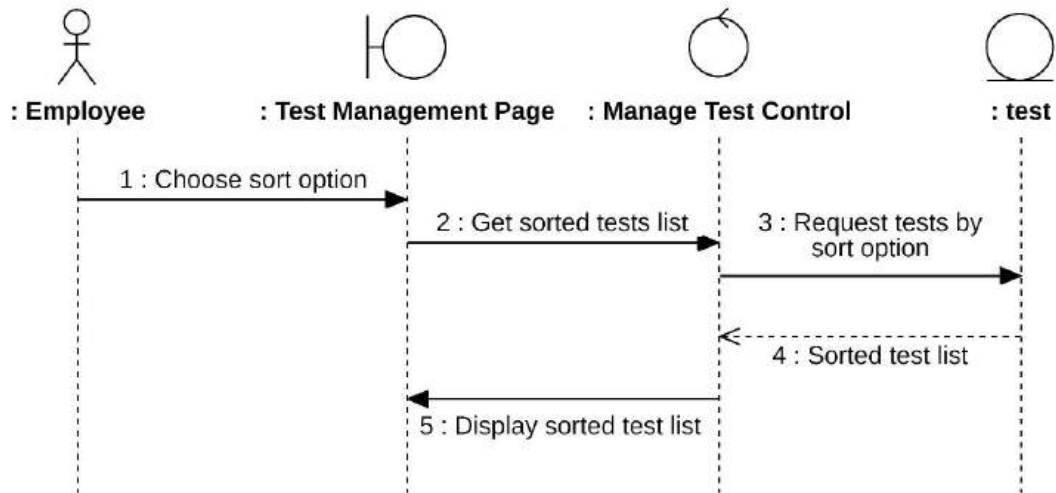


Figure 3.20. Sort test for employee - Sequence Diagram

3.1.7. Sort Test For User:

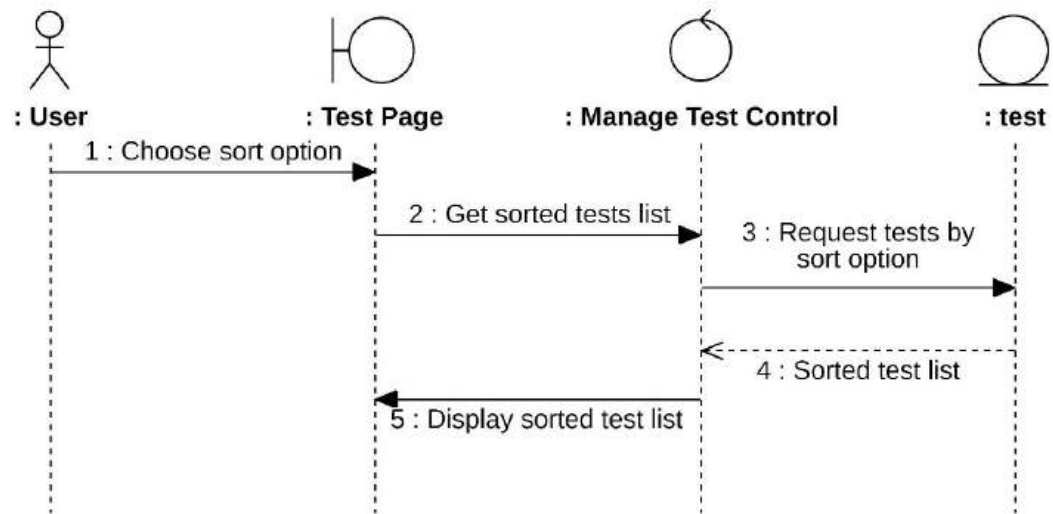


Figure 3.21. Search test for user - Sequence Diagram

3.1.8. View Test Detail:

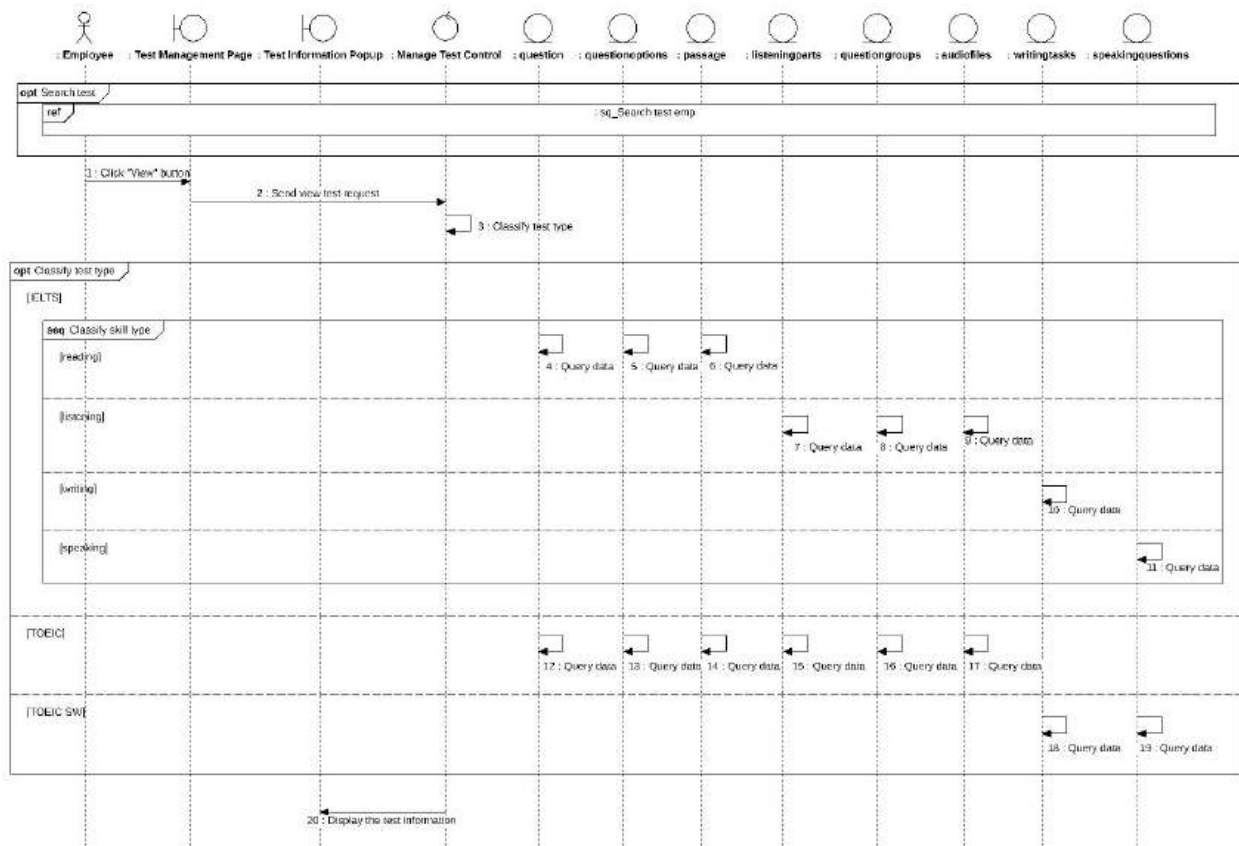


Figure 3.22. View test detail - Sequence Diagram

3.1.9. View All Test History:

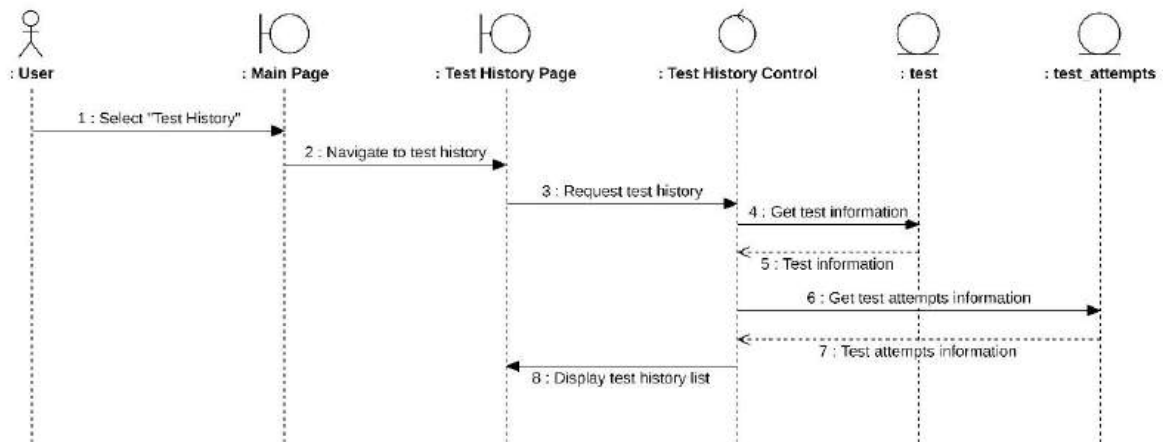


Figure 3.23. View all test history - Sequence Diagram

3.1.10. View User Profile:

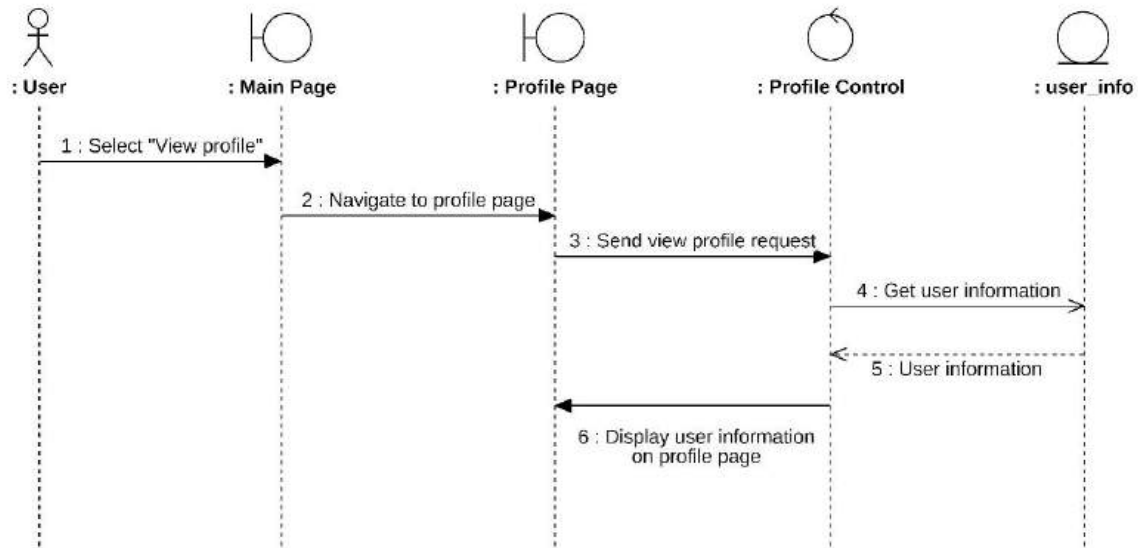


Figure 3.24. View user profile - Sequence Diagram

3.1.11. Take the Test:

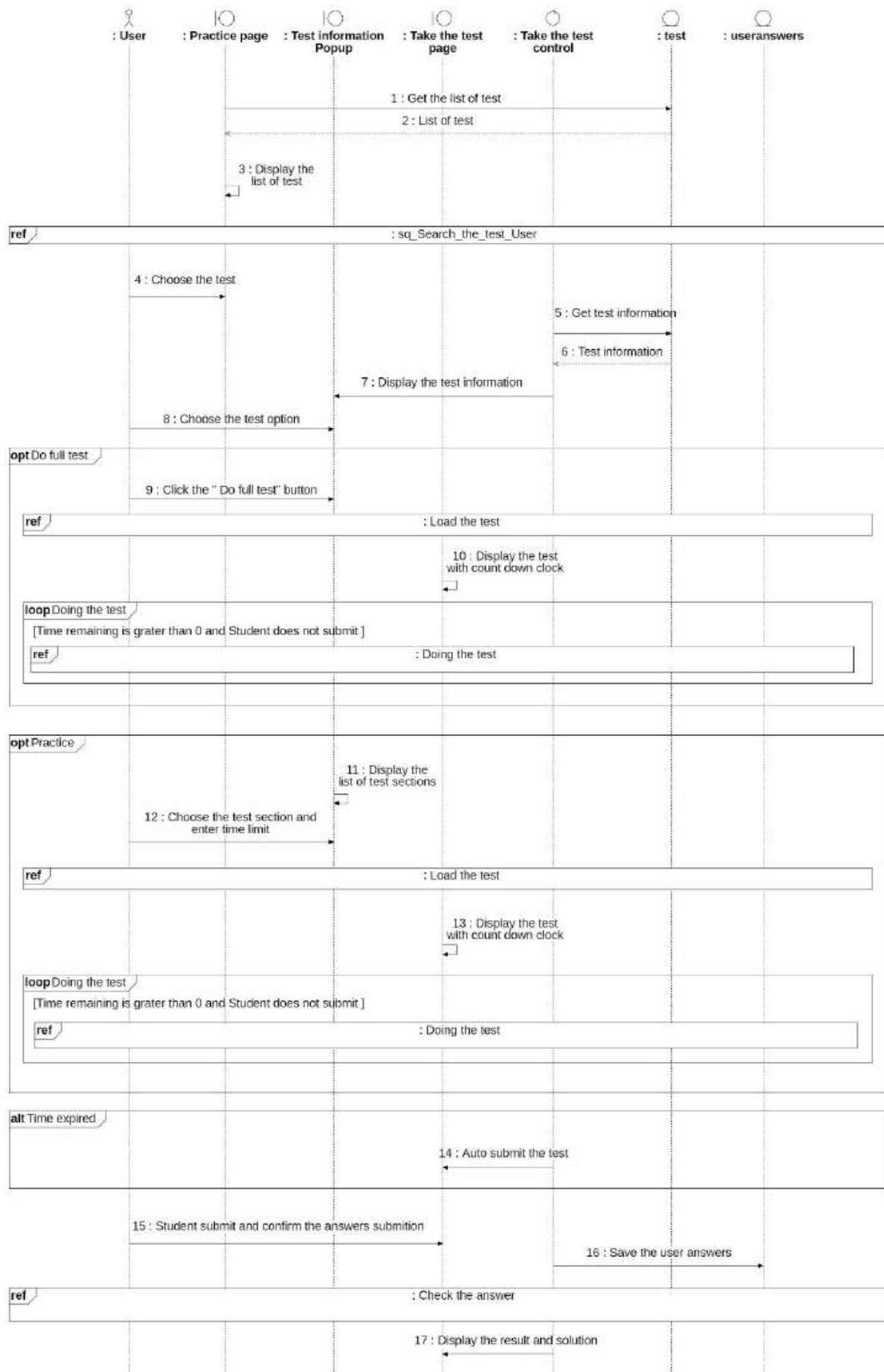


Figure 3.25. Take the test - Sequence Diagram

3.1.12. Load The Test:

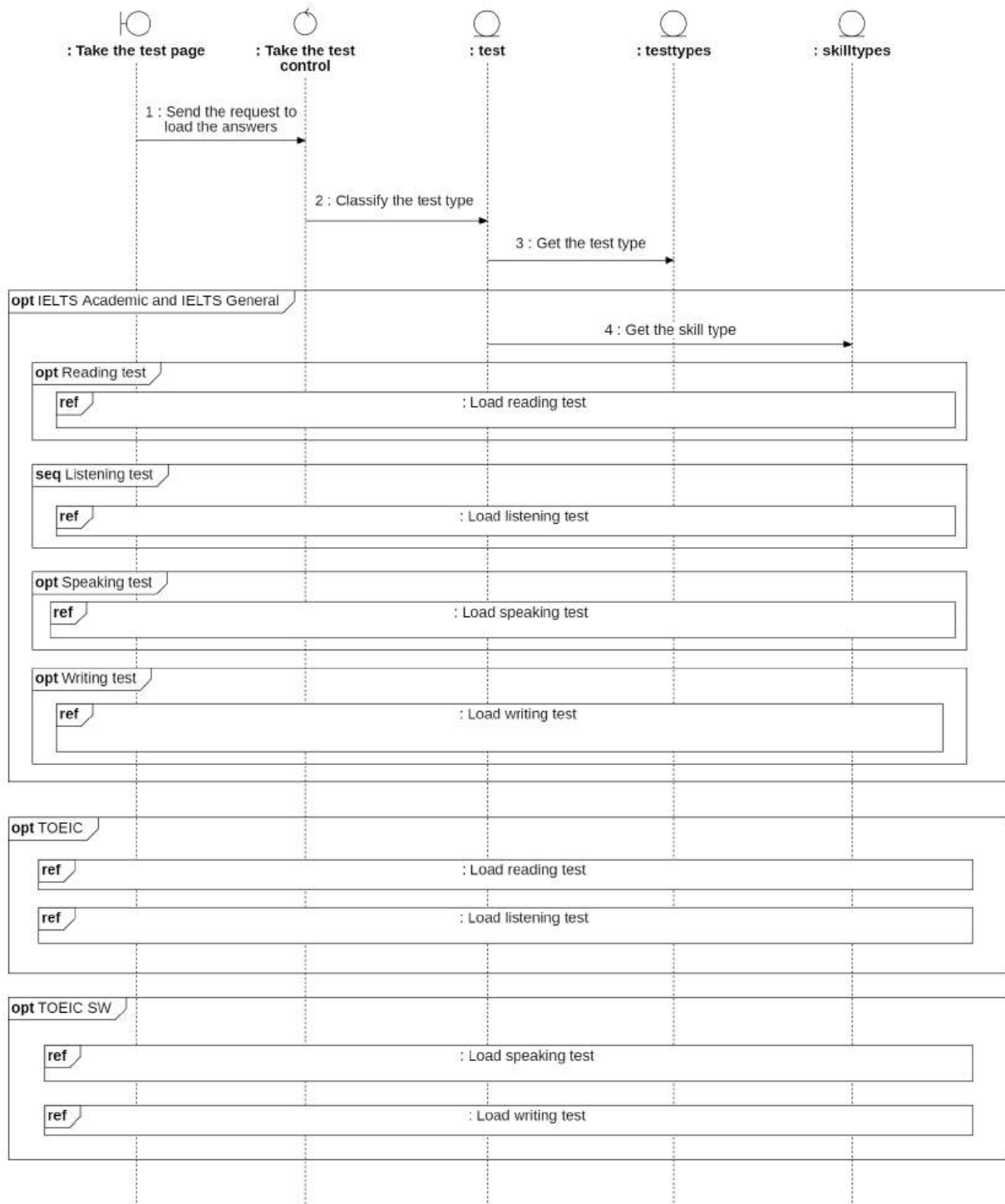


Figure 3.26. Load the test - Sequence Diagram

3.1.13. Load Reading Test:

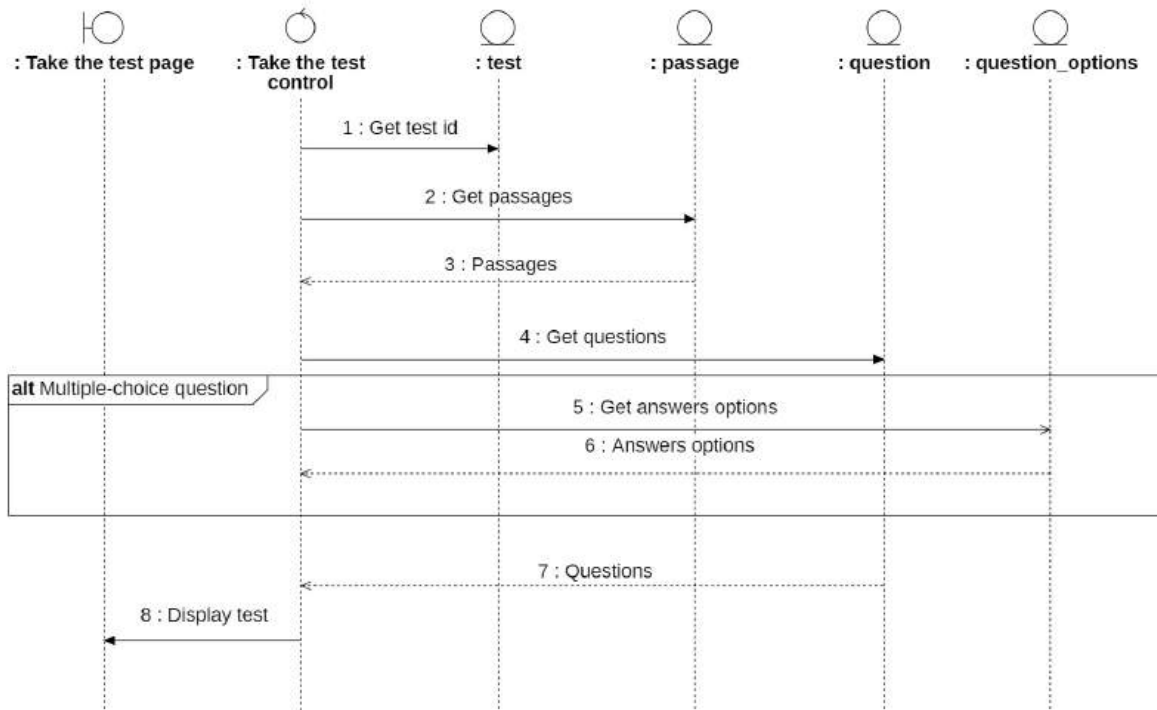


Figure 3.27. Load Reading test - Sequence Diagram

3.1.14. Load Listening Test:

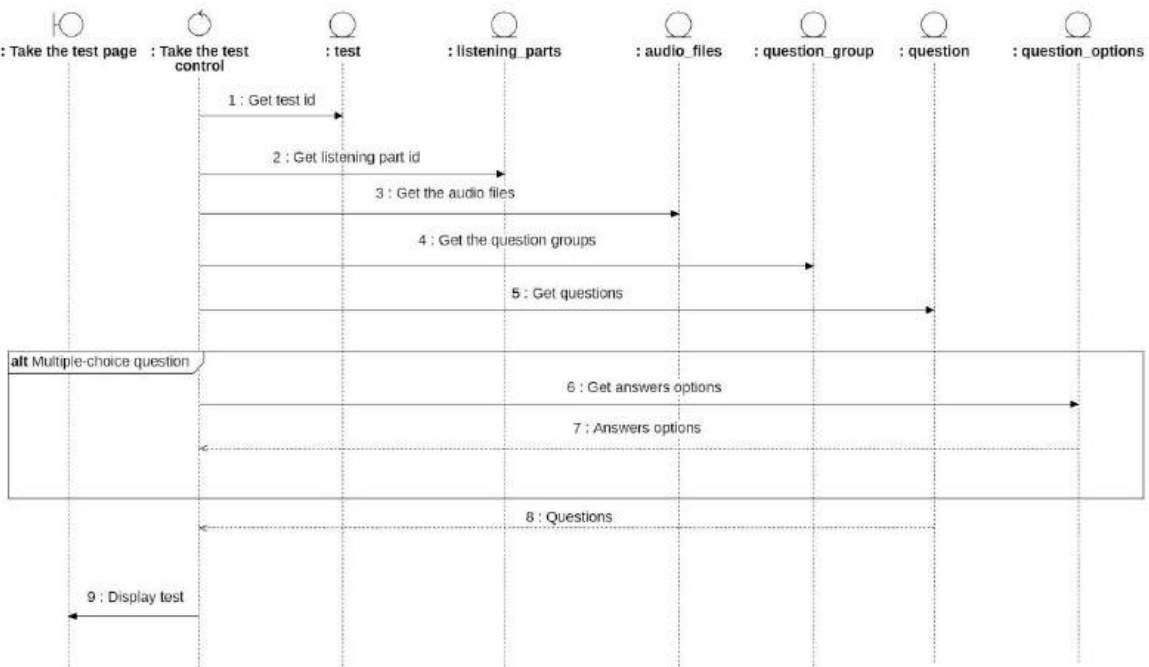


Figure 3.28. Load Listening test - Sequence Diagram

3.1.15. Load Speaking Test:

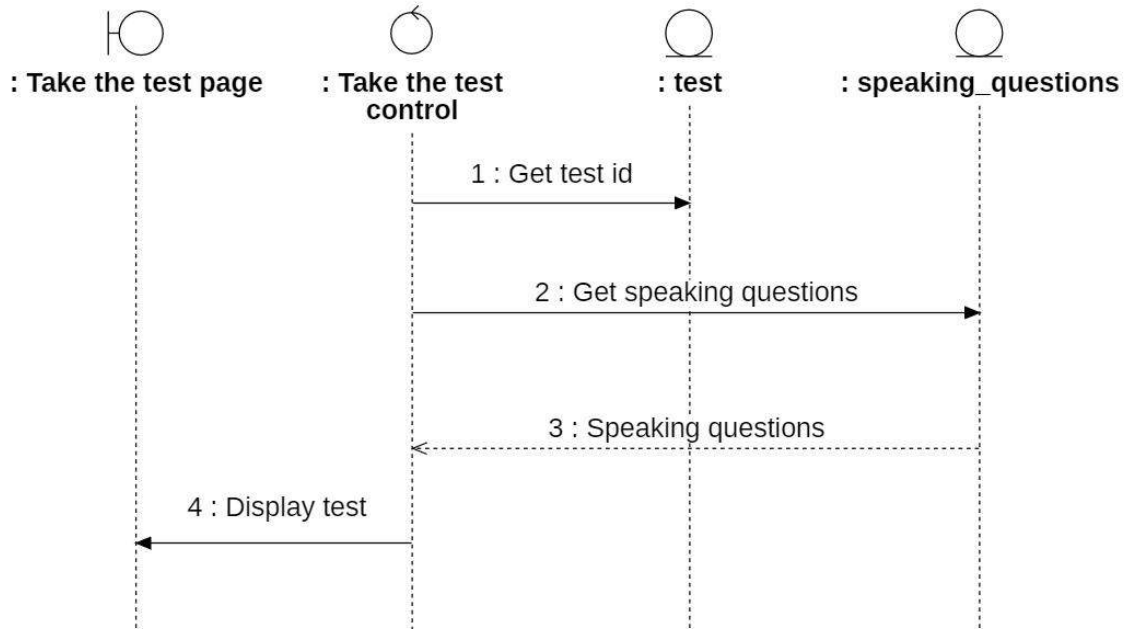


Figure 3.29. Load Speaking test

3.1.16. Load Writing Test:

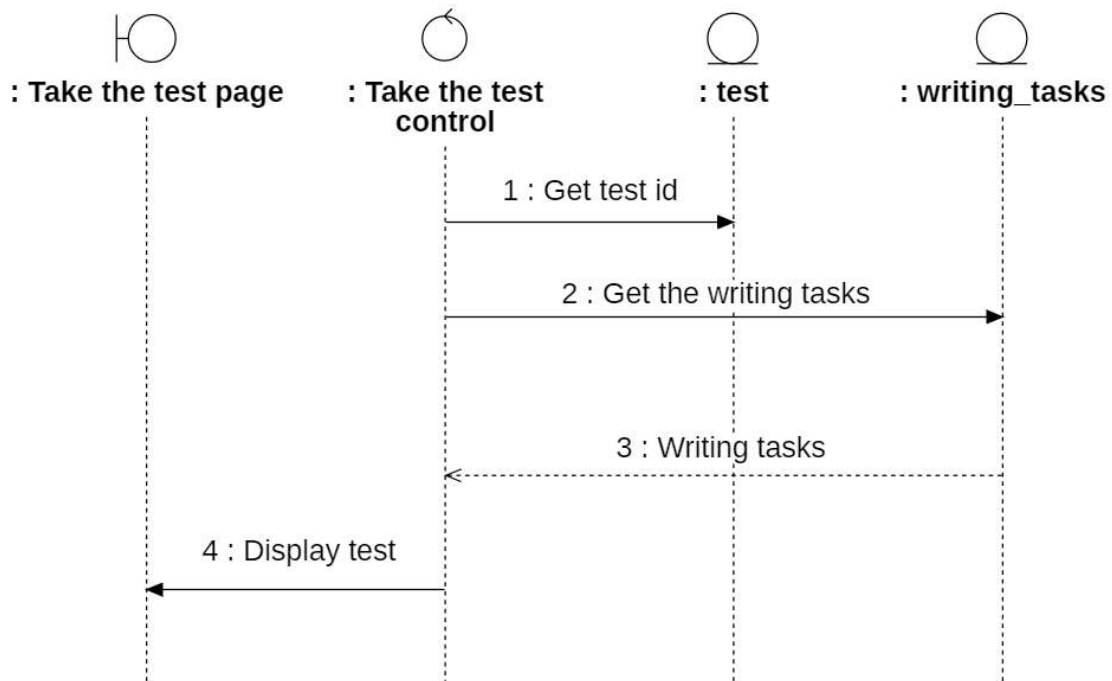


Figure 3.30. Load Writing test - Sequence Diagram

3.1.17. Doing The Test:

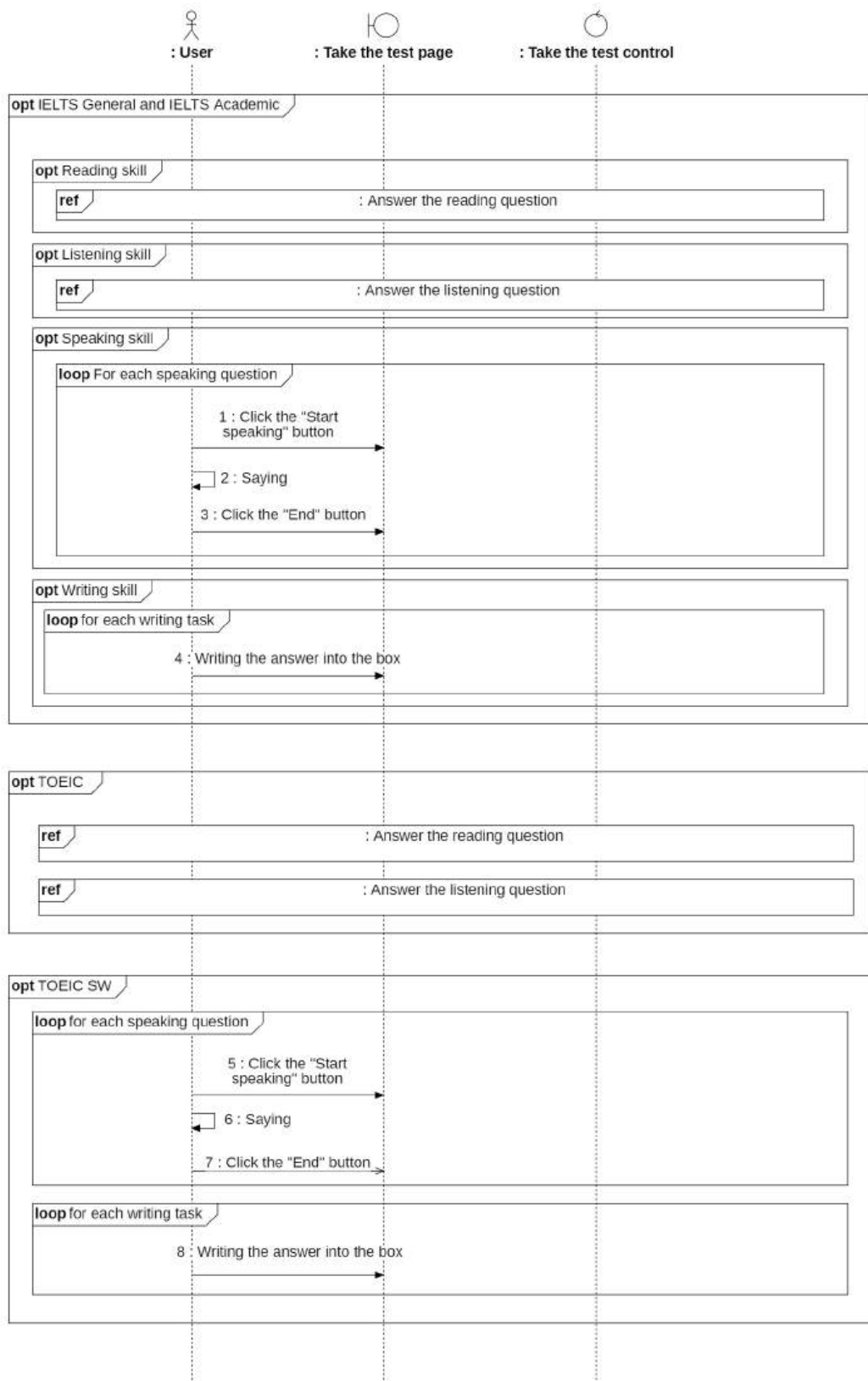


Figure 3.31 Doin. Doing the test - Sequence Diagram

3.1.18. Answer Reading Question:

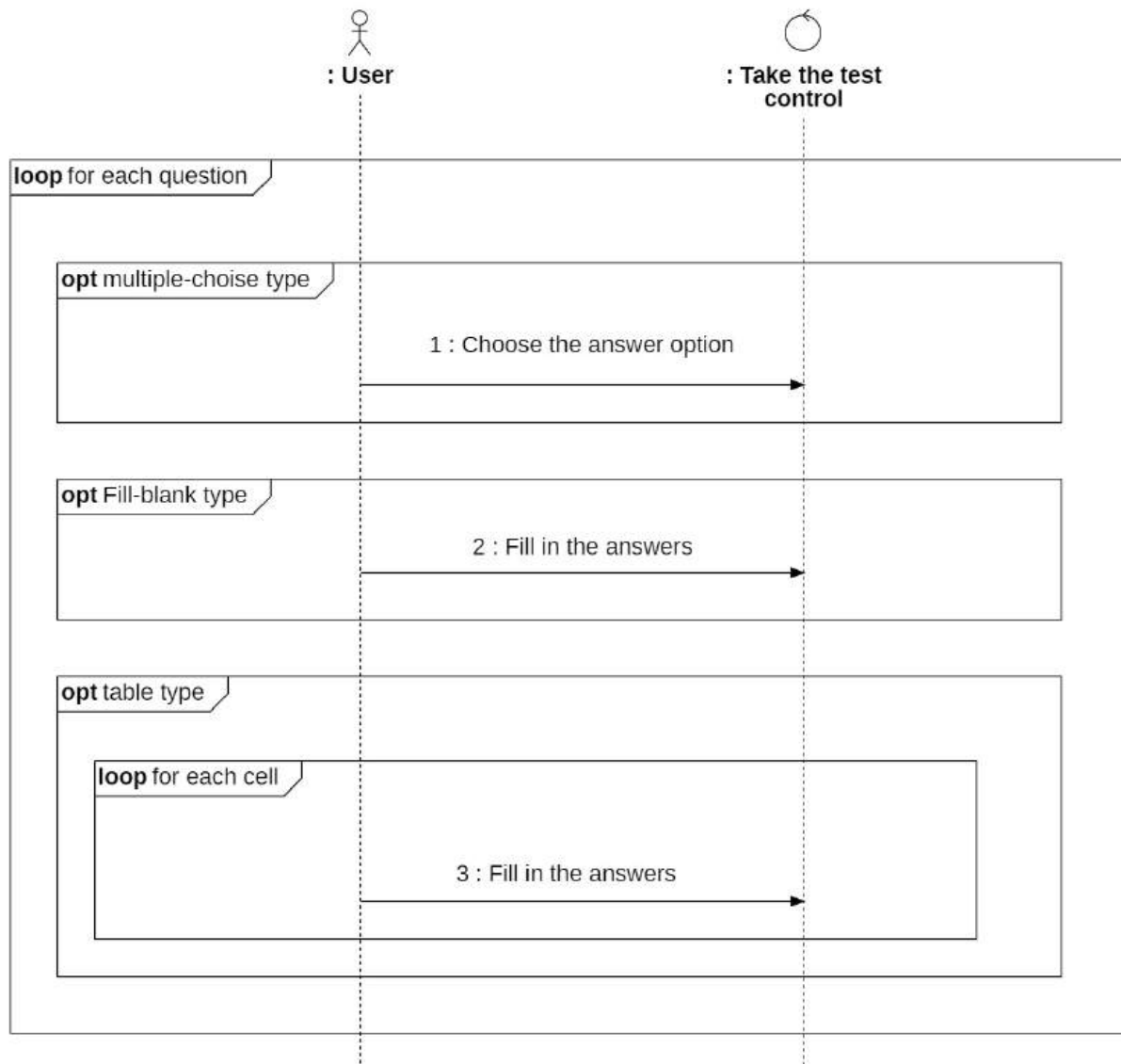


Figure 3.32. Answer Reading question - Sequence Diagram

3.1.19. Answer Listening Question:

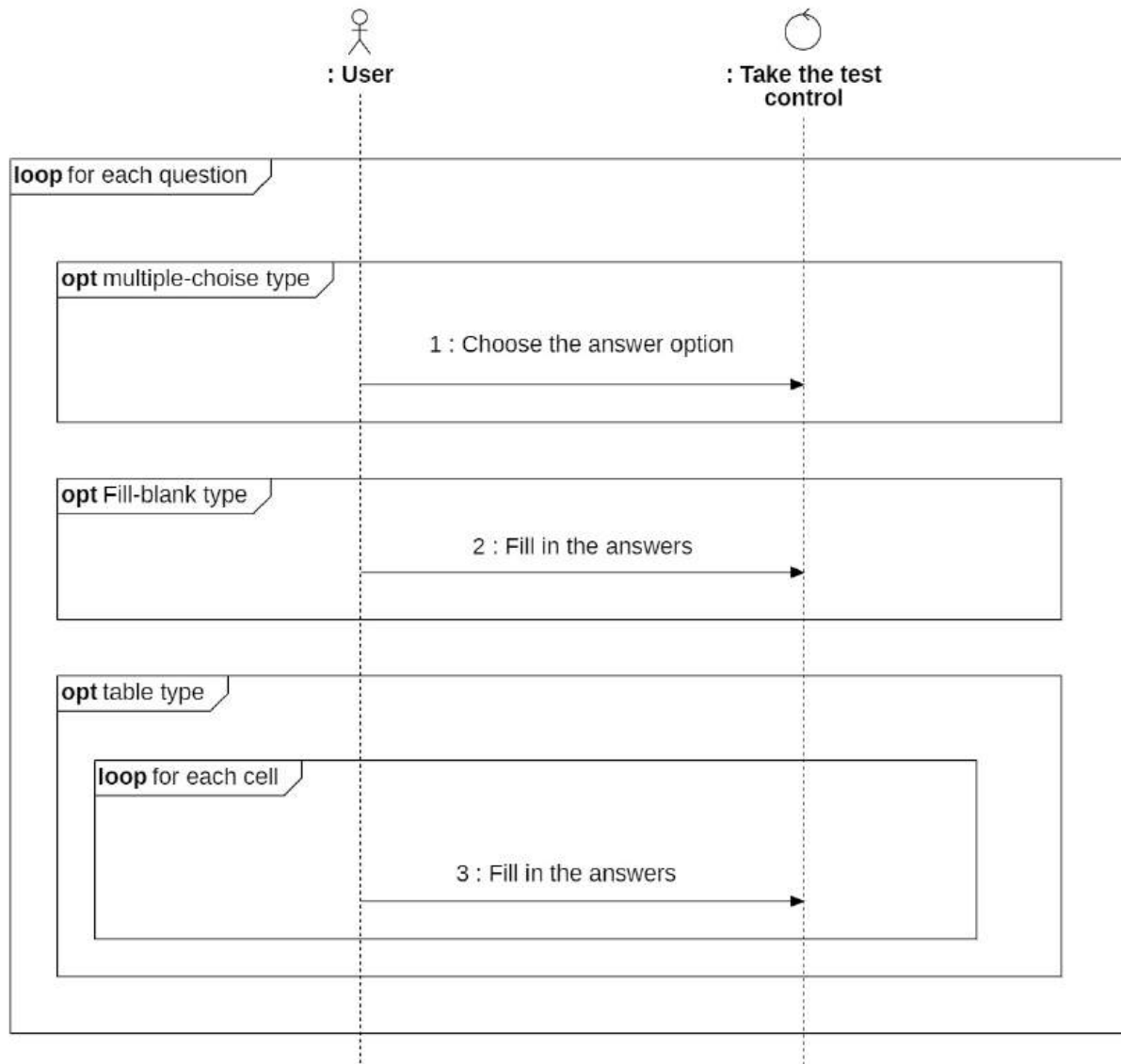


Figure 3.33. Answer Listening question - Sequence Diagram

3.1.20. Check The Answer:

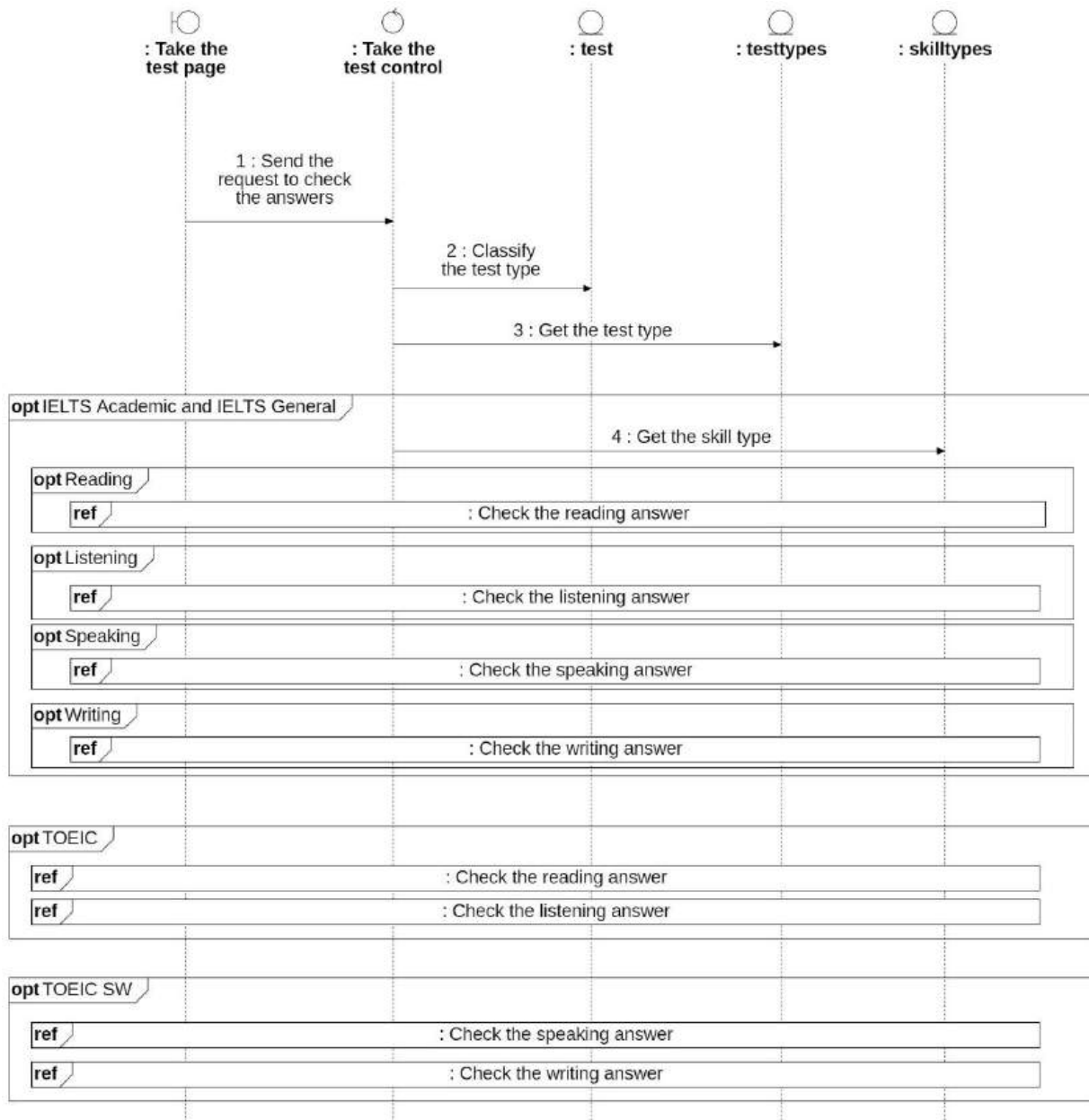


Figure 3.34. Check the answer - Sequence Diagram

3.1.21. Check Listening Answer:

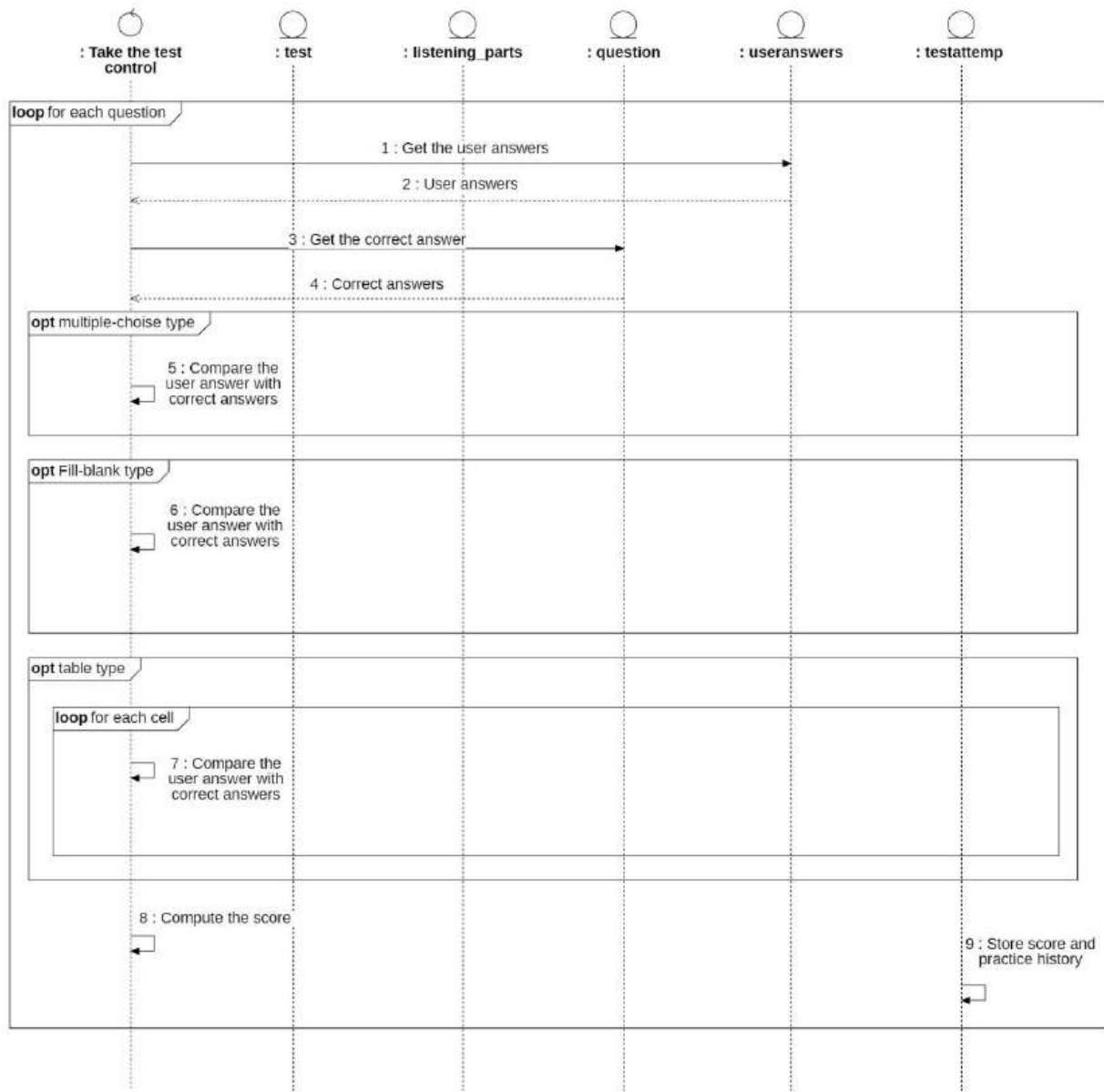


Figure 3.35. Check listening answer - Sequence Diagram

3.1.22. Check Reading Answer:

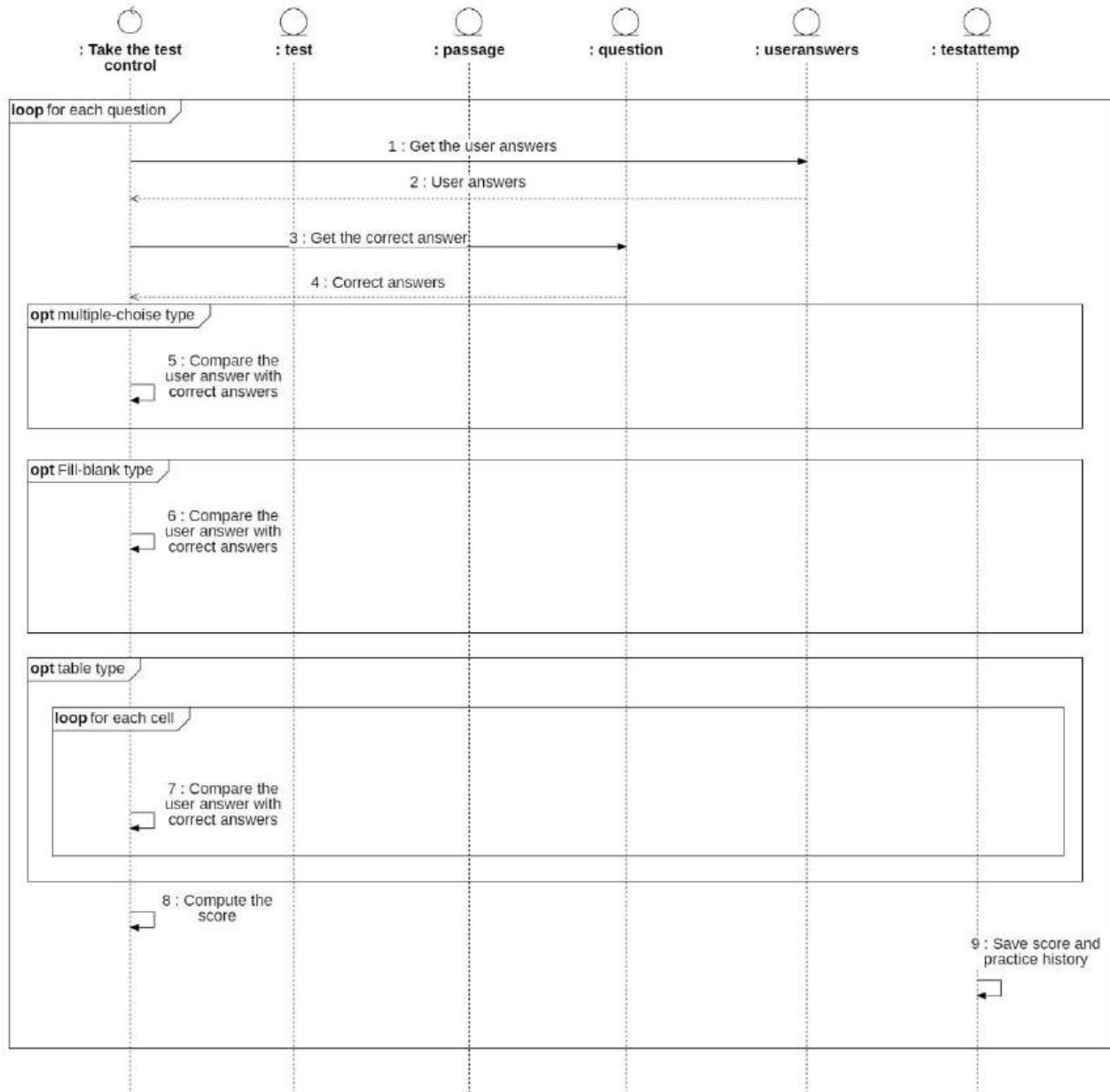


Figure 3.36. Check reading answer - Sequence Diagram

3.1.23. Check Listening Answer:

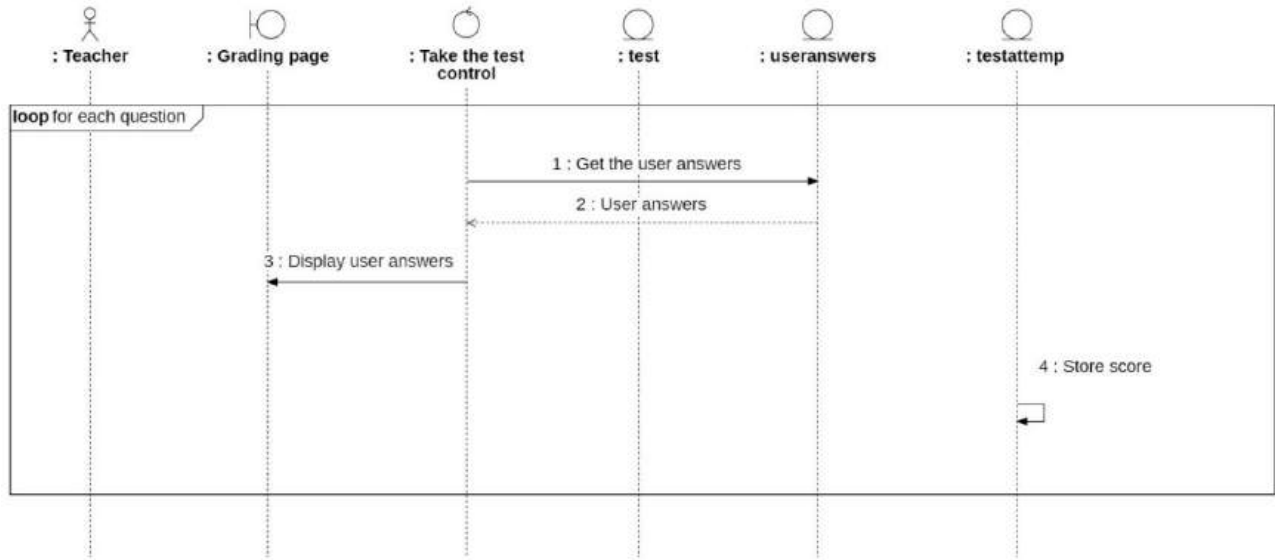


Figure 3.37. Check speaking answer - Sequence Diagram

3.1.24. Check Writing Answer:

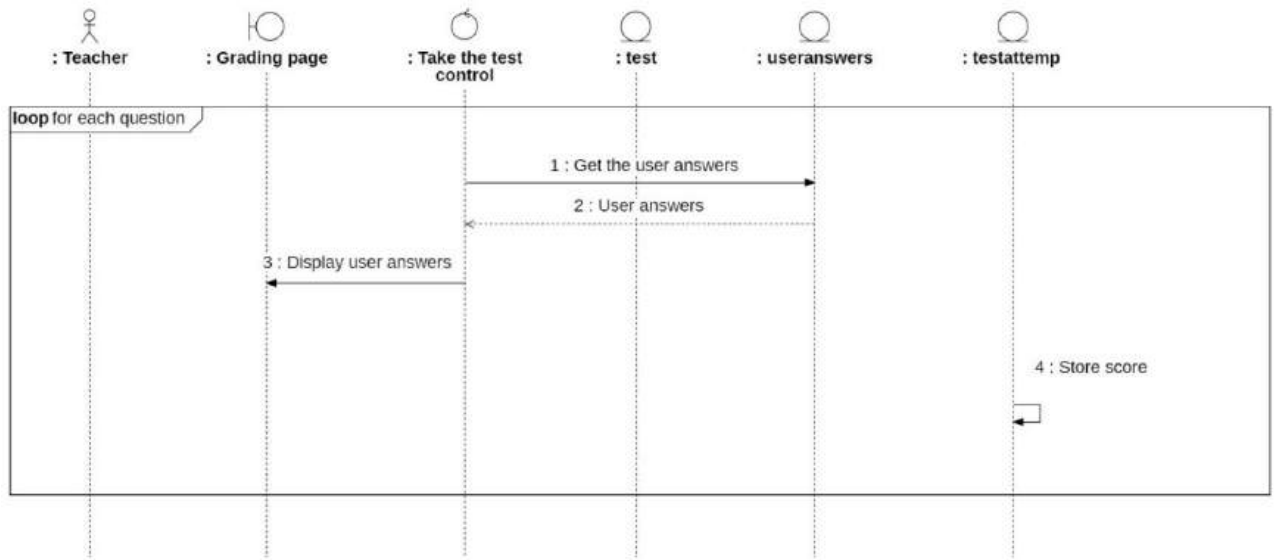


Figure 3.38. Check writing answer - Sequence Diagram

3.1.25. Create Test:

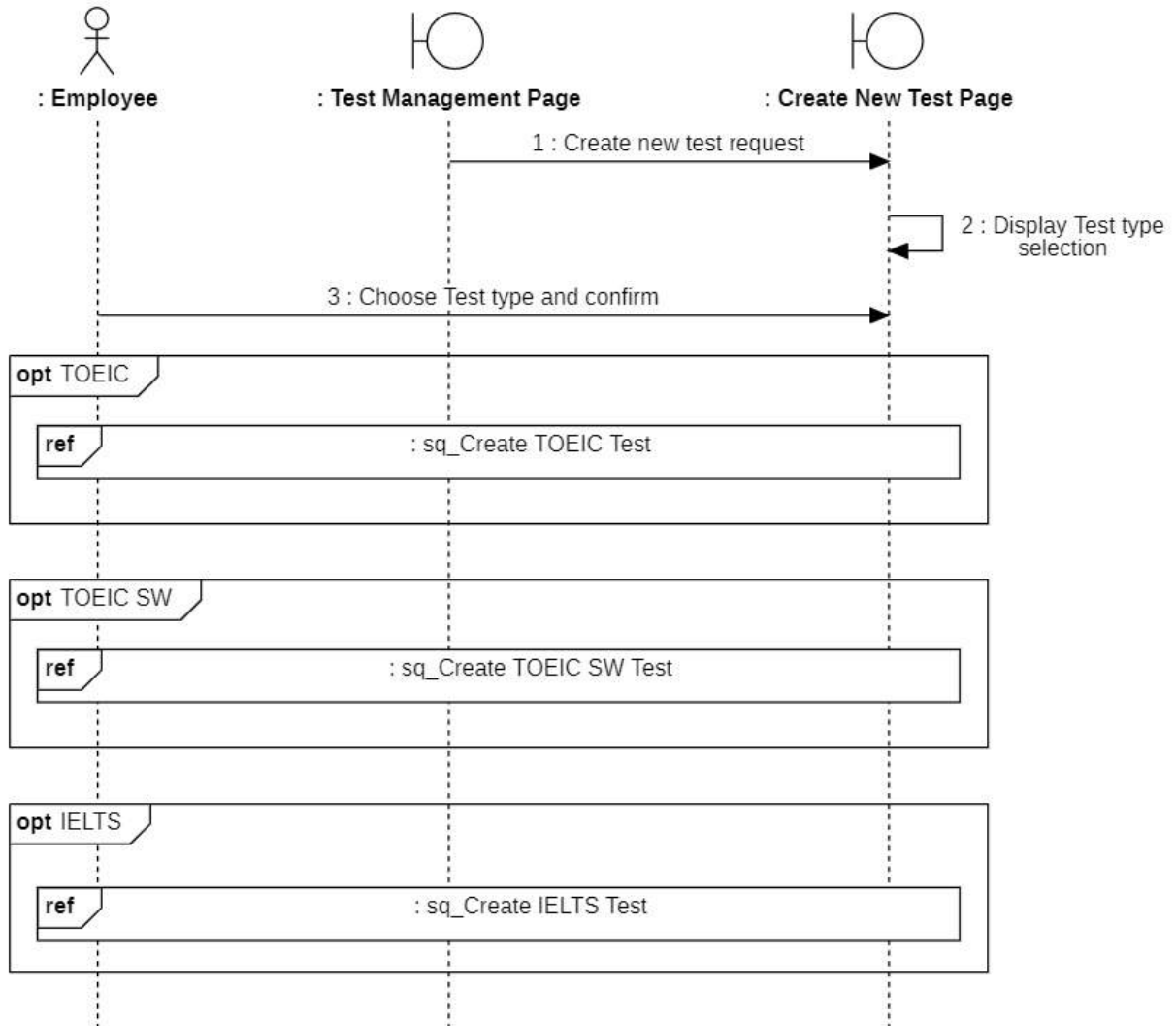


Figure 3.39. Create test - Sequence Diagram

3.1.26. Create TOEIC Test:

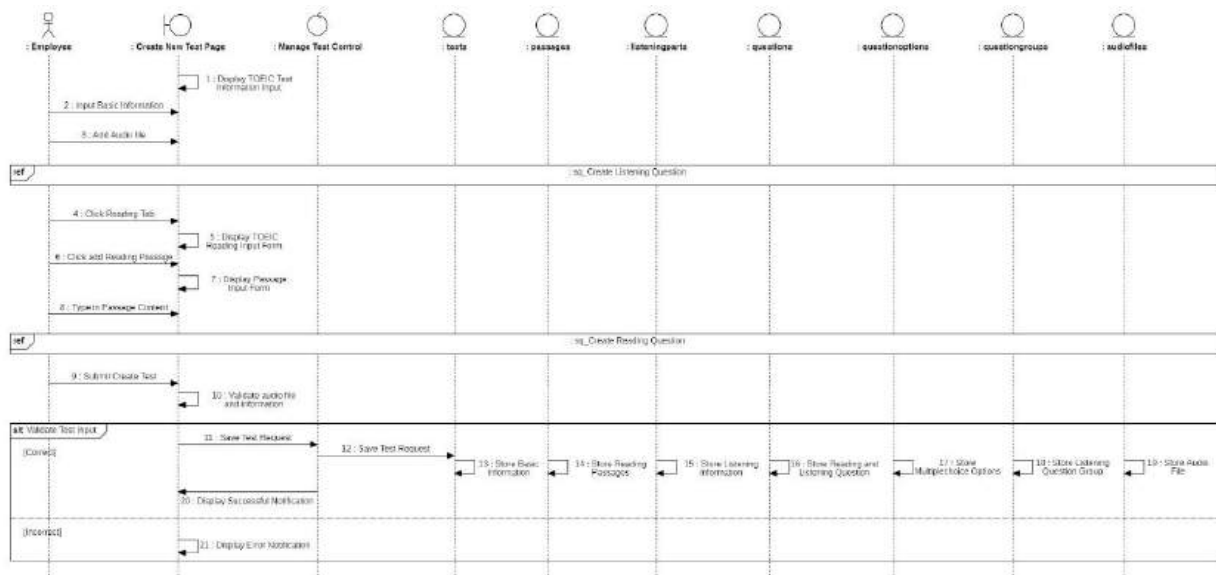


Figure 3.40. Create TOEIC Test - Sequence Diagram

3.1.27. Create TOEIC SW Test:

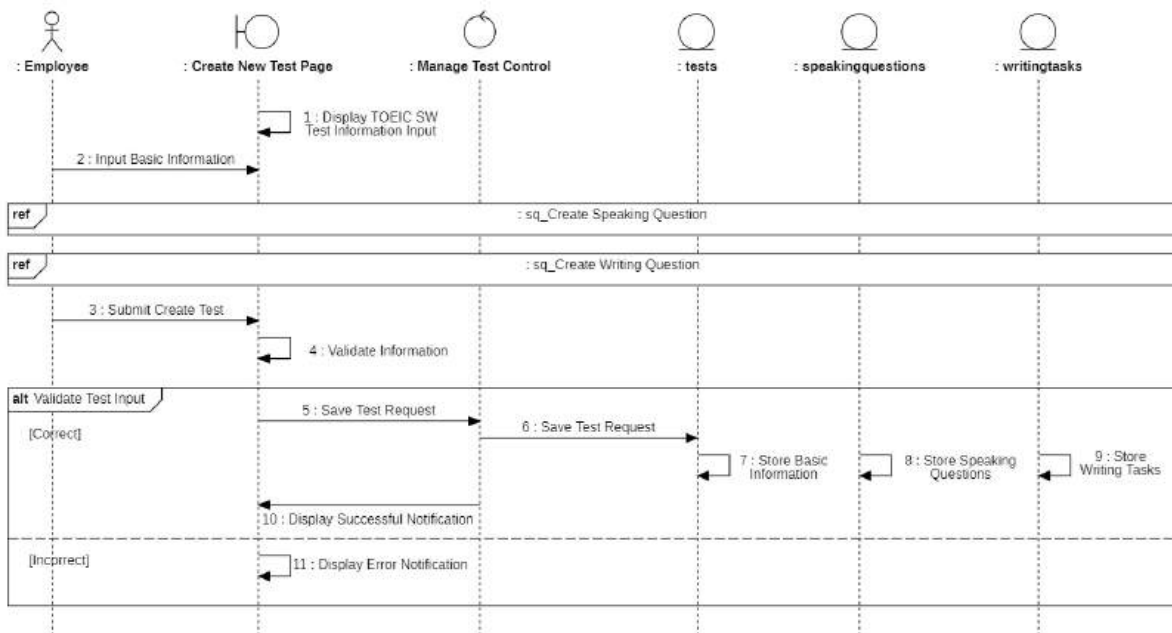


Figure 3.41. Create TOEIC SW Test - Sequence Diagram

3.1.28. Create IELTS Test:

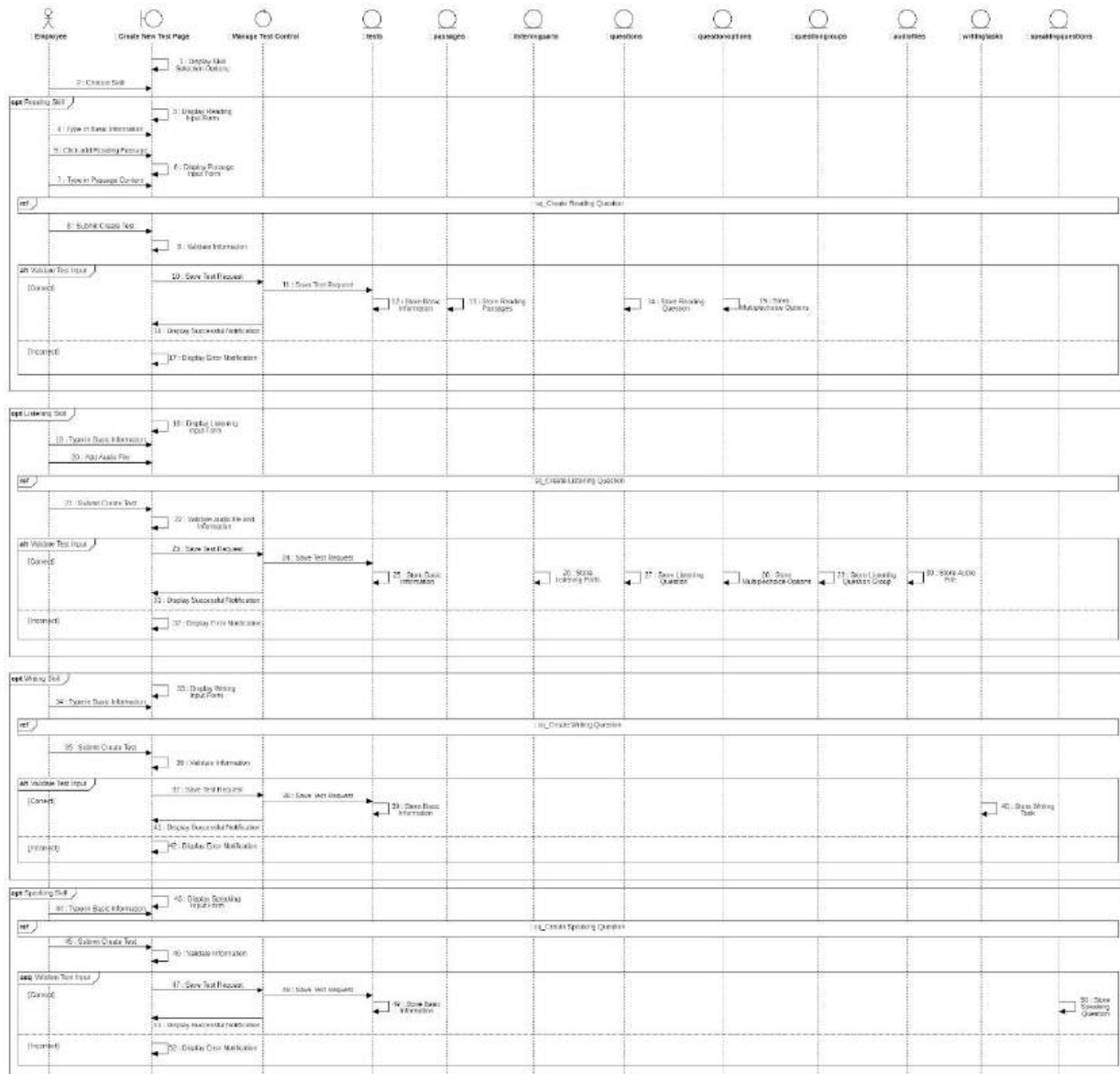


Figure 3.42. Create IELTS Test - Sequence Diagram

3.1.29. Create Reading Question:

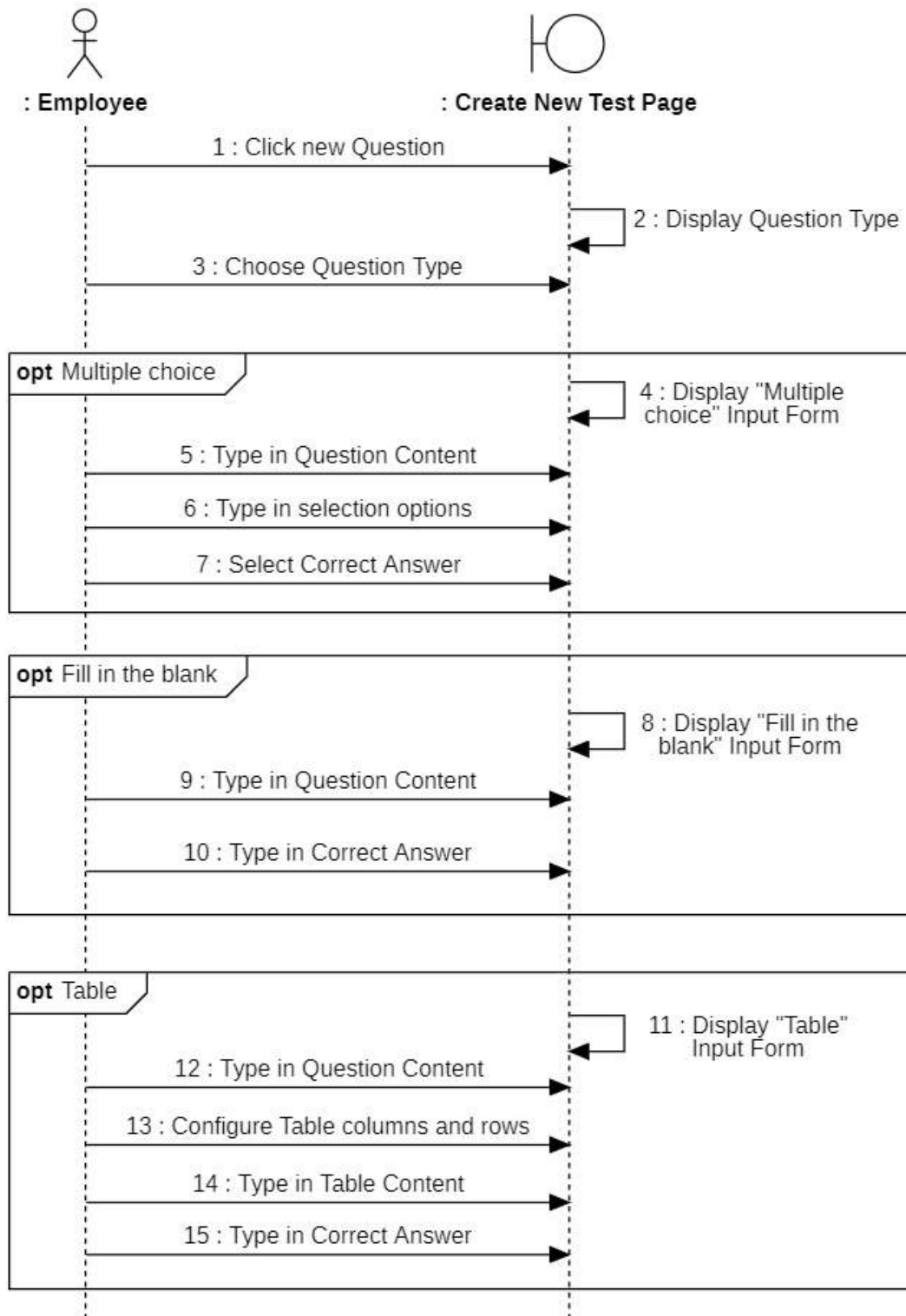


Figure 3.43. Create Reading Question - Sequence Diagram

3.1.30. Create Listening Question:

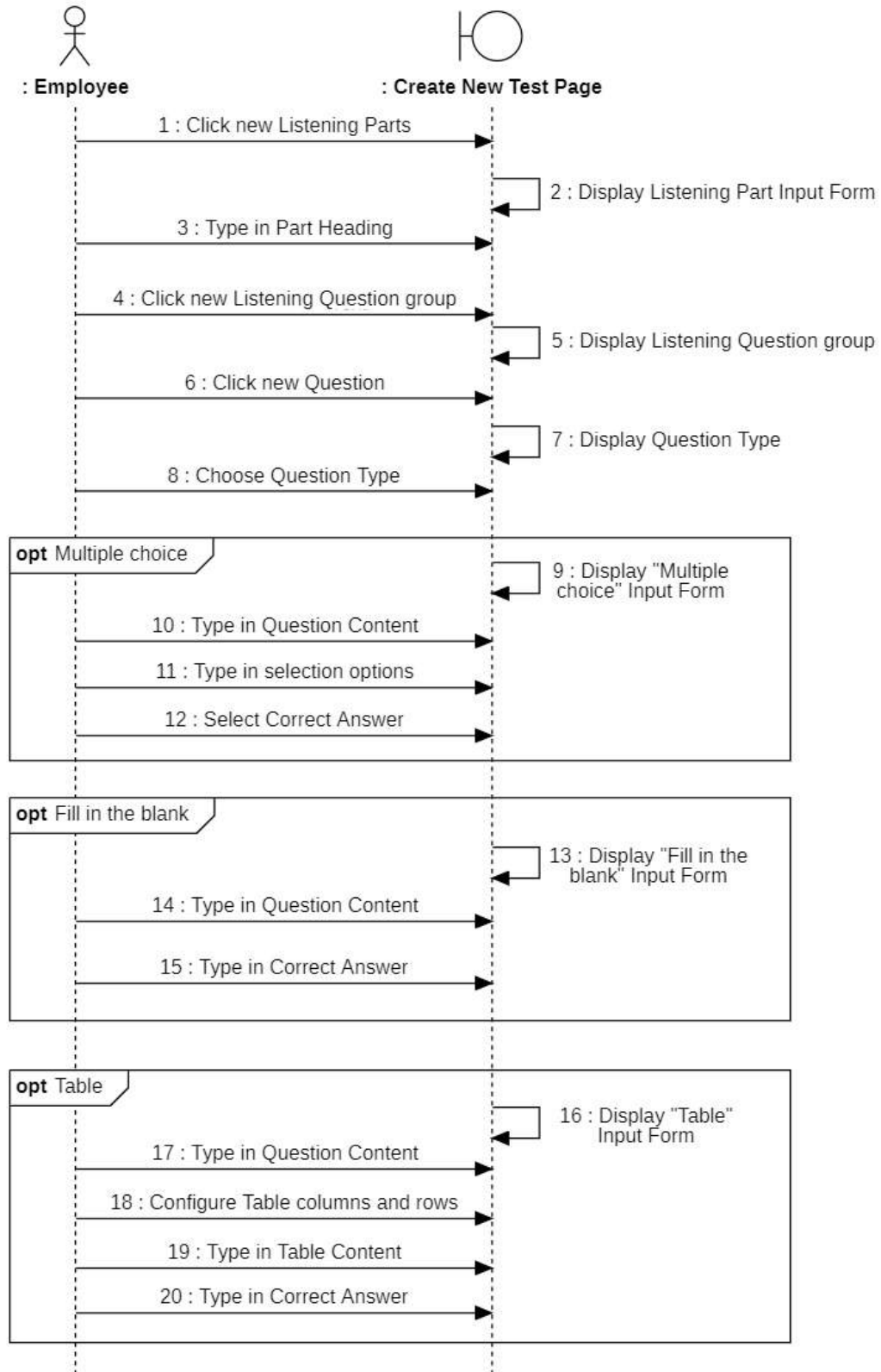


Figure 3.44. Create Listening Question - Sequence Diagram

3.1.31. Create Speaking Question:

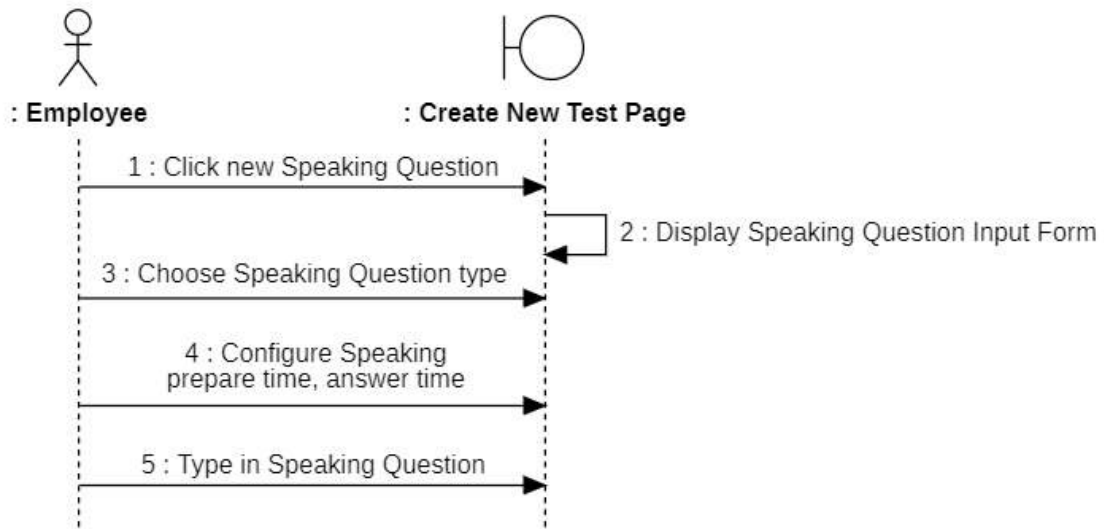


Figure 3.45. Create Speaking Question - Sequence Diagram

3.1.32. Create Writing Question:

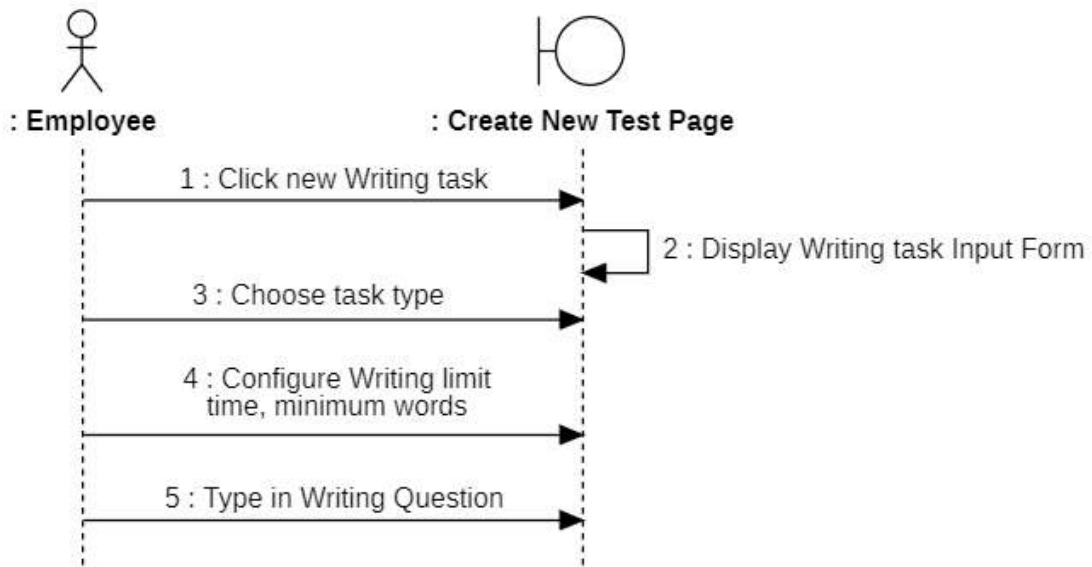


Figure 3.46. Create Writing Question - Sequence Diagram

3.1.33. Update TOEIC Test:

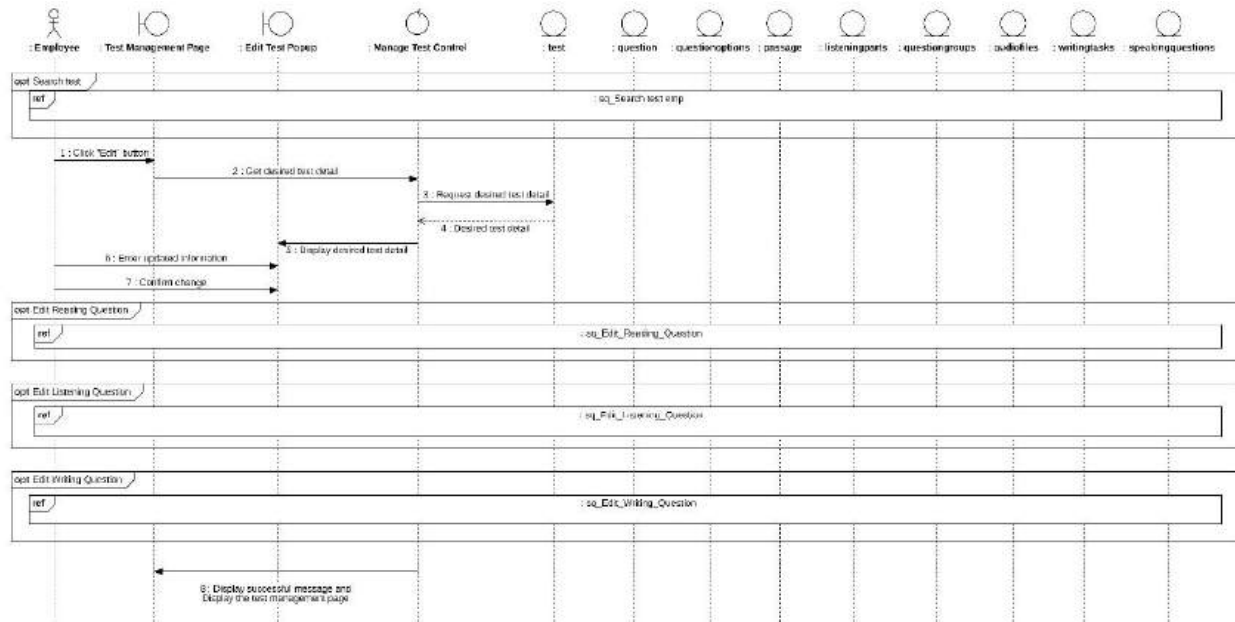


Figure 3.47. Update TOEIC Test - Sequence Diagram

3.1.34. Update TOEIC SW Test:

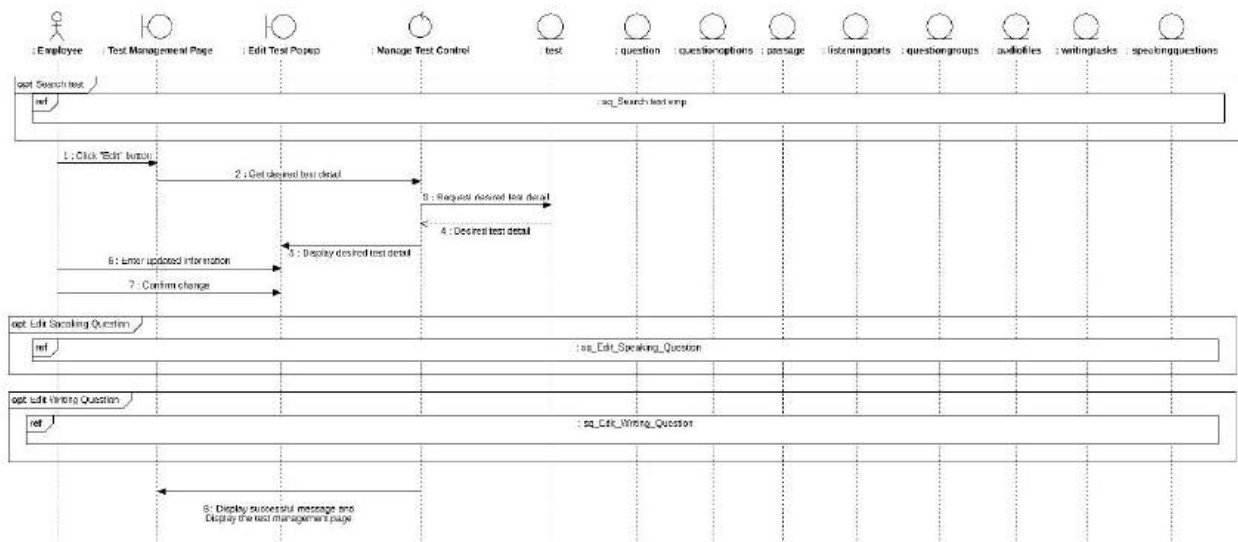


Figure 3.48. Update TOEIC SW Test - Sequence Diagram

3.1.35. Update IELTS Test:

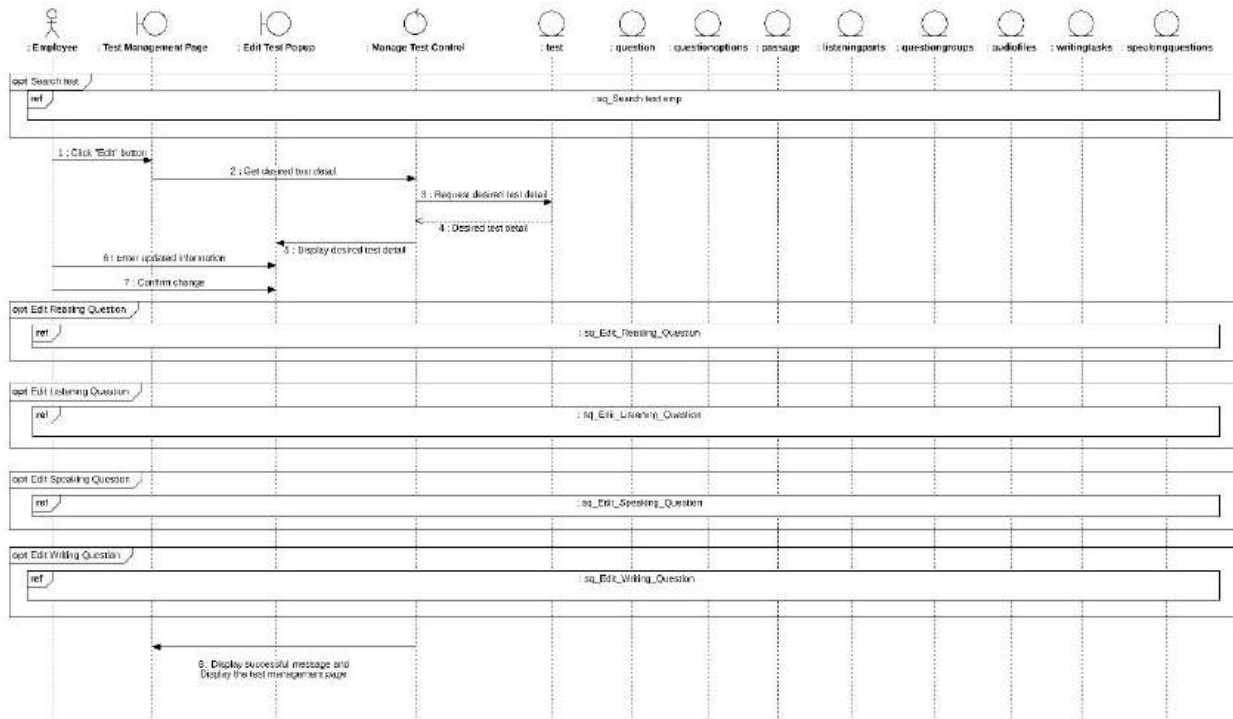


Figure 3.49. Update IELTS Test - Sequence Diagram

3.1.36. Update Reading Question:

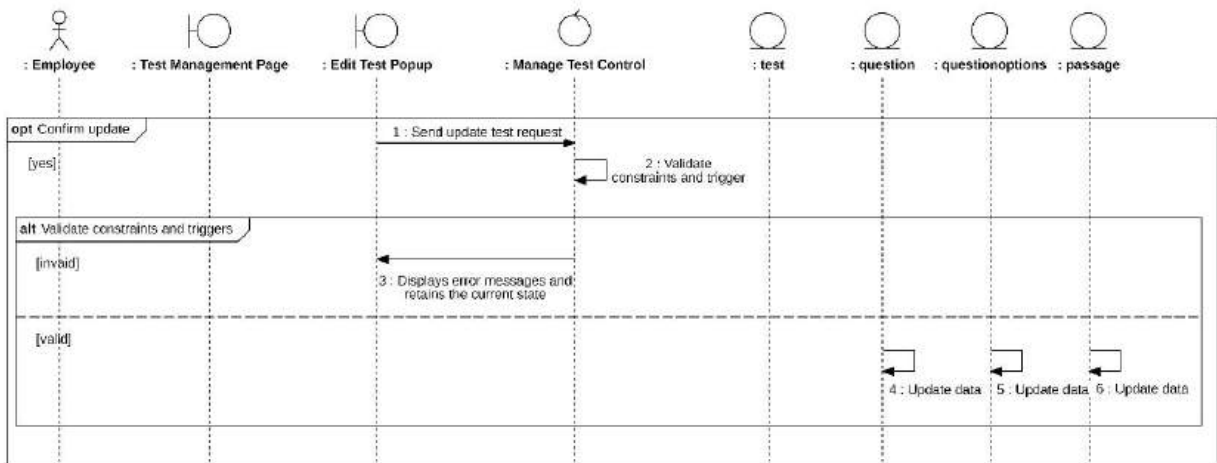


Figure 3.50. Update Reading Question - Sequence Diagram

3.1.37. Update Listening Question:

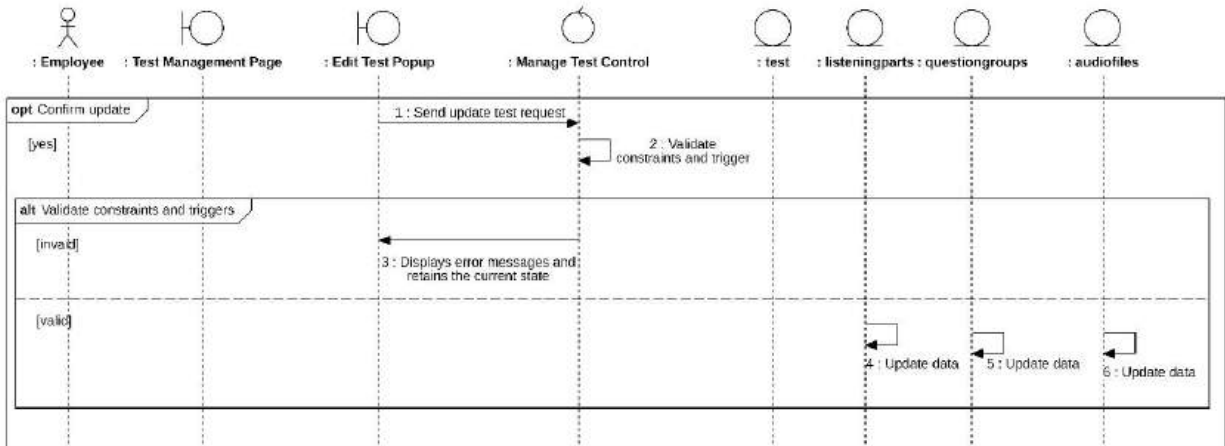


Figure 3.51. Update Listening Question - Sequence Diagram

3.1.38. Update Speaking Question:

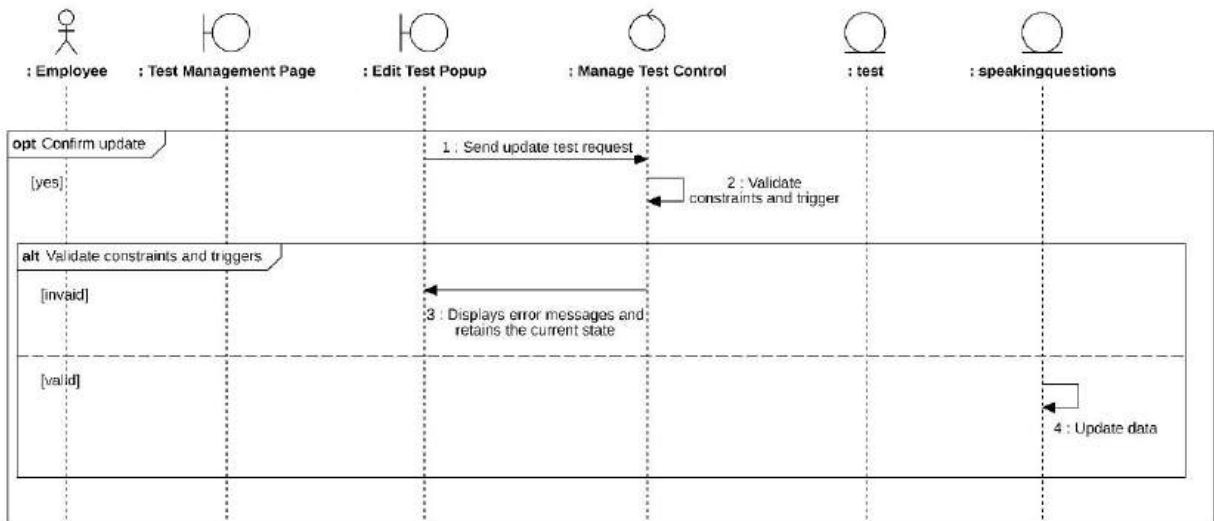


Figure 3.52. Update Speaking Question - Sequence Diagram

3.1.39. Update Writing Question:

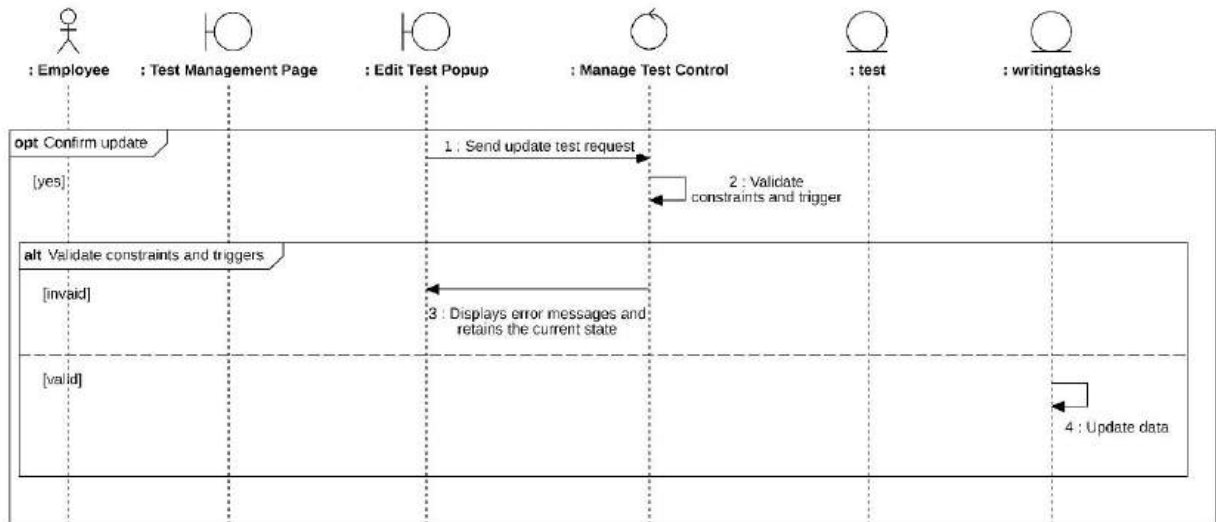


Figure 3.53. Update Writing Question - Sequence Diagram

3.1.40. Delete Test:

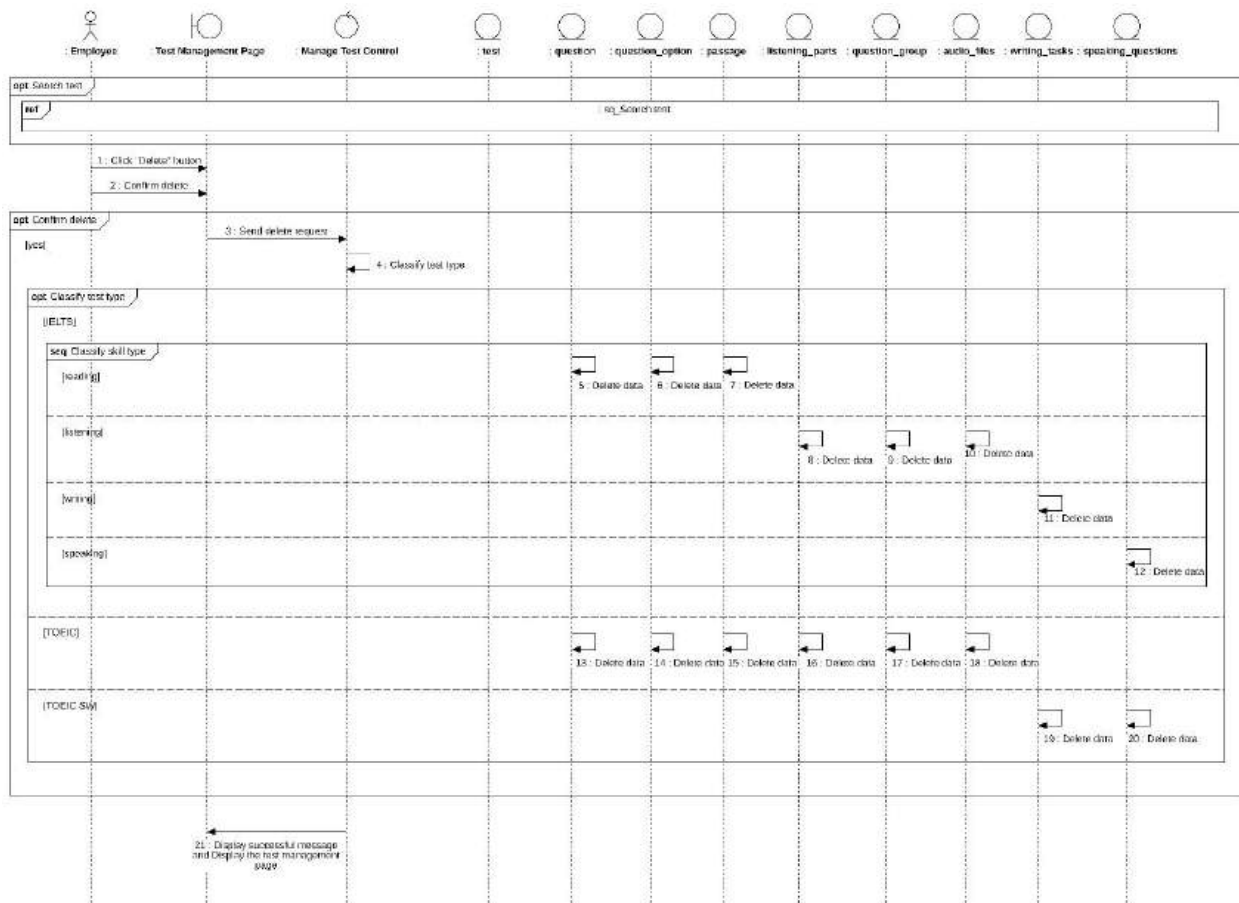


Figure 3.54. Delete test - Sequence Diagram

3.2. Class Diagram

3.2.1. Register:

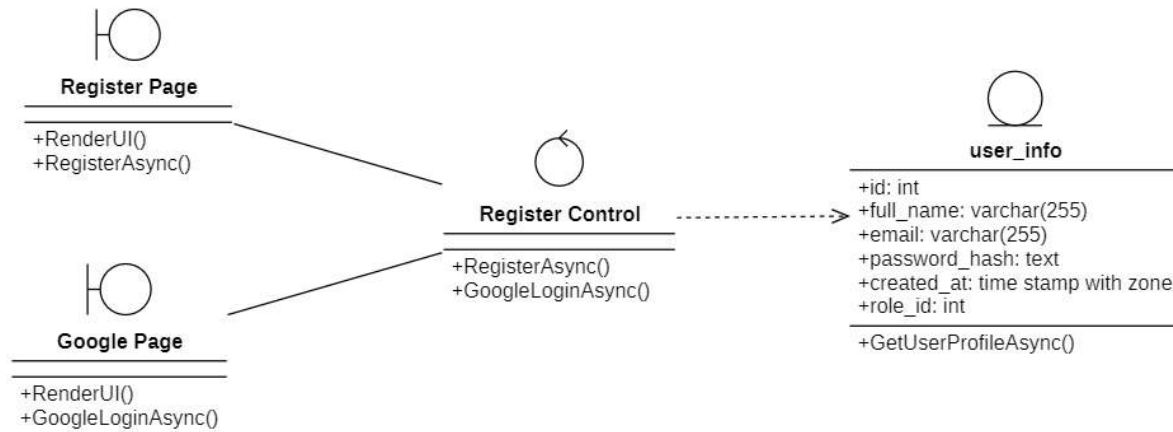


Figure 3.55. Register - Class diagram

3.2.2. Login:

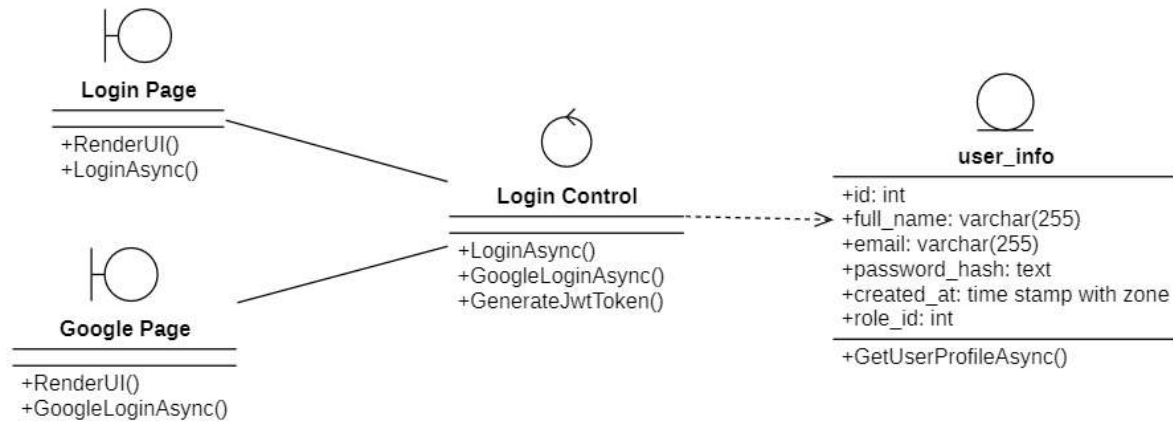


Figure 3.56. Login - Class diagram

3.2.3. Take the Test:

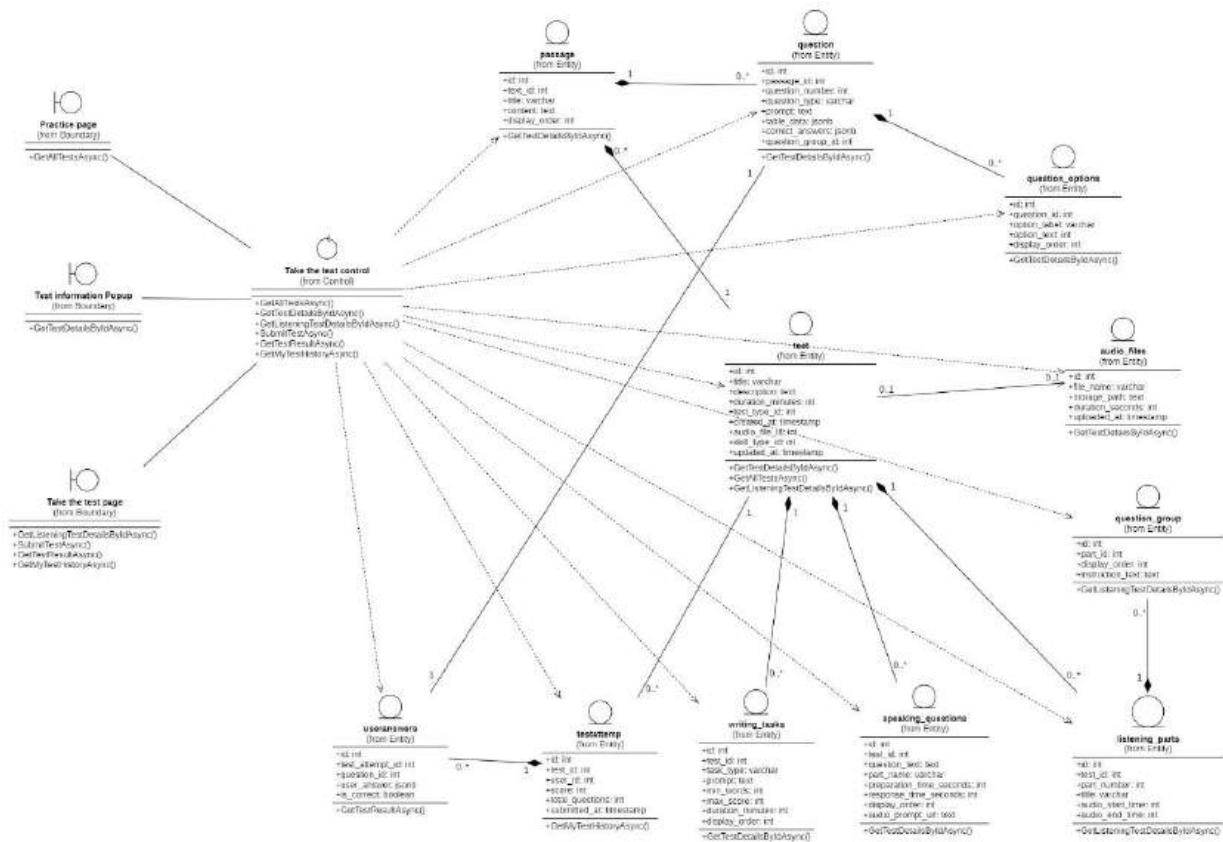


Figure 3.57. Take the test - Class diagram

3.2.4. Create Test:

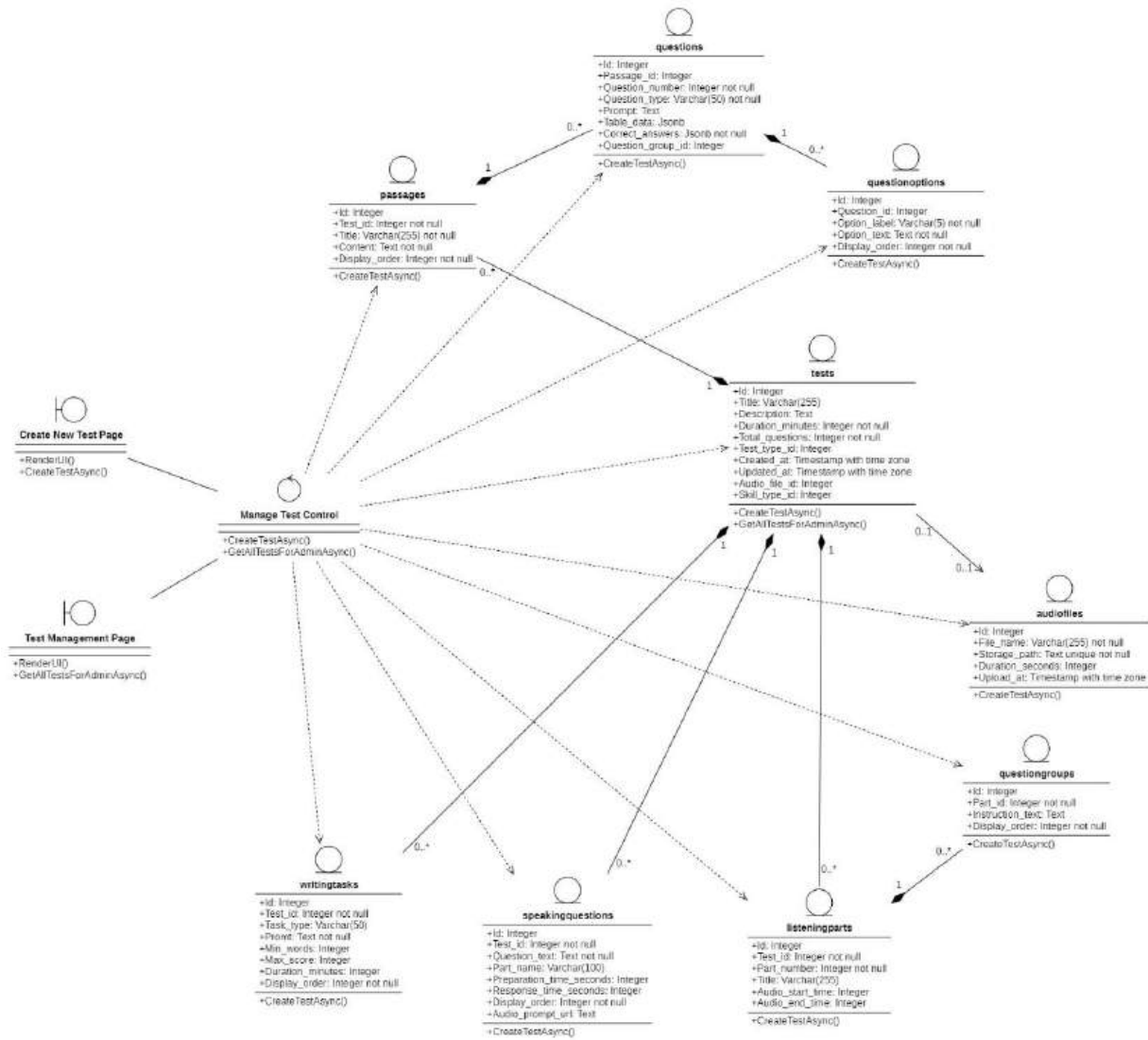


Figure 3.58. Create the test - Class diagram

3.2.5. Update Test:

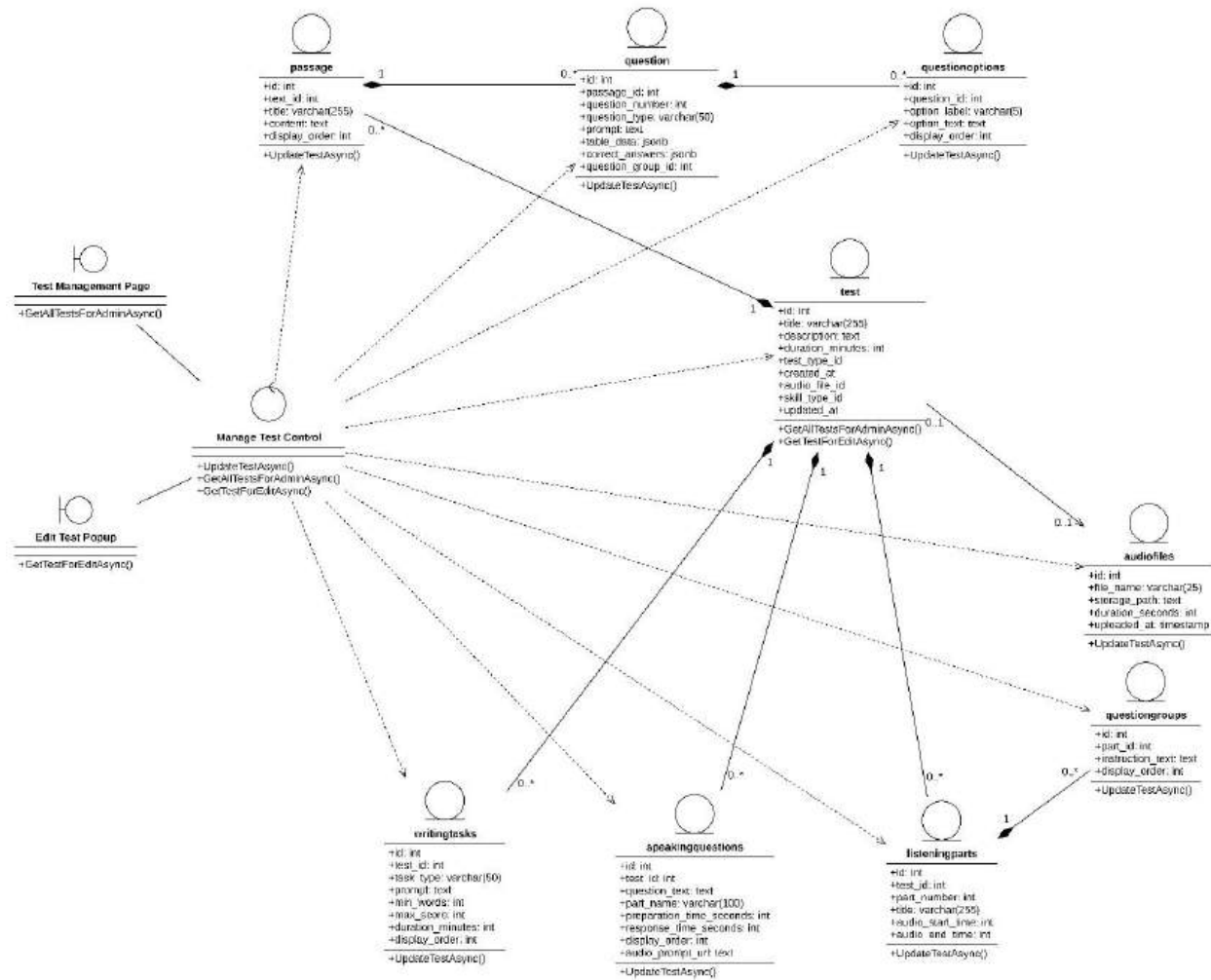


Figure 3.59. Update test - Class diagram

3.2.6. Delete Test:

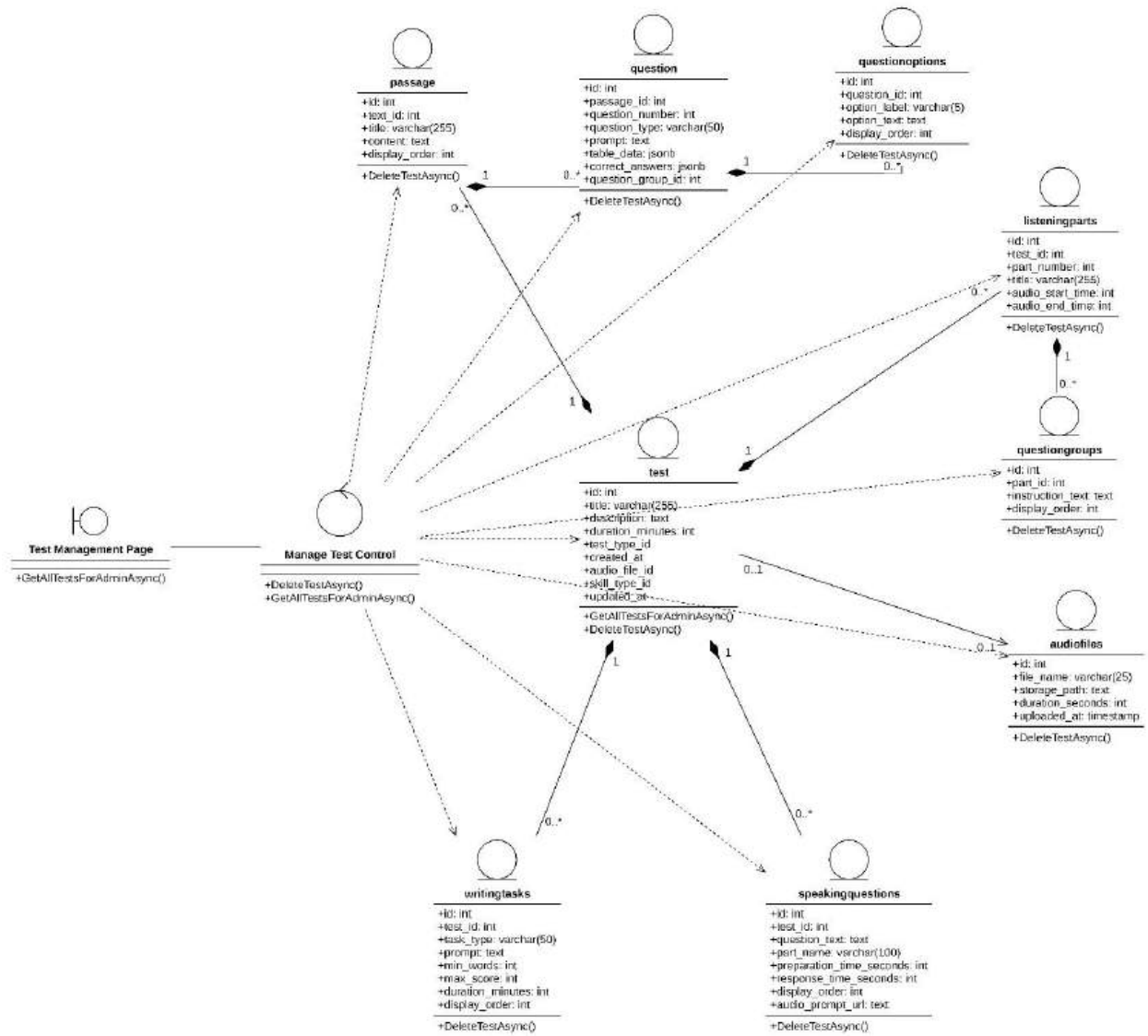


Figure 3.60. Delete - Class diagram

4. State diagram

4.1.1. Take the Test:

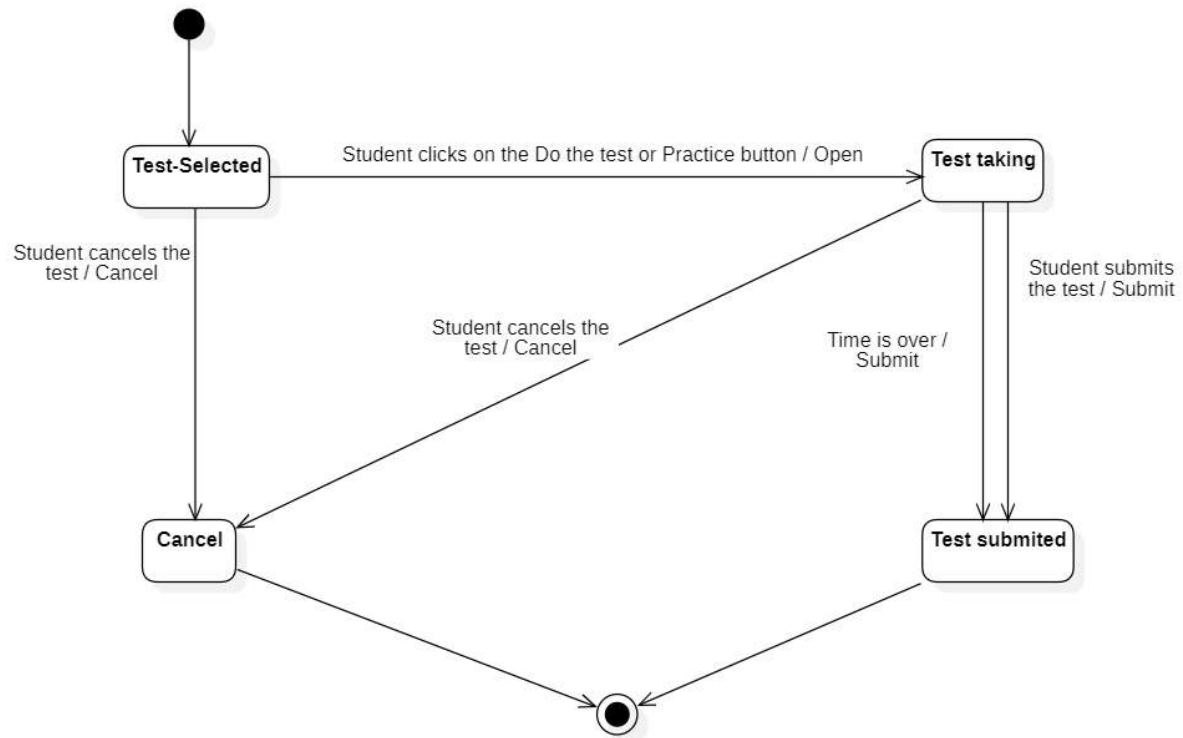


Figure 3.61. Take the Test - State diagram

5. Entity class diagram

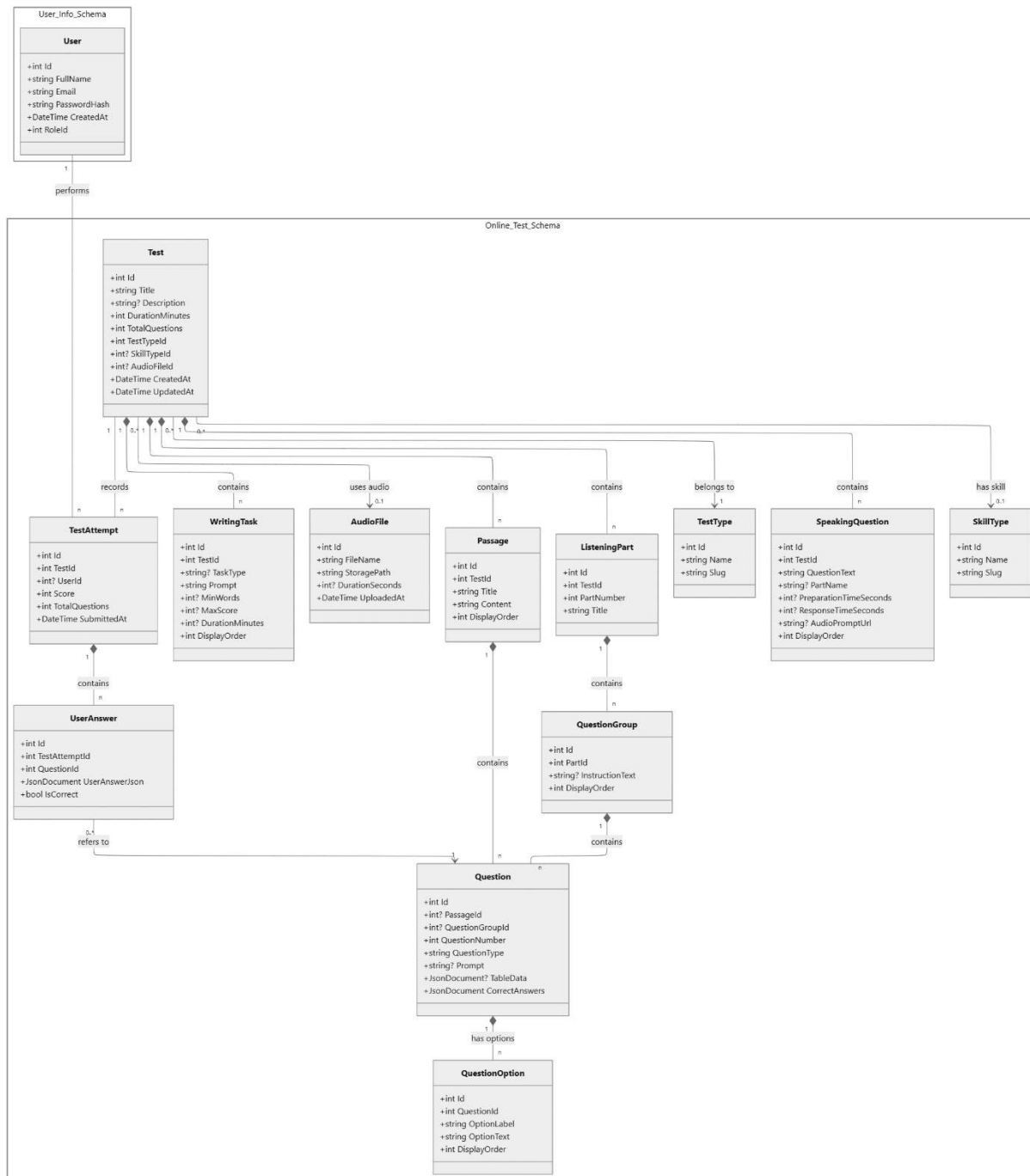


Figure 3.625. Entity class diagram

6. Mapping to Relational Model from Entity class diagram

Relational Model:

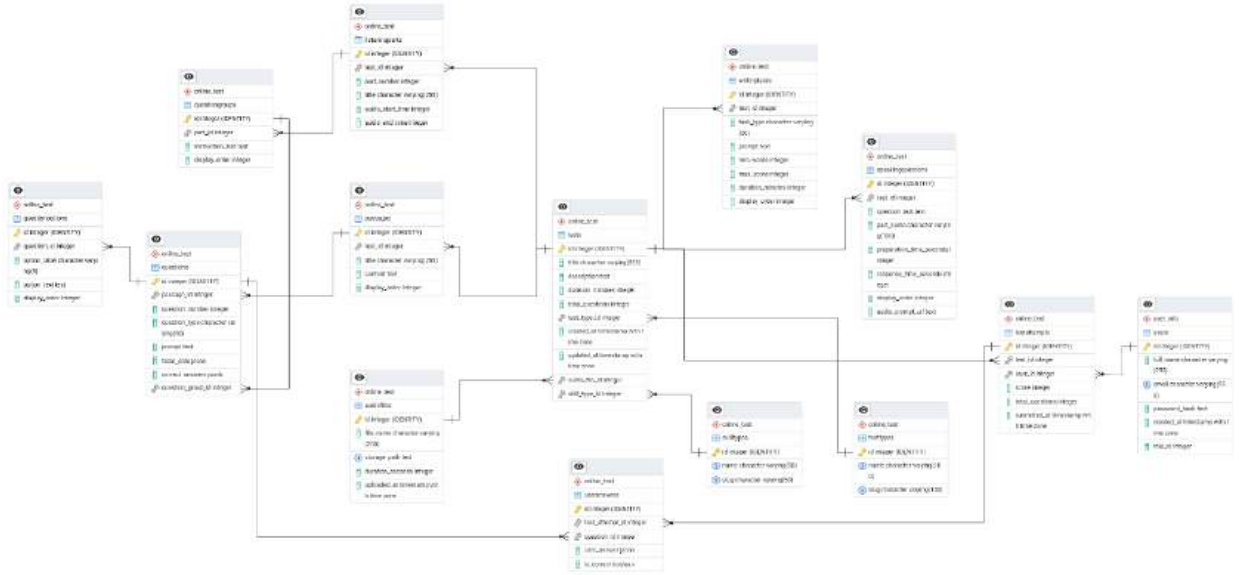


Figure 3.63. Relational Model

The transformation from the Entity Class Diagram to the Relational Model represents the instantiation of the logical design into a physical storage structure. In the Enly Online Test System, this mapping is executed via the Object-Relational Mapping (ORM) mechanism provided by Entity Framework Core. The process adheres to standard data normalization principles while incorporating specific optimizations to leverage the advanced capabilities of PostgreSQL.

Specifically, each Entity Class, representing a business domain object, is mapped to a corresponding Table in the database. Primitive properties such as Id, Title, and DurationMinutes are directly converted into Columns with appropriate data types. Notably, the system utilizes PostgreSQL's JSONB data type to map complex, semi-structured attributes like CorrectAnswers or UserAnswerJson. This approach preserves the flexibility of the object-oriented design without violating the rigidity of the relational schema. Associations such as Aggregation and Composition are translated into Foreign Key (FK) constraints, ensuring the referential integrity of the examination data.

The following table summarizes the transformation rules applied to the system, illustrated by key entities such as Test, Question, and UserAnswer:

Table 3.15. Table summarizes the transformation rules

Object Oriented Component (Entity Class Diagram)	Relational Component (Relational Model)	Mapping Rules and Rationale
Class	Table	- Each Class is mapped to a distinct Table. - Table names are converted from PascalCase to pluralized snake_case.
Identity Attribute	Primary Key	The Id property becomes the Primary Key
Primitive Attribute	Column	- String: VARCHAR(n) or TEXT - Int: INTEGER - DateTime: TIMESTAMP WITH TIME ZONE
Dynamic Structure Attribute	Binary JSON Column	- Instead of normalizing into auxiliary tables (EAV pattern), semi-structured attributes are mapped to JSONB columns. - This facilitates the storage of polymorphic answer schemas (e.g., multiple choice, gap-fill, matching) within a single column.
One-to-Many Relationship	Foreign Key	In the relational model, the child table (the "Many" side) holds the Foreign Key referencing the parent table (the "One" side).

Zero-to-One Relationship	Nullable Foreign Key	Optional associations (0..1) are mapped to Foreign Key columns that allow NULL values. Example: For Reading tests, the audio_file_id column remains NULL.
Polymorphic Association	Mutually Exclusive Nullable Foreign Keys	- A single questions table serves both Reading and Listening skills. - Two nullable FK columns (passage_id, question_group_id) are used to determine the context of the question.

CHAPTER 4. IMPLEMENTATION AND TESTING

1. Introduction to the implementation environment

On the frontend, the project uses Vue.js, a progressive JavaScript framework well-suited for building reactive and component-based user interfaces. Vue allows the application to efficiently render dynamic content such as test questions, timers, score displays, and interactive forms. Standard CSS is used for styling, ensuring a clean, responsive, and user-friendly interface compatible with both mobile and desktop devices.

The backend is developed using C# with the .NET platform, chosen for its robustness, strong typing, high performance, and scalability. The backend follows a RESTful API architecture, handling user authentication, test processing, scoring logic, and communication with the database. The system integrates Swagger (OpenAPI) to auto-generate interactive API documentation, enabling easier testing, debugging, and collaboration during development.

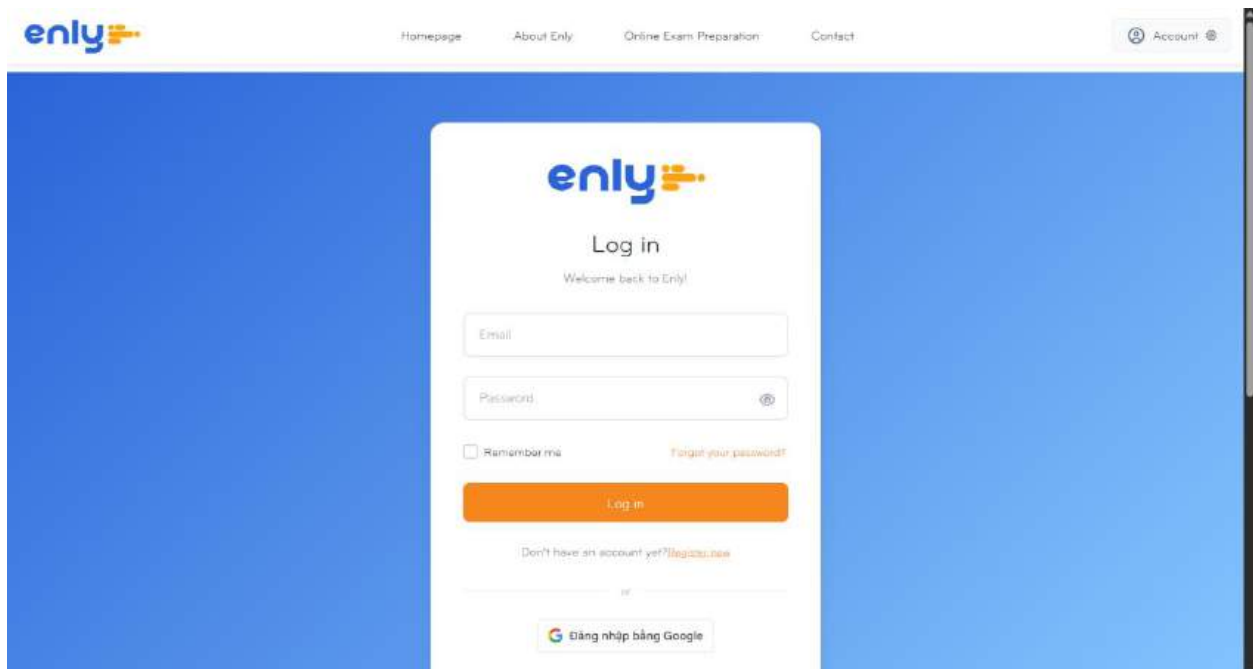
For data storage, the platform uses PostgreSQL, a powerful, open-source relational database known for stability, strong ACID compliance, and advanced indexing features. PostgreSQL manages core data such as user accounts, tests, questions, results, and contact messages. The database schema is designed based on the Entity Class Diagram developed during the analysis phase, ensuring normalization and referential integrity.

Development tools such as Visual Studio / Visual Studio Code, Git/GitHub for version control, and Postman or Swagger UI for API testing are utilized throughout the implementation process. UML modeling tools like StarUML or Draw.io are used during the design phase to create the system diagrams (use-case, sequence, class, state, and activity diagrams).

Overall, this implementation environment provides a reliable, high-performance foundation for developing the Enly system. It ensures smooth user interaction, maintainable backend logic, secure data storage, and the ability to extend the platform with new test modules or learning features in the future.

2. User Interface

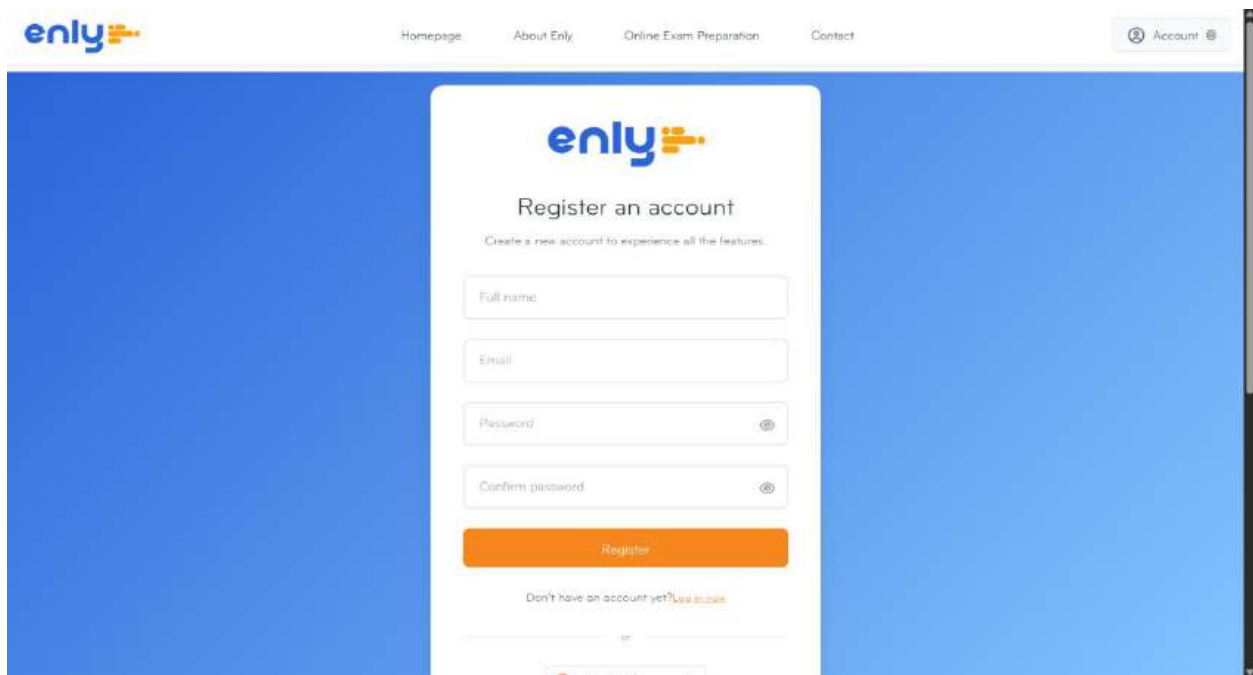
2.1. Login page



The screenshot shows the login page of the Enly website. The page has a blue background with a white central form. At the top, there is a navigation bar with the Enly logo on the left and links for 'Homepage', 'About Enly', 'Online Exam Preparation', and 'Contact' on the right. In the top right corner, there is an 'Account' link with a user icon. The central form features the Enly logo at the top, followed by the text 'Log in' and 'Welcome back to Enly!'. Below this, there are input fields for 'Email' and 'Password'. The password field has an eye icon to toggle visibility. There is a checkbox for 'Remember me' and a link for 'Forgot your password?'. A large orange 'Log in' button is positioned below the input fields. At the bottom of the form, there is a link 'Don't have an account yet? [Register now](#)' and a 'or' separator. Below the separator is a button for 'Đăng nhập bằng Google' with the Google logo.

Figure 4.1. Login page

2.2. Register page



The screenshot shows the register page of the Enly website. The page has a blue background with a white central form. At the top, there is a navigation bar with the Enly logo on the left and links for 'Homepage', 'About Enly', 'Online Exam Preparation', and 'Contact' on the right. In the top right corner, there is an 'Account' link with a user icon. The central form features the Enly logo at the top, followed by the text 'Register an account' and 'Create a new account to experience all the features.'. Below this, there are input fields for 'Full name', 'Email', 'Password', and 'Confirm password'. The password and confirm password fields have eye icons to toggle visibility. A large orange 'Register' button is positioned below the input fields. At the bottom of the form, there is a link 'Don't have an account yet? [Log in now](#)' and a 'or' separator. Below the separator is a button for 'Đăng nhập bằng Google' with the Google logo.

Figure 4.2. Register page

2.3. Home page

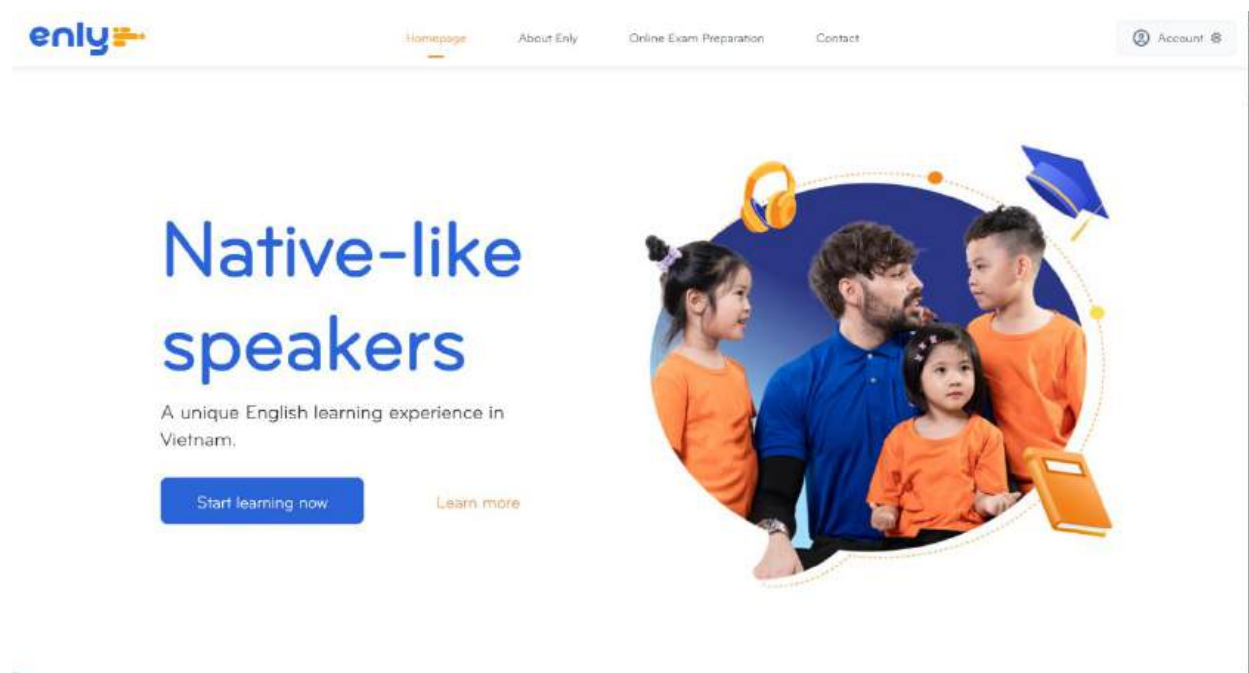


Figure 4.3. Home page

2.4. About page



Figure 4.4. About page

2.5. Practice page

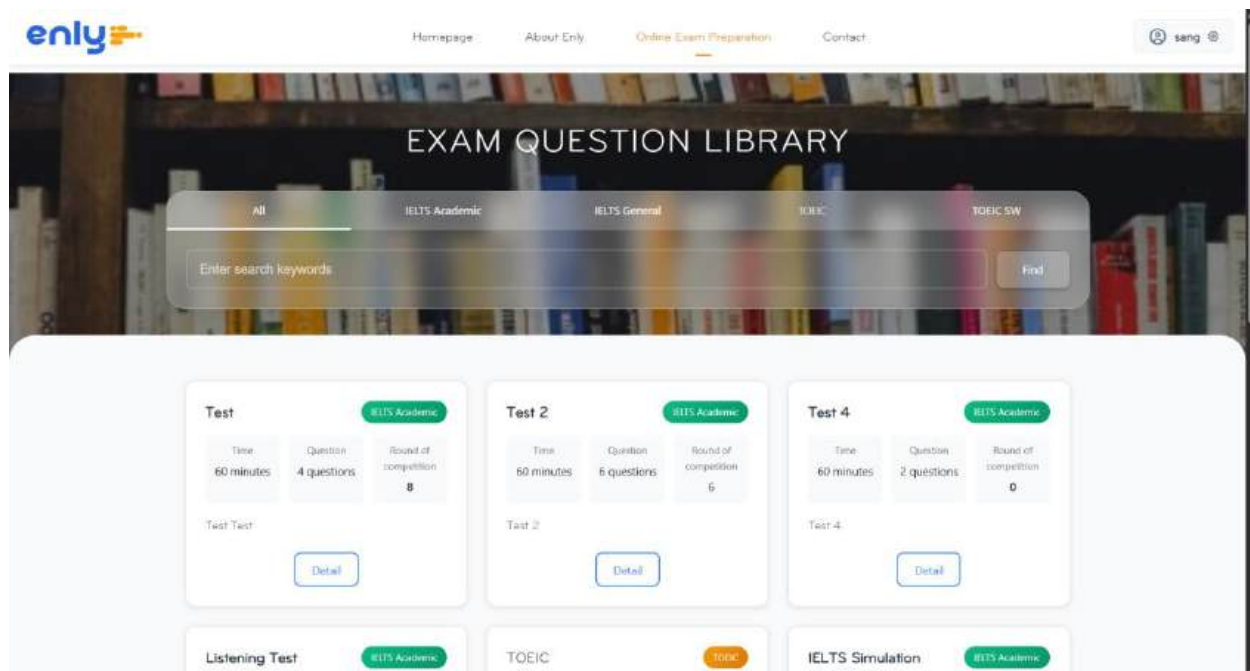


Figure 4.5. Practice page

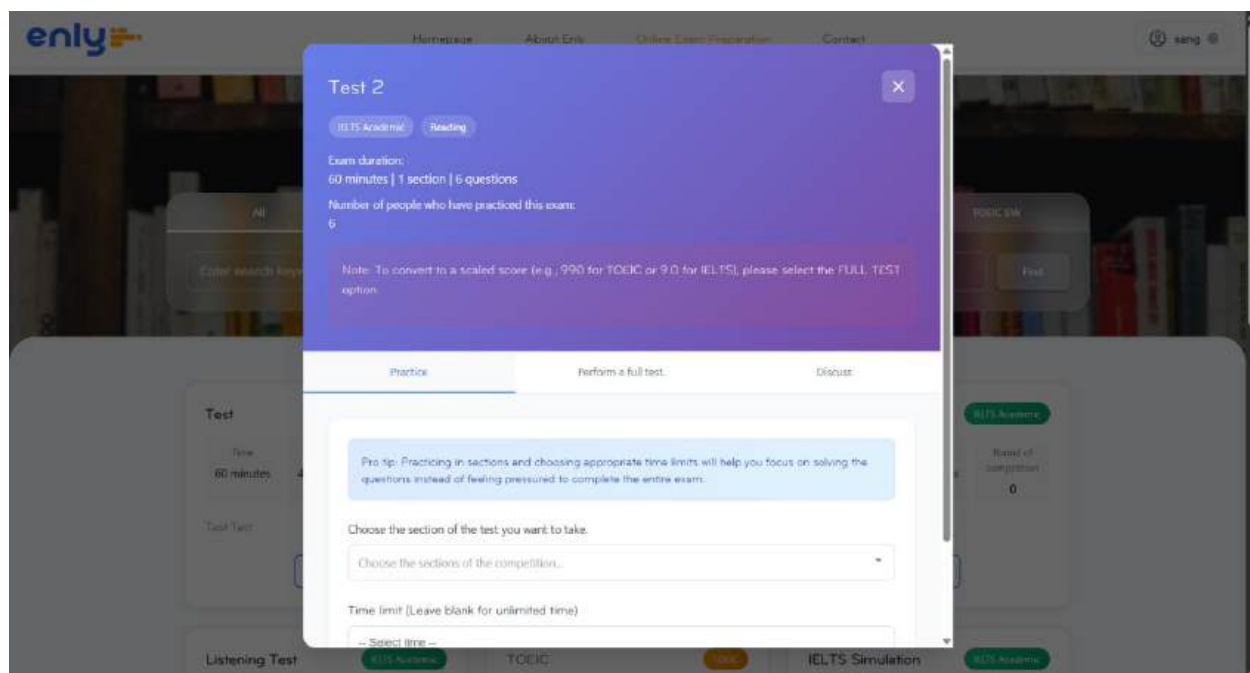


Figure 4.6. Test description pop up

2.6. Contact page

enly

Homepage About Enly Online Exam Preparation **Contact**

Account

Hello

Connect with us to begin your English learning journey.

Surname Name

Email Phone number

Message

Enter your message content...

Send a message

Figure 4.7. Contact page

2.7. Test manager page

enly

Homepage About Enly Online Exam Preparation **Exam management** Contact

ADMIN

EXAM MANAGEMENT

Create, edit, and manage your exams.

Search for exam papers... All types Create a new exam

ID	Exam title	Type	Time	Question	Creation date	Update	Operation
1	b b	IELTS GENERAL	0 minutes	1 sentence	26/12/2025	26/12/2025	 
2	a a	IELTS GENERAL	0 minutes	1 sentence	26/12/2025	26/12/2025	 
3	IELTS Simulation Reading test 7 No description	IELTS ACADEMIC	0 minutes	3 sentences	23/12/2025	23/12/2025	 
4	reading test 2 No description	IELTS ACADEMIC	0 minutes	1 sentence	21/12/2025	21/12/2025	 

Figure 4.8. Test manager page

2.7.1. Create test

enly

Homepage About Enly Online Exam Preparation Exam management Contact

ADMIN

+ Choose the type of exam you want to create.

Step 1: Choose the type of exam.

IELTS Academic
Academic IELTS test for studying abroad and research.

IELTS General
IELTS General Exam for Immigration and Work

TOEIC
Complete TOEIC Listening & Reading test

TOEIC SW
TOEIC Speaking & Writing Test

Lựa chọn của bạn:
Loại đề thi: IELTS Academic

Next

Figure 4.9. Create test page 1

enly

Homepage About Enly Online Exam Preparation Exam management Contact

ADMIN

+ Choose the type of exam you want to create.

Step 2: Choose a skill

Reading
Create a reading comprehension test with a passage and questions.

Listening
Create a listening test with audio files and questions.

Writing
Create a written exam with tasks and requirements.

Your choice
Exam type: IELTS Academic
Skill: Reading

Come back Start creating

Figure 4.10. Create test page 2

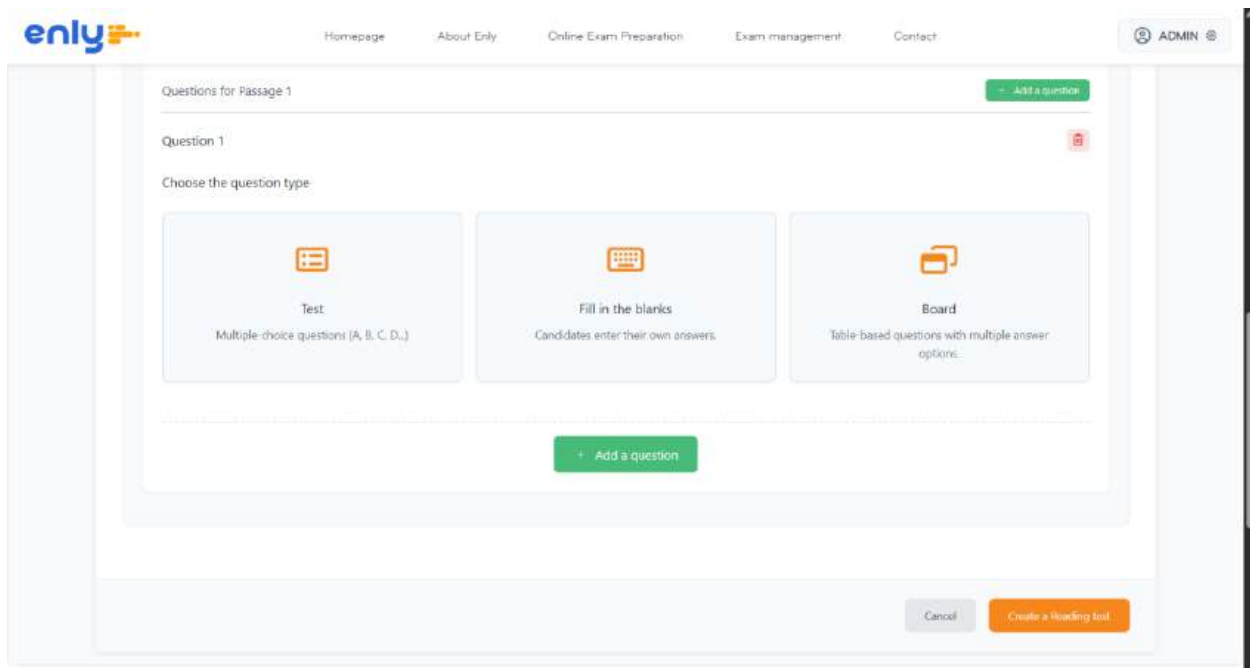


Figure 4.11. Create test page 3

2.7.2. Update the test

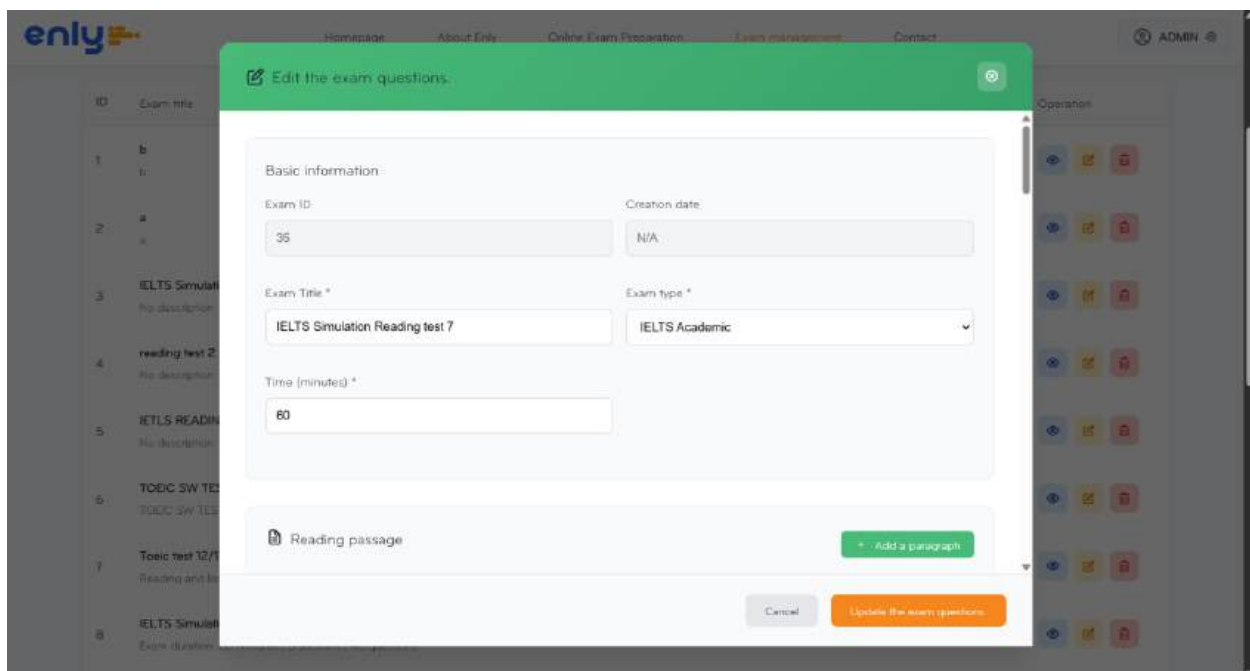


Figure 4.12. Update the test - UI

2.7.3. View test detail

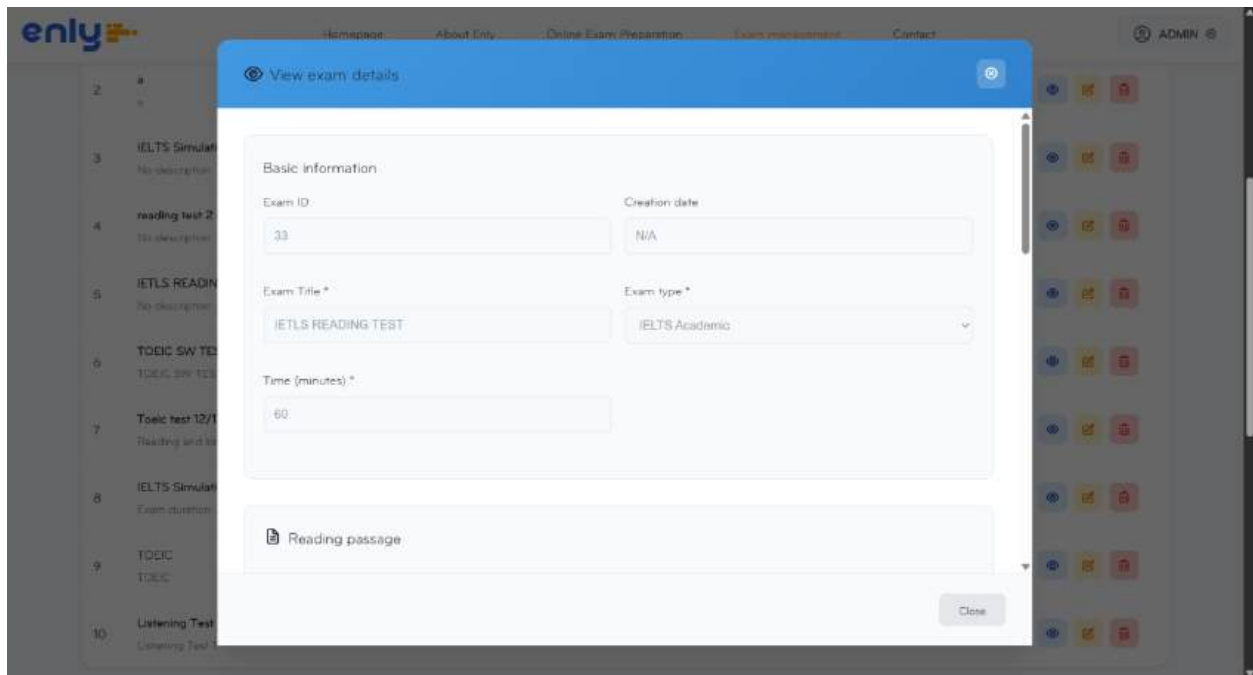


Figure 4.13. View test detail - UI

2.7.4. Delete the test

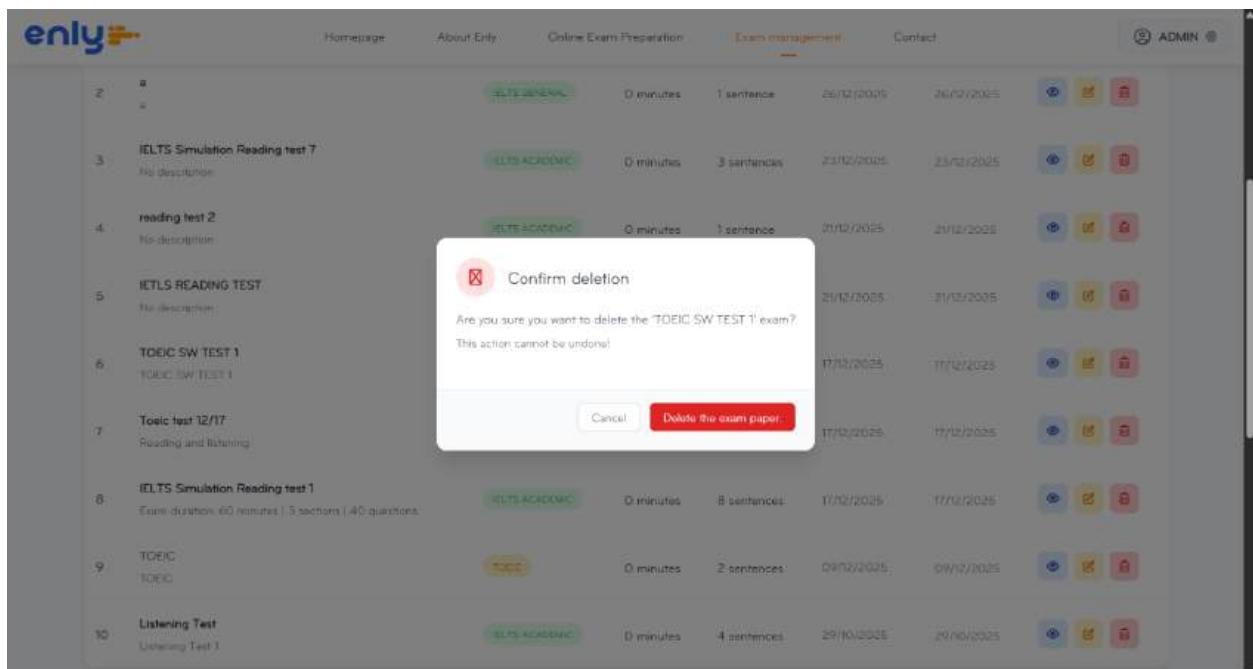


Figure 4.14. Delete the test - UI

3. Test case

3.1. Test case 1: Create a test successfully

Table 4.1. Test case 1 table

Field	Description
Test Case ID	TC-CT-001
Test Name	Create reading test successfully
Pre-condition	User is logged in as Teacher/Admin
Steps	1. Navigate to “Test manage” page 2. Click on the “Create a new test” button 3. Choose the type of test such as; IELTS, TOEIC,... 4. Choose the reading skill of test. 5. Enter valid test information (Title, Time, Description). 6. Enter the paragraph (include title, content) 7. Enter the question. 8. Click the “Create” button.
Expected Result	System saves the test, displays “Test created successfully”, and test appears in the test list

3.2. Test case 2: Duplicate test name

Table 4.2. Test case 2 table

Field	Description
Test Case ID	TC-CT-002
Test Name	Duplicate test title
Pre-condition	A test with the same title exists
Steps	1. Enter an existing test title 2. Add questions 3. Click “Create Test”

Expected Result	System displays error: “Test title already exists”
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CHAPTER 5. CONCLUSION

This project presents a comprehensive analysis and design of an online English learning and test-preparation system aimed at improving the effectiveness of studying for standardized exams such as IELTS and TOEIC. Through the application of Object-Oriented Analysis and Design (OOAD) and the use of UML modeling techniques, the project establishes a clear and structured blueprint for a scalable, maintainable, and user-centered platform. The system addresses key challenges identified during the requirements analysis, including the limitations of online test-taking, inefficiencies in exam content management, fragmented evaluation workflows, and technical barriers to scalability.

The proposed models - ranging from use-case diagrams and activity diagrams to sequence diagrams, class diagrams, and state diagrams - offer detailed insights into system behavior, interactions, and architecture. These models serve as a foundation for future implementation, ensuring that core functionalities such as test creation, test execution, automated scoring, user authentication, and teacher evaluation workflows are supported in a coherent and robust manner. Additionally, the relational database design derived from the entity class diagram ensures data consistency, integrity, and flexibility for storing diverse test formats and multimedia content.

While the study provides a solid conceptual framework, it acknowledges several limitations. Advanced AI capabilities, full UI/UX specifications, performance testing, and complete technical implementation lie outside the scope of this report. Nonetheless, the work establishes essential groundwork for further development. Future enhancements may include integrating AI-based scoring systems, improving analytical dashboards for learners, expanding the test library, and implementing performance optimization strategies.

In conclusion, this project demonstrates the importance of systematic analysis and design in building a modern educational platform. By identifying existing challenges and proposing a detailed solution architecture, the study contributes to the foundation of a reliable, efficient, and scalable system capable of supporting learners, teachers, and administrators in a rapidly evolving digital learning environment.